Post-Capitalist Labs Imprint · Working Draft Monograph

DISTRIBUTED PARTICIPATORY ECONOMICS PROTOCOL

A Rigorous, Source-Grounded Feasibility Study and Post-Quantum Technical Architecture for Democratic Planning, Self-Management, and the Productive Commons



DESIGNED FOR THREE INTERDISCIPLINARY AUDIENCES:

- Advocates & Cooperative Practitioners: Council democracy, balanced job complexes, and historical lessons from Catalonia (1936), the Paris Commune (1871), and Classical Athens.
- Economists & Mechanism Designers: Formal welfare theorems, Sandström's double-entry accounting, pollution demand revelation (CAPs), and candidate ADMM regularisation under convex restrictions.
- Systems Programmers & Cryptographers: Local-first CRDTs, Two-Root BLAKE3 aggregation, deterministic WASM IFB engines, and NIST FIPS 203/204/205 post-quantum primitives.

POST-CAPITALIST LABS IMPRINT · https://github.com/Post-Capitalist-Labs Working Draft · Nov 2026

Leveraging technology to amplify social movements · Decentralised consensus, mechanism design & democratic coordination

Post-Capitalist Labs Imprint · Open-Source Research Monograph Series

Distributed Participatory Economics Protocol (DPEP)

A Rigorous, Source-Grounded Feasibility Study and Post-Quantum Technical Architecture for Democratic Planning, Self-Management, and the Productive Commons

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About Post-Capitalist Labs (PCL)

At Post-Capitalist Labs, we harness the transformative power of technology to bolster social movements and build directly democratic alternatives to capitalism. Our work spans decentralised consensus, applied cryptography, operations research, and democratic mechanism design.

Our Vision: We envisage a world where solidarity, care, classlessness, diversity, self-reflection, and direct democracy are not just ideals, but the foundations of everyday life. Our goal is to make this vision a reality through the thoughtful application of open-source technology, where people remain in control.

Our Approach:

- Learning from History: Analyzing past and present democratic movements and self-managed collectives to guide institutional design.
- Emancipatory Modeling: Developing rigorous mathematical and computational models for equitable, non-market resource coordination.
- Open-Source Engineering: Turning theoretical models into verifiable, accessible, community-driven proofs of concept and protocols.

Organization Profile: https://github.com/Post-Capitalist-Labs
Repository: https://github.com/Post-Capitalist-Labs/distributed-participatory-economics-protocol

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[1] Full list: research/SOURCE_AUDIT.md in the project repository.

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Interdisciplinary Navigation Guide

How to Read This Book Across Disciplines

This monograph bridges libertarian-socialist institutional theory, formal welfare and planning economics, and modern distributed systems cryptography. Depending on your primary background, we recommend the following reading paths:

FOR ADVOCATES, COOPERATORS & ORGANIZERS — Institutional Self-Management Without Hierarchy

Start with Chapter 1 (Executive Summary) and Chapter 2 (Historical & Institutional Inspirations) to see how lessons from the 1936–1939 Spanish Revolution, the 1871 Paris Commune, Classical Athens, Occupy Wall Street, Mondragón, Porto Alegre Participatory Budgeting, Rojava, and Elinor Ostrom's commons principles are translated into concrete protocol safeguards against bureaucratic capture, meeting fatigue, and inter-regional inequality. Then read Chapter 3 for the complete traceability matrix linking Michael Albert and Robin Hahnel's 1991 and 2021 books to our design.

FOR ECONOMISTS & OPERATIONS RESEARCHERS — Formal Models, Accounting & Non-Convex Convergence

Focus on Chapter 4 (Formal Economic Model), which unites Albert and Hahnel's Formal Model of a Participatory Economy (FMPE) with Anders Sandström's double-entry Society Account and Robin Hahnel's Pollution Demand-Revealing Mechanism (PDRM). See Section 4.4 for our source-code audit of pequod-cljs and the four research-candidate solvers (Algorithms 3a–3d), alongside Chapter 8 and Chapter 10 for engagement with John O'Neill (2024) and the non-convex experiments still required.

FOR SYSTEMS PROGRAMMERS & CRYPTOGRAPHERS — Local-First P2P, Two-Root Aggregation & Post-Quantum Security

Jump directly to Chapter 4 (Section 4.5 State Machine & Canonical CBOR Schemas), Chapter 5 (Four-Layer Architecture D & 6-Role Mobile Taxonomy), Chapter 6 (Weighted Stack Comparison), Chapter 7 (NIST FIPS 203/204/205, Sybil Resistance & Prio3 VDAF Aggregation), and Chapter 9 (Experimental Programme, Kill Gates & Multi-Scale Estimates).

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List of Abbreviations and Mathematical Notation

Because this monograph integrates participatory economics, labor and cooperative history, operations research, distributed systems, and post-quantum cryptography, this reference table defines every abbreviation and mathematical symbol used across Chapters 1–10.

A. Participatory Economics, Accounting, and Bibliographic Abbreviations

Abbreviation	Full Term	Definition & Role in Monograph
BJC	Balanced Job Complex	Bundle of workplace tasks balanced for empowerment and desirability within and across workplaces (PEPE Ch. 2).
CAP	Community of Affected Parties	Ecological/geographic collective of residents impacted by a specific pollutant in an airshed or watershed (DEP Ch. 7).
CC (CC_h)	Consumers' Council	Democratic neighbourhood consumer council or nested ward/city/regional consumer federation.
CFB	Consumption Facilitation Board	Informational board assisting consumer councils and federations in evaluating collective consumption projects (LF Ch. 6).
CLUEA	Consejo Levantino Unificado de Exportación Agrícola	Unified Levantine Council for Agricultural Export organised by the CNT and UGT in Valencia (1936–1939).
CNT	Confederación Nacional del Trabajo	Anarcho-syndicalist National Confederation of Labour in the 1936–1939 Spanish Revolution.
CPR	Common-Pool Resource	Shared natural or human-made resource system governed by collective-choice rules (Ostrom 1990).
DEP	Democratic Economic Planning	Primary reference volume by Robin Hahnel et al. (Routledge, 2021).
DPEP	Distributed Participatory Economics Protocol	The open-source decentralised coordination protocol specified in this monograph.
EFB	Employment Facilitation Board	Facilitation board assisting workers transferring between councils and balancing cross-workplace BJCs (LF Ch. 6).
EJPE	Erasmus Journal for Philosophy and Economics	Academic journal hosting the Winter 2024 symposium on Hahnel's Democratic Economic Planning.
FMPE	Formal Model of a Participatory Economy	Mathematical formalisation of participatory planning analyzed in PEPE Chapter 5 (pp. 73–106).
FTE	Full-Time Equivalent	Unit of engineering or research staffing effort across roadmap scenarios.
GA	General Assembly	Direct plenary deliberative assembly of a workplace, village collective, or horizontal movement.

Continued from previous page

Abbreviation	Full Term	Definition & Role in Monograph
GDP	Gross Domestic Product	Aggregate social value of net final output (Y^{GDP}) cleared in the Society Account.
GHG	Greenhouse Gas	Atmospheric emissions (CO_2-equivalent) subject to hard biophysical ceilings ($E_z^{\max}$).
HFB	Household Facilitation Board	Facilitation board assisting individuals and families with housing and neighbourhood membership (LF Ch. 6).
IF (IF_k)	Industry Federation	Federation of worker councils within industry k managing sector capacity, R&D, and SB/SC appeals.
IFB	Iteration Facilitation Board	Informational board/engine that aggregates supply and demand and updates indicative prices via public rules.
LF	Looking Forward	Primary reference volume by Michael Albert and Robin Hahnel (South End Press, 1991).
Parecon / PE	Participatory Economics	Post-capitalist economic model developed by Michael Albert and Robin Hahnel.
PB	Participatory Budgeting / Planning Bureau	Municipal democratic budgeting (Porto Alegre, 1989); also denotes the abstract Planning Bureau in FMPE Ch. 5.
PCL	Post-Capitalist Labs	Open-source research and engineering collective publishing this monograph (https://github.com/Post-Capitalist-Labs).
PDRM	Pollution Demand-Revealing Mechanism	Iterative procedure pricing local pollutant damages via CAP emission tolerances and credits (DEP Ch. 7).
PEP	Participatory Economy Project	Research collective maintaining participatoryeconomy.org and the 2023 User Story Mapping.
PEPE	The Political Economy of Participatory Economics	Primary technical reference volume by Michael Albert and Robin Hahnel (Princeton University Press, 1991).
PLAC	Parecon: Life After Capitalism	Comprehensive institutional and daily-life synthesis of Participatory Economics by Michael Albert (Verso Books, 2003).
PFB	Production Facilitation Board	Board assisting worker councils with production externalities, scale economies, and technology choice (LF Ch. 6).
PIPA	Practically Identical Production Activities	Equivalence classes $\{PIPA\}_i$ partitioning workplace tasks by qualitative empowerment effect (PEPE Ch. 2, p. 26).
SB / SC	Social Benefit / Social Cost	Ratio $SB_j(P)/SC_j(P)$ measuring the social value of a worker council's outputs relative to the opportunity cost of its inputs and endowments (≥ 1 constraint).
SKU	Stock Keeping Unit	Granular product variant (size, color, style) resolved at intra-year fulfilment from coarse planning categories.

Continued from previous page

Abbreviation	Full Term	Definition & Role in Monograph
UFB	Updating Facilitation Board	Facilitation board mediating intra-year plan adjustments when demand shifts or supply shocks exceed slack (LF Ch. 6).
UGT	Unión General de Trabajadores	Socialist General Union of Workers that co-organised industrial and agrarian collectives in Spain (1936–1939).
WC (WC_j)	Workers' Council	Democratic workplace collective where each worker has one vote and holds a Balanced Job Complex.

B. Operations Research, AI, Distributed Systems, and Cryptography Abbreviations

Abbreviation	Full Term	Definition & Role in Monograph
ADMM	Alternating Direction Method of Multipliers	Augmented-Lagrangian primal-dual method; Algorithm 3b is the proximal Jacobian variant of Deng et al. (2017), a research candidate.
AP / CP	Availability / Consistency under Partition	CAP theorem properties: local-first drafting is AP (available offline), while Epoch BFT price finality is CP.
ARIMA / ETS	AutoRegressive Integrated Moving Average / Error-Trend-Seasonality	Classical statistical time-series forecasting models used for baseline demand and SKU disaggregation.
BFT	Byzantine Fault Tolerance	Distributed consensus property guaranteeing safety even when up to $f < n/3$ validator nodes act maliciously.
BLAKE3	BLAKE3 Cryptographic Hash	Hash function with 256-bit output and a 128-bit security target; used for the Two-Root aggregation trees and content addressing.
BSP	Bulk-Synchronous Parallel	Execution model alternating parallel local computation rounds with synchronised global aggregation checkpoints.
CBOR (CDE)	Concise Binary Object Representation	RFC 8949 Canonical Deterministic Encoding used for all signed DPEP messages with fixed 10^{-6} micro-unit scale.
CapCert	Capability Certificate	Chained, signed certificates encoding council membership and revocable delegation. Format to be chosen (UCAN 1.0 and Biscuit are candidates); signed under the active CryptoSuiteId (Section 7.2.1).
CID	Content Identifier	Self-certifying cryptographic hash address referencing a redactable qualitative List or plan blob.
DA	Data Availability	Verification that all leaf proposals of a round root are downloadable by auditing nodes (0.24–1.46 GB raw per round at $N = 10^6$; Table 5.1).
FLP	Fully Linear Proof	VDAF-22 proof on secret shares; binds report validity, not aggregator honesty.

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Abbreviation	Full Term	Definition & Role in Monograph
Prio3 / VDAF	Verifiable Distributed Aggregation Function	IETF CFRG draft-irtf-cfrg-vdaf-22 (Informational). Privacy needs one honest aggregator; correct totals need all of them.
CRDT	Conflict-Free Replicated Data Type	Data structure (Loro / Automerge) allowing offline concurrent edits in council workspaces that merge deterministically.
CRQC	Cryptographically Relevant Quantum Computer	Future fault-tolerant quantum computer capable of breaking classical elliptic-curve and RSA cryptography via Shor's algorithm.
DAO	Decentralised Autonomous Organisation	Blockchain governance construct critiqued in Chapter 2 for conflating token voting with council democracy.
DTN	Delay-Tolerant Networking	Store-and-forward networking (including LoRa mesh and USB data-mules) for intermittently connected regions.
EEIO / LCA	Environmentally Extended Input-Output / Lifecycle Assessment	Matrix accounting methods estimating cradle-to-gate resource and emission footprints across supply chains.
FIPS	Federal Information Processing Standards	NIST post-quantum cryptographic standards finalised in August 2024: FIPS 203, FIPS 204, and FIPS 205.
FRI / STARK	Fast Reed-Solomon IOP / Scalable Transparent ARgument of Knowledge	Discarded household-budget design; not part of Architecture D.
IEEE-754	IEEE Floating-Point Standard	Hardware floating-point representation replaced in DPEP core rules by deterministic 64-bit fixed-point integers.
KKT	Karush-Kuhn-Tucker Conditions	First-order necessary and sufficient optimality conditions for constrained nonlinear/conic optimisation.
LP / QP / MILP	Linear / Quadratic / Mixed-Integer Linear Programming	Mathematical optimisation paradigms solved locally via Clarabel.rs and HiGHS.
MAPE	Mean Absolute Percentage Error	Accuracy metric used to audit and backtest statistical demand forecasts.
ML-DSA-65	Module-Lattice Digital Signature Algorithm	NIST FIPS 204 (Dilithium3) post-quantum digital signature scheme combined in hybrid mode with Ed25519.
ML-KEM-768	Module-Lattice Key-Encapsulation Mechanism	NIST FIPS 203 (Kyber-768) post-quantum key-establishment scheme combined in hybrid mode with X25519.
MLS	Messaging Layer Security	IETF RFC 9420 protocol providing end-to-end encrypted group messaging inside councils.
MPC	Secure Multiparty Computation	Candidate hardening of aggregator correctness (Table 7.4); not the Phase-1 construction.
Two-Root	Two-Root BLAKE3 aggregation	Proposal Merkle root plus category-totals root (Section 5.2.1). Replaces internal-vector Sparse Merkle-Sum Trees.
NASA-TLX	NASA Task Load Index	Standardised workload questionnaire proposed for measuring participation fatigue in Phase 1H and Phase 3.

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Abbreviation	Full Term	Definition & Role in Monograph
HRU	Household Reporting Unit	Per-category integer unit for household Prio3 reports, converted exactly to protocol micro-units (Section 7.4.1).
NMT / TTS	Neural Machine Translation / Text-to-Speech	Local accessibility models translating qualitative Lists across languages and assisting visually impaired users.
OR	Operations Research	Discipline of mathematical optimisation, linear programming, and supply-chain clearing.
P2P	Peer-to-Peer	Decentralised networking architecture without central client-server gatekeepers.
PoC	Proof of Concept	A proposed, unbuilt reference implementation (Phases 1–3, Chapter 9); no implementation exists.
PQC	Post-Quantum Cryptography	Cryptographic algorithms resistant to both classical and quantum (Shor/Grover) attacks.
PWA	Progressive Web Application	Offline-capable cross-platform web client running the Rust/WASM Role-LP engine on smartphones and desktops.
QUIC	Quick UDP Internet Connections	Modern encrypted, multiplexed transport protocol (quinn / iroh) used for Layer 2 federation sync.
Role-LP / CC / SR / PCW / BFT / GW	DPEP Node Roles 1–6	Lightweight Participant (LP), Council Coordinator (CC), Storage Replica (SR), Planning Compute Worker (PCW), Epoch Validator (BFT), and Delay-Tolerant Gateway (GW).
SLH-DSA	Stateless Hash-Based Digital Signature Algorithm	NIST FIPS 205 (SPHINCS+) signature standard relying solely on hash functions, used for long-term root anchors.
TPS	Transactions Per Second	Throughput metric shown in Chapter 5 to be irrelevant to epoch-synchronised participatory planning.
TTL	Time-To-Live	Hard expiration timer on delegate capability certificates and emergency replanning playbooks.
UniFFI	Universal Foreign Function Interface	Mozilla binding generator exposing the Rust dpep-core library to native Kotlin (Android) and Swift (iOS).
WASM	WebAssembly	Deterministic, sandboxed binary instruction format (Wasmtime) executing IFB price-update rules.

C. Summary of Core Mathematical Notation (Chapter 4)

Symbol	Mathematical Meaning
Ω	Master catalogue of all $G = \lvert\Omega\rvert$ priced economic categories: $\mathcal{G}^{\text{priv}} \sqcup \mathcal{G}^{\text{pub}} \sqcup \mathcal{G}^{\text{int}} \sqcup \mathcal{R} \sqcup \mathcal{L} \sqcup \mathcal{K} \sqcup \mathcal{Z}$.
$P^{(t)} \in \mathbb{R}_{+}^{\lvert\Omega\rvert}$	Vector of indicative social opportunity-cost prices at planning iteration t.
$\bar{a}_j, \underline{a}_j, r_j, e_j$	Worker council WC_j's proposed outputs ($\bar{a}_j$), intermediate inputs ($\underline{a}_j$), natural resources (r_j), and emissions (e_j).

$l_j^0, k_j^0, \Delta l_j, \Delta k_j$	Initial labour (l_j^0) and capital (k_j^0) endowments of WC_j, and proposed net additions/releases ($\Delta l_j, \Delta k_j$).
$r^*, l^*, k^*, E_z^{\max}$	Economy-wide available natural resources (r^*), labour supplies (l^*), capital stocks (k^*), and hard biophysical emission caps ($E_z^{\max}$).
$w_{\text{base}}, \epsilon_u, B_u(P)$	Base hourly remuneration rate (w_{base}), worker effort multiplier (ϵ_u), and citizen u's warranted annual consumption budget (B_u).
$S_i^{(t)}, D_i^{(t)}, SL_i^{(t)}$	Aggregate supply ($S_i^{(t)}$), aggregate demand ($D_i^{(t)}$), and planned slack buffer ($SL_i^{(t)} = \eta_i S_i^{(t)}$) for category i.
$\rho_i^{(t)}, \tau$	Normalized relative excess demand imbalance $\rho_i^{(t)} = \frac{2\lvert S_i^{(t)} - D_i^{(t)} \rvert}{S_i^{(t)} + D_i^{(t)}}$ and two-sided clearing tolerance $\tau \in [0.01, 0.05]$ (Definition 4.2).

CHAPTER 1

Executive Summary: Feasibility, Architecture, and Strategic Roadmap

> "No central agency composes the final plan. No market competition generates the final plan. A decentralised, social, iterative, communication process allows all actors to democratically formulate the final plan."
>
> — Michael Albert and Robin Hahnel, Looking Forward: Participatory Economics for the Twenty First Century (1991), Ch. 6[1]

1.1 What Is Being Proposed

This monograph sets out a source-grounded feasibility assessment, a gated research programme, and a technical architecture proposal for the Distributed Participatory Economics Protocol (DPEP): an open-source, local-first, cryptographically verifiable coordination protocol intended to enable democratic worker councils, neighbourhood consumer councils, and nested federations to formulate, iterate, ratify, and execute comprehensive economic plans without capitalist markets or authoritarian central planning.

The intellectual foundation of this protocol is Participatory Economics (Parecon), originally developed by Michael Albert and Robin Hahnel in The Political Economy of Participatory Economics (1991) and Looking Forward (1991),[2] elaborated across daily-life institutions and critic rebuttals in Michael Albert's Parecon: Life After Capitalism (2003),[3] and mathematically extended in Robin Hahnel's Democratic Economic Planning (2021),[4] Anders Sandström's Anarchist Accounting (2020),[5] and Mitchell Szczepanczyk's computer simulation work (pequod-cljs,

[1] Michael Albert and Robin Hahnel, Looking Forward: Participatory Economics for the Twenty First Century (Boston, MA: South End Press, 1991), Ch. 6 ("Participatory Allocation"), accessed via ZNetwork authorized online edition (https://znetwork.org/wp-content/uploads/zbooks/htdocs/ParEcon/lookingforward/lf6_1.htm).

[2] Michael Albert and Robin Hahnel, The Political Economy of Participatory Economics (Princeton, NJ: Princeton University Press, 1991), https://doi.org/10.2307/j.ctv131bvr2; Albert and Hahnel, Looking Forward (1991).

[3] Michael Albert, Parecon: Life After Capitalism (London & New York: Verso Books, 2003), full text archived at ZCommunications/ZNetwork (http://books.zcommunications.org/books/pareconv/).

[4] Robin Hahnel (with contributions by Peter Bohmer, Savvina Chowdhury, Allison Kerkhoff, Mitchell Szczepanczyk, and Michael Weisdorf), Democratic Economic Planning (Abingdon & New York: Routledge, 2021), https://doi.org/10.4324/9781003173700.

[5] Anders Sandström, Anarchist Accounting: Accounting Principles for a Democratic Economy (Abingdon & New York: Routledge, 2020), https://doi.org/10.4324/9781003036005; see also Sandström, "An Accounting System for a Libertarian Socialist Economy," Participatory Economy Project (Feb. 14, 2023), https://participatoryeconomy.org/an-accounting-system-for-a-libertarian-socialist-economy/.

2023).[6]

DPEP provides the computational substrate for four core institutional pillars:

1. Self-Managed Work and Consumption: Democratic worker councils (WC_j) and consumer councils (CC_h) in which participants exercise decision-making input proportional to the degree they are affected by an economic choice.[7]
2. Equitable Remuneration and Balanced Job Complexes (BJCs): Non-transferable consumption allowances calibrated to the duration, intensity, and onerousness (effort and sacrifice) of socially valued labour, supplemented by universal free public goods and council-approved special-need allowances, paired with job complexes balanced for empowerment and desirability within and across workplaces.[8]
3. Decentralised Iterative Planning: A multi-round coordination protocol in which councils submit production and consumption proposals, federated aggregators compute excess demands across seven economic categories (private goods, public goods, intermediate inputs, natural resources, labour categories, capital stocks, and pollutant emissions), and a deterministic Iteration Facilitation Board (IFB) engine updates indicative social opportunity-cost prices until a feasible, ecologically bounded plan is ratified.[9]
4. Intra-Year Operational Clearing and Scoped Replanning: Continuous tracking of deliveries and inventory drawdowns against the ratified plan, absorbing routine consumer preference shifts locally within neighbourhood and city slack buffers while mediating larger supply shocks across affected industry federations via Updating Facilitation Boards (UFBs).[10]

1.2 Why It Matters: Breaking the "No Third Way" Dogma

For over a century, mainstream and socialist economists alike have been haunted by Alec Nove's famous assertion in The Economics of Feasible Socialism (1983):

> "In a complex industrial economy the interrelation between its parts can be based in principle either on freely chosen negotiated contracts [i.e., markets],

[6] Mitchell Szczepanczyk, "Pseudocode and Algorithms for Computer Simulations of Democratically Planned Economies," Journal of Information Economics 1, no. 3 (2023): 43–54, https://doi.org/10.58567/jie01030004; open-source repository at https://github.com/msszczep/pequod-cljs.

[7] Albert and Hahnel, Political Economy of Participatory Economics, Ch. 1, pp. 8, 16; Ch. 2, p. 24; Hahnel, Democratic Economic Planning, Ch. 1, pp. 14–16.

[8] Albert and Hahnel, Political Economy of Participatory Economics, Ch. 2, pp. 25–30; Ch. 3, pp. 44–55.

[9] Albert and Hahnel, Political Economy of Participatory Economics, Ch. 4, pp. 57–72; Ch. 5, pp. 73–106; Hahnel, Democratic Economic Planning, Ch. 7, pp. 130–162.

[10] Albert and Hahnel, Political Economy of Participatory Economics, Ch. 4, pp. 66–67; Looking Forward, Ch. 6 (lf6_1.htm); Hahnel, Democratic Economic Planning, Ch. 8, pp. 167–170.

> or on a system of binding instructions from planning offices [i.e., command planning]. There is no third way."[11]

Yet historical experiments in democratic self-management—most notably the industrial and agrarian collectives of the Spanish Revolution (1936–1939), the Paris Commune (1871), federated cooperatives like Mondragón, and horizontal movements like Occupy Wall Street (2011)—repeatedly encountered two structural barriers when attempting to coordinate beyond single workplaces or rural villages:

- The Accounting and Valuation Barrier: Without a unified social accounting and indicative-price system, wealthier collectives hoarded superior inherited machinery (what Gaston Leval and Frank Mintz termed "collective egoism" or neocapitalismo colectivo in 1936 Barcelona),[12] while inter-industry coordination of complex intermediate inputs broke down into cumbersome bilateral barter.
- The Deliberative Scaling Barrier: Without structured, asynchronous computational protocols, horizontal deliberation tended towards exhausting multi-hour plenary meetings (speaking time grows at least linearly with attendance, and possible pairwise exchanges quadratically; this is arithmetic, not an empirical threshold, Section 2.5) or unaccountable informal cliques (Jo Freeman's "tyranny of structurelessness").[13]

Contemporary distributed computing, post-quantum cryptography, and mathematical optimisation make it conceivable to attack both barriers at once. Whether millions of participants can coordinate industrial supply chains and ecological boundaries democratically, and at what wall-clock cost per round, is an experimental claim: Category B estimates in Table 1.1 and Table 9.3 are schema-derived, not measured at that scale.

1.3 Technical Feasibility: Demonstrated Results, Estimates, and Open Targets

A core mandate of this monograph is to separate what has been shown, what has been estimated, and what is unknown.

[11] Alec Nove, The Economics of Feasible Socialism (London: George Allen & Unwin, 1983), pp. ix–x, 44; quoted directly at the opening of Chapter 4 in Albert and Hahnel, Political Economy of Participatory Economics, p. 57.

[12] Gaston Leval, Collectives in the Spanish Revolution, trans. Vernon Richards (London: Freedom Press, 1975); Frank Mintz, Anarchism and Workers' Self-Management in Revolutionary Spain, trans. Paul Sharkey (Oakland, CA: AK Press, 2013); Sam Dolgoff, ed., The Anarchist Collectives: Workers' Self-Management in the Spanish Revolution, 1936–1939 (New York: Free Life Editions, 1974), Ch. 6.

[13] Jo Freeman, "The Tyranny of Structurelessness," The Second Wave 2, no. 1 (1972) / Berkeley Journal of Sociology 17 (1972–1973): 151–164; David Graeber, The Democracy Project: A History, a Crisis, a Movement (New York: Spiegel & Grau, 2013).

What is established, estimated and unknown (plain statement). Established: the 1991 model and its later formalisations exist and are internally coherent at the level of accounting identities; the pequod-cljs Cobb–Douglas simulations reach a 5% imbalance in about 12 rounds (fixed incomes, smooth convex technology); the unmodified price rule fails on Scarf's cycle and on a single indivisible plant; the NIST post-quantum primitives are standardised, and every function in the stack except threshold signatures, anonymous credentials, the sortition beacon, capability-certificate formats and post-quantum HPKE has a standardised, deployed primitive (Table 6.3), although almost every way DPEP combines them is original; and Prio3SumVec report sizes and desktop timings have been measured with libprio. Estimated (not measured at scale): round latency, archive sizes, mobile bandwidth, hosting cost, and aggregator CPU at 10^6 participants. Unknown: whether any candidate solver clears economies with Leontief inputs, lumpy plants, endogenous incomes and strategic reporting; whether ordinary people can and will take part in multi-round planning without a technical elite forming around the software; how household budgets can be checked privately when prices change between rounds; and whether a credential scheme can enforce one report per person without a population register. Phase 1 and Phase 1H (Chapter 9) are designed to settle the first two unknowns, or to stop the programme if they cannot be settled.

1.4 The Most Promising Architecture: Why We Reject a Global Blockchain

We explicitly reject building a per-transaction global blockchain (Architecture C), a centralised cloud database (Architecture A), or an unstructured gossip network without finality (Architecture B).

Participatory planning is not a stream of high-frequency financial trades; it is an epoch-synchronised, hierarchical bulk-synchronous coordination process followed by localized intra-year fulfilment. Furthermore, putting personal consumption requests on a public ledger violates personal privacy, and broadcasting 3.3 KB post-quantum signatures for every household action across all global nodes wastes bandwidth and energy.

Instead, our weighted evaluation (Chapter 6: 4.07 / 5.00 for D against 3.10 for a federated SQL service and 2.54 for a blockchain; expert judgements, not measurements, and A overtakes D when simplicity, throughput and cost carry 70% of the weight) selects Architecture D — A Four-Layer Hybrid Local-First, Federated P2P, and Selective Epoch-BFT Architecture:

1. Layer 4 (Human Deliberation & Local Solver Edge): Accessible mobile/desk-

[14] Rows 1 and 2 of Category A are computed by research/analysis/verify_r2.py (Scarf's cycle by research/analysis/verify_economics.py) in the project repository.

[15] Leonid Hurwicz, "On Informationally Decentralized Systems," in Decision and Organization, ed. C. B. McGuire and R. Radner (Amsterdam: North-Holland, 1972), pp. 297–336; Hurwicz, "The Design of Mechanisms for Resource Allocation," American Economic Review 63, no. 2 (1973): 1–30.

Table 1.1: Three-Tier Evidence Classification: Demonstrated Results, Analytical Estimates, and Research Targets

Category A: Empirically Demonstrated (Code / Data Audited)	Category B: Analytical Estimates (Schema / Complexity Models)	Category C: Open Research Targets (Require Phase 1–3 Validation)
1. Smooth Cobb-Douglas Convergence (pequod-cljs CSVs): Across the 40 large-run logs the book cites (dep1ex61–100.csv),[14] Year 1 reaches $\rho_i \leq 5\%$ in 11.85 rounds and $\rho_i \leq 3\%$ in 19.20 rounds; Year 2 reaches 5% in 6.58 rounds. The 50 small-run files (dep1ex01–50.csv) average 11.18 Year-1 5% rounds. All 70 dep1ex*.clj generators (20 stored via Git LFS) were audited.	1. Distributed Round Latency ($N = 10^6$): A round moves a 0.24–1.5 GB raw archive of 40k council proposals (Table 5.1) plus a BFT checkpoint, and verifying 10^6 household reports costs each aggregator ≈ 4 (coarse) to ≈ 61 (dense) desktop core-minutes. No round-latency figure is claimed; Kill Gate 2 measures it against a pre-registered threshold.	1. Non-Convex & Leontief Convergence (Kill Gate 1): Whether any of Algorithms 3a–3d clears (not merely satisfies one-sided feasibility; Definition 4.2) Leontief + lumpy setup-cost economies in ≤ 25 rounds on pre-registered instances, including Scarf cycles, infeasible plans and endogenous income.
2. Refutation of Unregularised Tâtonnement on Scarf & Single-Plant Jumps: Fails Scarf's 3-good Leontief cycle ($\min \rho = 10.5\%$ over 3,000 rounds) and single indivisible plants ($\min \rho = 50.4\%$).	2. One-Good Shapley–Folkman Stress Test ($K = 20$): The minimum imbalance across 500 rounds is 0.40%, but the maximum in the last 100 rounds is 8.4% at $a = 70$; a sweep $a \in [60, 80]$ has median last-100-round max 6.0% and worst 9.5%. This says nothing about interdependent Leontief markets.	2. Human Override & Agency (Kill Gate 1H): Whether 30–60 self-selected volunteers modify solver drafts, and what that can and cannot show about democratic agency.
3. NIST PQC Standards and measured Prio3 sizes: FIPS 203/204/205 are standardised. draft-irtf-cfrg-vdaf-22 is an Informational IRTF draft with no post-quantum claim. Prio3SumVec sizes and desktop timings were measured with libprio 0.18.1 (Table 7.5): a dense 3,000-category, 20-bit report is 975,760 B to the leader and costs ≈ 3.5 ms to verify per aggregator on one desktop core.	3. Mobile Round Bandwidth (Role-LP): Quorum signatures at $n = 128$ are ≈ 290 kB ($86 \times 3{,}373$ B). A proposed coarse 200-category, 16-bit household report is 55 kB; a dense 3,000-category report does not fit the budget. Kill Gate 3 sets separate light-client and full-verification caps (Table 9.2).	3. Incentive & Soft-Budget Resistance: Mitigating capacity understatement (Weitzman 1980), soft-budget appeals (Kornai 1980), and carousel fraud (Hurwicz 1972) remains an open mechanism-design problem.[15]

top Progressive Web App (PWA) and native clients running an embedded interior-point conic/quadratic solver (Clarabel.rs / HiGHS in WASM) for instant local proposal drafting and "what-if" collective-good exploration.

2. Layer 3 (Local-First Council & Household State): Encrypted local SQLite (SQLCipher) for structured vectors and accounting ledgers + Loro/Automerge CRDTs for offline-capable workplace and neighbourhood council deliberation, authorised via revocable capability certificates (CapCert; format to be chosen, signed under the active CryptoSuiteId).
3. Layer 2 (Federated Aggregation & Content-Addressed Data Plane): Peer-to-peer QUIC (quinn / iroh) transport organising councils into nested Ward → City → Region and Industry Federation Two-Root BLAKE3 authenticated aggregation (proposal Merkle root + category-totals root; Section 5.2.1), paired with content-addressed storage for qualitative council narratives (Lists).
4. Layer 1 (Selective Epoch-BFT Checkpoint & Deterministic IFB Engine): A rotating BFT committee of $n = 128$ validators (drawn by sortition; an interim classical threshold beacon is adopted and a post-quantum beacon is open, Sections 7.2.1 and 7.6) that commits one checkpoint per planning round, plus daily operational rollups and scoped replans—425 blocks/year at a 10-round plan $(10 + 365 + 50)$, or 435 at 20 rounds—anchoring the two aggregation roots, executing the deterministic WASM price-update rule, and settling Sandström's macro Society Account. "10–20 checkpoint blocks" refers only to the annual planning rounds, not the Layer-1 archive.

1.5 Where AI Adds Value—and Where It Must Be Excluded

In strict adherence to Parecon's anti-coordinator principles,[16] AI must never become an unaccountable planning authority:

- Deterministic Core (by design, not yet implemented): Indicative price adjustments, SB/SC ratios, and Society Account clearing are specified as deterministic fixed-point arithmetic and convex operations-research solvers with no neural-network weights. Local IEEE-754 drafts are advisory only (Section 4.5).
- Accountable Edge Support: Classical statistical models (hierarchical ETS/ARIMA) forecast SKU size/color splits; local conic QP solvers (Clarabel.rs) automatically draft price-adjusted council revisions; and optional small local open-weight LLMs are restricted to multi-lingual translation, screen-reader accessibility, and summarising qualitative Lists with mandatory links to raw council source text.

[16] Albert and Hahnel, Political Economy of Participatory Economics, Ch. 1, pp. 15–17; Looking Forward, Ch. 9 (lf9_2.htm) and Ch. 10 (lf10_1.htm).

1.6 Estimated Effort, Budget Scenarios, and the Five Key Decisions

Table 1.2: Proposed budget scenarios at one assumed loaded rate (for author ratification)

Phase	Staff & Time	Non-labour	Estimate & Gate
Phase 1 Economic simulator	1.0 FTE 4–6 months	Compute \$1k–\$3k	\$41k–\$93k. Pre-registration in month 1. Kill Gate 1.
Phase 1H Low-tech human probe	1.0 FTE 3–4 months (in parallel)	Ethics review, consent, compensation, data protection: \$8k–\$25k	\$38k–\$85k. Kill Gate 1H.
Phase 2 Distributed coordination	2.5 FTE 4–6 months	WAN cluster \$2k–\$5k	\$102k–\$230k. Kill Gate 2.
Phase 3 PQC, Prio3, co-op pilot	4.0 FTE 9–12 months	Audit \$50k–\$150k; hosting \$10k–\$40k; pilot ethics and data protection \$20k–\$40k	\$440k–\$950k for Phase 3 alone. Kill Gate 3.
Programme			\$0.62M–\$1.36M if every gate passes.

Basis: an assumed fully loaded cost of \$120k–\$180k per FTE-year (low end of each range uses the shorter duration and lower rate). This is a planning assumption, not a sourced market rate; it should be replaced by the author's actual pay scale, which in a cooperative may be flat and much lower. Figures from research/analysis/verify_r3.py. Two to three months is not realistic for building a simulator, pre-registering instances and implementing an endogenous-income solver, so Phase 1 is planned at 4–6 months.

The Five Most Important Decisions for Project Leadership

1. Treat Architecture D as the leading long-term hypothesis (local-first + Two-Root aggregation + epoch checkpoints) and reject a per-transaction blockchain. Use Architecture A (a disclosed trusted server) for Phase 1H and the earliest pilots: it is simpler to operate and explain, and it wins whenever simplicity, throughput and cost dominate the weighting (Section 6.1.1).

2. Replace unconstrained Cobb-Douglas with constrained local solvers, and treat Algorithms 3a–3c as research candidates to be benchmarked in Phase 1 rather than as a committed production solver.

3. Enforce Hard Biophysical Caps $(r^*, E_z^{\max})$ Lexicographically Above PDRM Pollution Pricing.

4. Design for a "3-Milestone Hybrid Deliberation Schedule" separating au-

tomated intermediate input-clearing sub-iterations from human value-deliberation milestones.

5. Target Dual Utility: Support both full-scale Parecon simulation research and immediate deployment for federated worker cooperatives and participatory budgeting communities ("Experiment 2: Developing a Parallel Economy").[17]

Final Recommendation: PROCEED WITH PHASE 1 AND PHASE 1H AS KILL GATES—the Python-first Economic Coordination & Non-Convexity Simulator (Phase 1) and the Low-Technology Human Participation Probe (Phase 1H) first, each with pre-registered pass and fail conditions (Table 9.2), holding Phase 2 (distributed Rust nodes) and Phase 3 (post-quantum and Prio3 field pilots, with a mandatory $50k–$150k external audit) contingent on passing Kill Gates 1 and 1H. A failed gate stops or narrows the programme; it is not reported as a partial success.

[17] Albert and Hahnel, Political Economy of Participatory Economics, Ch. 6, pp. 129–130.

CHAPTER 2

Historical and Institutional Inspirations: From Council Practice to Protocol Design

> "In almost all the industries, factories, mills, workshops, transportation services, public services, and utilities, the rank and file workers, their revolutionary committees, and their syndicates reorganized and administered production, distribution, and public services without capitalists, high salaried managers, or the authority of the state… They instituted not bourgeois formal democracy but genuine grass roots functional libertarian democracy, where each individual participated directly in the revolutionary reorganization of social life."
>
> — Gaston Leval, Espagne Libertaire (1971), p. 11, translated in Sam Dolgoff, ed., The Anarchist Collectives: Workers' Self-Management in the Spanish Revolution, 1936–1939 (1974), Ch. 1, p. 6[1]

2.1 Epistemological Premise: Institutions Before Cryptography

A recurring failure mode of contemporary "decentralised governance" (DAO) architectures is the conflation of cryptographic trustlessness with democratic self-management. Recording a tally on an immutable ledger does not render an economy democratic if:

1. Proposal drafting and agenda-setting are monopolised by an unaccountable technical or managerial elite—what Albert and Hahnel diagnose as the coordinator class.[2]
2. Decision-making influence is proportional to token wealth or meeting endurance rather than proportional to the degree participants are affected.
3. Wealthier production collectives hoard superior inherited productive endowments and refuse to equalise resources across regions.

In this chapter, we examine five historical and contemporary institutional traditions as empirical laboratories of democratic coordination—analyzing both their institutional breakthroughs and their structural failures—to derive binding architectural rules for DPEP.

[1] Gaston Leval, Espagne Libertaire (1936–1939) (Paris: Éditions du Monde Libertaire, 1971), p. 11; verbatim English translation in Sam Dolgoff, ed., The Anarchist Collectives: Workers' Self-Management in the Spanish Revolution, 1936–1939 (New York: Free Life Editions, 1974), Ch. 1, p. 6.

[2] Albert and Hahnel, Political Economy of Participatory Economics, Ch. 1, pp. 14–17, 20–22; Looking Forward, Prologue ("Coordinatorism" and "The Origins of Coordinatorism").

2.2 The Spanish Revolution (1936–1939): Industrial and Agrarian Collectives

Following the defeat of the Nationalist military coup in July 1936 across Catalonia, Aragon, Valencia, and parts of Castile and Andalusia, workers and peasants affiliated with the anarchist CNT (Confederación Nacional del Trabajo) and socialist UGT (Unión General de Trabajadores) collectivised a large share of Catalan industry (figures of 70–80% are commonly repeated in the secondary literature; this monograph has not traced them to a primary count) and several hundred agrarian collectives in Aragon. Estimates of participation vary widely and come mostly from participants and sympathisers: Frank Mintz gives a minimum of about 1.84 million collectivists (1.08 million industrial, 0.76 million agrarian), revised downwards in 1977 from his earlier 2.44–3.2 million, while the eyewitness and CNT militant Gaston Leval gave figures of up to 3 million. Mintz's figures are cited here from the bibliographic record and were not checked against the volume.[3]

2.2.1 Internal Self-Management, Delegation, and Recall

Across both Barcelona factories (tramways, textiles, metalworks, water/gas/electric utilities) and Aragonese villages (Alcañiz, Calanda, Binéfar, Graus, Mas de las Matas), supreme authority resided in the weekly or biweekly General Assembly (Asamblea General) of all workers or villagers.[4] Assemblies elected specialized administrative committees (Comités de Administración) to manage daily coordination, input procurement, and statistical reporting. As documented by Bookchin, Peirats, and Leval, these delegates operated under strict anti-bureaucratic constraints:

> "Where such popular assemblies were possible, their decisions were executed by a committee elected from the assembly. Apparently, the right to recall committee members was taken for granted and they certainly enjoyed no privileges, emoluments, or institutional power… It remained a cardinal principle of Spanish anarchists never to pay their delegates, even when the CNT numbered a million members. Normally, the responsibilities of elected delegates had to be discharged after working hours."[5]

[3] Frank Mintz, Anarchism and Workers' Self-Management in Revolutionary Spain, trans. Paul Sharkey (Oakland, CA: AK Press, 2013 [1970/1977]); Gaston Leval, Collectives in the Spanish Revolution (London: Freedom Press, 1975); Sam Dolgoff, ed., The Anarchist Collectives: Workers' Self-Management in the Spanish Revolution, 1936–1939 (New York: Free Life Editions, 1974), available in full at https://theanarchistlibrary.org/library/sam-dolgoff-editor-the-anarchist-collectives.

[4] Dolgoff, The Anarchist Collectives, Ch. 2 and Ch. 9; Murray Bookchin, "Introductory Essay," in Dolgoff, The Anarchist Collectives, pp. xxvii–xxx.

[5] Bookchin, "Introductory Essay," in Dolgoff, The Anarchist Collectives, p. xxviii.

2.2.2 Inter-Collective Coordination, Reorganization, and Accounting

- Regional Equalisation in Aragon: The Regional Defence Council of Aragon was created at a plenary of unions and militia columns in Bujaraloz on 6 October 1936. The Aragon Federation of Collectives was constituted later, at the Caspe congress of 14–15 February 1937. Accounts of its size differ by a factor of five, and all come from sympathetic sources: Leval, who attended, lists 24 cantonal federations covering 275 villages and 141,430 member families; Peirats, citing Souchy, gives 500 collectives with 433,000 members; a widely reproduced summary gives 500 delegates representing 80,000 collectivists. The federation pooled surpluses at cantonal depots, moved machinery and seed between villages, and traded directly with Catalan syndicates. The Council was dissolved by decree on 10–11 August 1937, after Republican troops under Líster entered Aragon.[6]
- Unified Export Coordination in Valencia (CLUEA): In the Levante, the CNT and UGT established the Consejo Levantino Unificado de Exportación Agrícola (CLUEA), federating ∼270 local orange and citrus committees to standardize grading, rail/maritime transport, and export clearing.[7]
- Industrial Recomposition in Catalonia: In Barcelona, the CNT's industrial unions (Sindicatos Únicos de Ramo) consolidated hundreds of small, unsanitary workshops in woodworking, baking, tanning, and optic/glass production into modern integrated plants, eliminating redundant middlemen while preserving full employment.[8]
- Accounting and the "Family Wage": Many Aragonese villages abolished official pesetas internally, distributing abundant staples free according to need (Maxim 4) and issuing a family consumption booklet (libreta de consumo) or local points calibrated to family size for scarce goods.[9] Inter-collective trade relied on standardised valuation tables (indexed in wheat equivalents or shadow pesetas) cleared through cantonal ledgers.

2.2.3 Historical Limitations and Structural Failures

Despite remarkable wartime productivity in munitions, transport, and agriculture, the Spanish collectives suffered four critical weaknesses that directly inform DPEP's design:

[6] Leval, Collectives in the Spanish Revolution, Ch. V (Caspe congress); José Peirats, in Dolgoff, The Anarchist Collectives, Ch. 9 (giving Souchy's estimate); the 80,000 figure appears without a source in the Wikipedia article "Regional Defence Council of Aragon" and is reported only as a range bound. The October 1936 date belongs to the Council, not the Federation.

[7] Mintz, Anarchism and Workers' Self-Management in Revolutionary Spain, Ch. 4.

[8] Augustin Souchy, "Collectivizations in Catalonia," in Dolgoff, The Anarchist Collectives, Ch. 7.

[9] Leval, Collectives in the Spanish Revolution, pp. 165–210.

1. "Collective Egoism" (Neocapitalismo Colectivo) Between Rich and Poor Enterprises: In urban Catalonia, prior to the socialisation decrees of late 1937, collectives remained financially autonomous inside a quasi-market environment. Wealthy collectives that inherited modern machinery or lucrative monopolies (such as the Barcelona tramways, power utilities, and luxury hotels) enjoyed large surpluses, while raw-material-starved textile and construction collectives struggled to pay base wages.[10] Protocol Lesson: This failure is why Albert and Hahnel (1991a, Ch. 5, pp. 84–87) and Sandström (2020) insist that productive assets belong to the Productive Commons, and that every Worker Council be charged an opportunity-cost rental fee ($P_k \cdot k_j$) on its inherited capital stock.

2. Burnout from Unpaid Evening Governance: Requiring delegates to perform all federation administration after a full day of physical factory labour caused severe exhaustion and favoured childless militants over workers with caregiving duties. Protocol Lesson: Governance and facilitation work must be counted inside paid work hours as part of every citizen's Balanced Job Complex.

3. Lack of Unified Shadow Pricing for Complex Intermediate Chains: While cantonal barter cleared simple agricultural staples, urban industries lacked an iterative mechanism to compute social opportunity costs across hundreds of interdependent intermediate inputs once pre-war market prices grew stale.

4. Wartime militia coercion, smallholder conflict, and "workers against work" (Casanova 2005; Seidman 1991): Internal weaknesses were not merely accounting bugs. Casanova (2005) and Seidman (1991, 2002), consulted at bibliographic level only, are the standard critical accounts; they and other secondary sources describe Aragonese collectives formed under armed militia columns, conflict with "individualist" smallholders, absenteeism in Barcelona factories, credit starvation by the central government, dissolution in Aragon by Líster's 11th Division in August 1937, and Franco's victory in 1939. No page-level claim is made for either author.

2.3 The Paris Commune (March 18 – May 28, 1871)

During its 72 days of existence before its bloody suppression during La Semaine Sanglante, the Paris Commune established foundational principles of council democracy celebrated in Karl Marx's The Civil War in France (1871).[11] A rigorous institutional analysis must distinguish what the Commune actually implemented from unimplemented proposals and later theoretical extrapolations.

[10] José Peirats, in Dolgoff, The Anarchist Collectives, Ch. 6; Mintz, Anarchism and Workers' Self-Management.

[11] Karl Marx, The Civil War in France (London: General Council of the International Workingmen's Association, 1871); John Merriman, Massacre: The Life and Death of the Paris Commune of 1871 (New York: Basic Books, 2014).

Table 2.1: Paris Commune (1871): Actually Implemented Institutions vs. Proposals & Extrapolations

Dimension	Actually Implemented (March–May 1871)	Proposed on Paper or Later Extrapolation
Representation & Recall	Municipal elections across 20 arrondissements (March 26, 1871) by universal male suffrage; councilors declared revocable at any time; merged legislative and executive work across 10 Commissions.[12]	Formal recall was rarely exercised during the 72-day military siege; women were excluded from formal voting despite leading grass-roots defence and cooperative organising via the Union des Femmes (Élisabeth Dmitrieff, Nathalie Lemel).
Remuneration Cap	Decree of April 2, 1871: Capped maximum salaries of all Commune officials and municipal administrators at 6,000 francs/year (equivalent to a skilled Parisian workman's wage) and banned multiple stipends.	Did not abolish wage hierarchies inside private Parisian enterprises or establish balanced job complexes inside workshops.
Workplace Self-Management	Decree of April 16, 1871 (drafted by Léo Frankel's Commission of Labour and Exchange): Ordered a census of workshops abandoned by fleeing owners and their transfer to worker cooperative associations; abolished night baking (April 20) and employer wage fines (April 27).[13]	Only a few dozen cooperative workshops (e.g., the Louvre armaments workshop electing its own council and foremen; women's uniform cooperatives) actually began operating before Versailles troops entered Paris on May 21.
National Federation & Credit	Operated strictly within besieged Paris; left the Banque de France intact (negotiating advances via Charles Beslay rather than socializing credit).	Declaration to the French People (April 19, 1871) (drafted by Proudhonian Pierre Denis) and Marx envisioned a national federation of autonomous communes coordinating via delegated national assemblies—never realized due to the crushing of provincial communes in Lyon and Marseille.
Crisis Centralisation	On May 1, 1871, under bombardment, the Jacobin/Blanquist majority voted (45 to 23) to create a 5-member Committee of Public Safety with sweeping powers over the objection of the Federalist minority (Frankel, Varlin, Longuet, Courbet).	Demonstrates the acute vulnerability of council democracies during crisis to revert to emergency command authority unless crisis-mode bounds (TTLs and post-incident audits) are constitutionally hard-coded.

2.4 Classical Athenian Democracy (∼508/7 – 322 BCE)

Over nearly two centuries from the reforms of Kleisthenes (508/7 BCE) to the Macedonian conquest (322 BCE), classical Athens developed the most sophisticated institutional machinery for direct citizen self-governance in antiquity, analyzed definitively by Mogens Herman Hansen.[14]

2.4.1 Core Institutional Mechanisms Adaptable to DPEP

1. Direct Assembly (Ekklesia) vs. Norm-Setting (Nomothetai): Open to all adult male citizens (∼30,000–40,000 eligible in the 4th century BCE, with ∼6,000 attending the 40 annual meetings on the Pnyx), the Ekklesia voted on decrees (psephismata), while permanent constitutional laws (nomoi) were ratified by a separate panel of 1,000 nomothetai selected by lot.

2. Sortition (Kleroterion) and Strict Rotation: Roughly 600 of the ∼700 domestic magistrates (archai) and the 6,000 annual jurors (dikastai) were selected by lot using bronze allotment machines (kleroteria) rather than election—following Aristotle's principle (Politics 1294b) that elections are intrinsically oligarchic (favoring the wealthy, famous, and rhetorically polished) whereas sortition is democratic.[15] Almost all offices were restricted to a non-repeatable one-year term (or repeatable only once in a lifetime for the Boule), preventing a permanent coordinator class.

3. Probouleutic Agenda Facilitation (Boule of 500): Five hundred citizens (50 from each of the 10 Kleisthenic tribes, chosen by lot from local demes) served for one year on the Council (Boule), with each tribe serving as the standing executive committee (prytaneis) for 1/10th of the year and a new chair (epistates) drawn by lot every 24 hours. Crucially, the Boule's role was probouleutic: it prepared structured proposals (probouleumata) for the Ekklesia to debate, amend, or reject, but could not impose major policies on its own.[16] This is the closest historical analogy (not a direct institutional ancestor) of Parecon's Facilitation Boards.

4. Public Scrutiny and Mandatory Audit (Dokimasia, Euthynai, Graphe Paranomon): Every magistrate underwent pre-office eligibility scrutiny (dokimasia) and a mandatory end-of-term financial and conduct audit (euthynai) before 10 logistai (accountants) and 10 euthynoi chosen by lot.[17]

5. Civic Remuneration (Misthophoria): Introduced by Pericles for jurors and councilors and expanded in the 4th century BCE to assembly attendance

[12] Marx, The Civil War in France, Section III.
[13] Merriman, Massacre: The Life and Death of the Paris Commune of 1871, Ch. 5.
[14] Mogens Herman Hansen, The Athenian Democracy in the Age of Demosthenes: Structure, Principles, and Ideology (Oxford: Blackwell / Norman: University of Oklahoma Press, 1991).
[15] Hansen, Athenian Democracy in the Age of Demosthenes, Ch. 9, pp. 230–245.
[16] Hansen, Athenian Democracy in the Age of Demosthenes, Ch. 10, pp. 246–265.
[17] Hansen, Athenian Democracy in the Age of Demosthenes, Ch. 9, pp. 218–224.

(misthos ekklesiastikos, 3 to 6 obols per day), ensuring working citizens (thetes) could afford to participate.

2.4.2 Explicit Examination of Structural Exclusions

Athenian democracy excluded women (who lacked political citizenship and independent property control), metics (resident non-citizens, ∼20,000–30,000 artisans and merchants), and enslaved people (∼80,000–150,000 forced laborers in the Laurion silver mines, agriculture, crafts, and households), leaving only ∼10–15% of the total population with full political rights.[18] Furthermore, as Moses Finley and G. E. M. de Ste. Croix demonstrate, Athens' fiscal capacity to fund misthophoria and public works depended heavily on the Laurion silver mines (worked by tens of thousands of enslaved miners) and 5th-century Delian League imperial tribute (phoros).[19] In a universal, non-extractive participatory economy, every hour spent in deliberation and euthynai audits carries a real opportunity cost in productive labour; governance time must therefore be budgeted inside paid Balanced Job Complexes and economised carefully, while sortition should draw from all citizens regardless of hardware ownership (the beacon design is open; Section 7.6).

2.5 Occupy Wall Street (2011) and Additional Precedents

- Occupy Wall Street (2011), Freeman's "Structurelessness," and Assembly Scaling: In small groups without formal rules, Jo Freeman (1972) showed that power concentrates in unaccountable informal friendship cliques ("the tyranny of structurelessness")—a sociological dynamic that asynchronous software alone does not eliminate.[20] Mansbridge (1980) and Dahl & Tufte (1973) are the classic treatments of assembly scale; they were not consulted directly, and no specific size threshold is attributed to them. The ∼50–200 neighbourhood-council bound and the 50–150 speaker figure in Table 2.2 are working design heuristics from speaking-time arithmetic (one hour shared among 60 speakers is one minute each), not sourced empirical findings. Albert (2003, Ch. 5) argues why unanimity vetoes fit small-team hiring while majority/supermajority rules with minority reconsideration delays fit workplace investments.
- Mondragón Cooperative Corporation (1956–Present): Across roughly 90 cooperatives and a group workforce commonly reported at 70,000–80,000, of whom a substantially smaller number are cooperative worker-owners (socios) because

[18] Hansen, Athenian Democracy in the Age of Demosthenes, Ch. 4, pp. 86–124.

[19] M. I. Finley, Economy and Society in Ancient Greece (London: Chatto & Windus, 1981), Ch. 5; G. E. M. de Ste. Croix, The Class Struggle in the Ancient Greek World (Ithaca, NY: Cornell University Press, 1981).

[20] Graeber, The Democracy Project, Chs. 1–3; Freeman, "The Tyranny of Structurelessness," pp. 151–164; Jane J. Mansbridge, Beyond Adversary Democracy (New York: Basic Books, 1980); Robert A. Dahl and Edward R. Tufte, Size and Democracy (Stanford: Stanford University Press, 1973); Albert, Parecon: Life After Capitalism (2003), Ch. 5.

much of the retail (Eroski) and overseas workforce are non-member employees (membership figures vary by year and source and are not independently checked here), Mondragón separates the elected Audit / Watchdog Committee (Comisión de Vigilancia) from the frontline workers' Social Council (Consejo Social),[21] illustrating both federated industrial self-management and the erosion of cooperative membership under global market competition.

- Porto Alegre Participatory Budgeting (1989–2017 Suspension): Combined regional assemblies and recallable delegates with an objective Quality of Life Need Matrix to redirect municipal investment to poor districts, before being suspended in 2017 following municipal coalition shifts, fiscal austerity, and ∼2,400 unexecuted citizen project requests.
- Rojava / DAANES (2012–2026 Damascus Integration Accord): Implemented nested communes, co-chair gender parity, and autonomous women's councils (Kongra Star),[22] before military and geopolitical pressure led to the 30 January 2026 SDF–Damascus integration agreement merging DAANES civil and security institutions into the Syrian state.
- Ostrom's Common-Pool Resource (CPR) Institutions (1990): Elinor Ostrom's study of enduring self-governed commons (e.g., Valencia's Tribunal de las Aguas) established eight design principles: defined boundaries, proportional equivalence, collective-choice arrangements, accountable monitoring, graduated sanctions, rapid local conflict resolution, right to organise, and nested enterprises.[23]

Synthesis: Comparative Institutional-to-Protocol Matrix

Table 2.2 synthesizes the eight core institutional mechanisms analyzed across Chapters 2.2–2.5, mapping their historical strengths and limitations directly to concrete capabilities in the Distributed Participatory Economics Protocol (DPEP).

[21] William Foote Whyte and Kathleen King Whyte, Making Mondragón, 2nd ed. (Ithaca, NY: ILR Press, 1991), consulted at bibliographic level.

[22] Hahnel, Bohmer, and Chowdhury, "Reproductive Labour," Ch. 10 in Hahnel, Democratic Economic Planning, p. 200.

[23] Elinor Ostrom, Governing the Commons (Cambridge: Cambridge University Press, 1990), Ch. 3, p. 90.

Table 2.2: Comparative Matrix: Institutional Mechanisms Mapped to DPEP Protocol Capabilities

Mechanism & Precedents	Strengths & Advantages	Historical Limitations	Concrete DPEP Protocol Capability
1. General Assembly Spain 1936; Athens Ekklesia; Occupy; Mondragón	Builds face-to-face solidarity; surfaces qualitative context.	Strains as attendance grows (working heuristic: ∼50–150 speakers; Section 2.5); meeting fatigue.	Scoped strictly to work teams (10–30) and neighborhood councils (50–200) with async pre-reading queues.
2. Revocable Delegation Paris Commune 1871; Spain CNT; Porto Alegre PB	Enables multi-tier federation scaling while tethering delegates to base.	Delegates ossify if recall is procedurally slow or difficult.	Scoped CapCert Delegate Certificates: Issued by council k-of-n multi-signature; revocable in a single council vote.
3. Sortition + Rotation Classical Athens (Boule, Logistai, Dikasteria)	Prevents factional elections, charismatic capture, and coordinator elites.	Unsuitable for rare specialist tasks unless drawn from qualified pool.	Sortition (beacon open): Intended to select rotating IFB oversight, audit panels, and dispute juries (Section 7.6).
4. Mandatory Audit Athens Euthynai (Logistai); Mondragón Comisión de Vigilancia	Systematically detects bias or unauthorised overrides.	Requires tamper-evident records and independent auditors. Mondragón's elected Social Council is a workers' voice body, not the audit committee.	Deterministic WASM + Two-Root Audit Logs: Any node with the leaf archive can recompute IFB price updates; mobile light clients verify headers only.
5. Endowment Equalization Aragon Depots (1936); Mondragón Funds	Prevents collectives with superior machinery from extracting rents.	Failed in 1936 Barcelona where rich collectives hoarded revenues.	Stand-Alone Capital Asset Ledger: Automatically debits each WC_j's SC_j for capital/resource user rights (P_k, P_r).
6. Proportional & Caucus Rojava Co-Chairs & Women's Councils	Protects disproportionately affected groups from majority tyranny.	Can deadlock if scope of veto is unbounded.	Impact-Weighted & Dual-Key Governance Rules (PEPE Ch. 2 p. 24; DEP Ch. 10 p. 200).
7. Iterative Tâtonnement IETF/Git; Lange-Arrow-Hurwicz; Parecon IFB	Scales quantitative clearing to millions asynchronously.	Pure prices strip out human context unless paired with qualitative text.	Two-Root BLAKE3 Aggregation + Content-Addressed Qualitative Lists.
8. Graduated Sanctions Ostrom CPRs (Valencia Tribunal de las Aguas)	Intended to keep routine disputes local (Ostrom 1990); no sourced "95%" rate is claimed here.	Requires clear invariants and fast local mediation.	Tiered Variance Alerts → UFB Mediation → Industry/Environmental Court.

CHAPTER 3

Source-Grounded Requirements of Participatory Economics

> "Until now, however, descriptions of alternatives have been unconvincing. Here Michael Albert and Robin Hahnel support the libertarian socialist tradition by presenting a rigorous, well-defined model of how producers and consumers could democratically plan their interconnected activities."
>
> — Michael Albert and Robin Hahnel, The Political Economy of Participatory Economics (1991), Front Flap / JSTOR Abstract[1]

3.1 Source Provenance and Classification Methodology

Before formulating mathematical equations or distributed protocols, we establish strict traceability to the foundational texts. Every requirement in Table 3.1 is classified into one of four provenance tiers:

- [ORIGINAL-1991]: Explicitly specified in Michael Albert and Robin Hahnel's two 1991 volumes, The Political Economy of Participatory Economics (PEPE) and Looking Forward (LF).[2]
- [INTERPRETATION-LATER]: Institutional elaborations, formalizations, and accounting procedures introduced in Michael Albert's Parecon: Life After Capitalism (PLAC, 2003), Robin Hahnel's Democratic Economic Planning (DEP, 2021) and A Participatory Economy (2022), Anders Sandström's Anarchist Accounting (2020/2023), Mitchell Szczepanczyk's simulation work (2023), and the Participatory Economy Project's User Story Mapping (2023).[3]
- [TECH-EXTENSION]: Technological extensions required to execute the model over a modern, decentralised, post-quantum, privacy-preserving computational network.
- [UNRESOLVED]: Open theoretical or institutional questions requiring simulation or empirical human research (examined in full in Chapter 10).

[1] Albert and Hahnel, Political Economy of Participatory Economics, book overview (https://www.jstor.org/stable/j.ctv131bvr2).

[2] Albert and Hahnel, Political Economy of Participatory Economics (1991), print pagination verified via library TOC scan (research/sources/pepe_dandelon.pdf, pp. 1–132) and full text analyzed via ZNetwork (intro.htm, 1.htm–6.htm); Albert and Hahnel, Looking Forward (1991), full text analyzed via ZNetwork (lfIntroduction.htm, lf1_1.htm–lf10_1.htm).

[3] Hahnel, Democratic Economic Planning (2021); Robin Hahnel, A Participatory Economy (Chico, CA: AK Press, 2022); Sandström, Anarchist Accounting (2020); Szczepanczyk, "Pseudocode and Algorithms" (2023); Participatory Economy Project, "Participatory Planning: User Story Mapping" (2023), https://participatoryeconomy.org/project/participatory-planning-user-story-mapping/.

3.2 Normative Foundations and Endogenous Preference Theorems

In Chapter 1 of The Political Economy of Participatory Economics (pp. 8–11), Albert and Hahnel evaluate economic institutions against five valuative criteria: Equity, Self-Management, Solidarity, Variety, and Efficiency.[4] For distributive equity, they distinguish four mutually exclusive maxims (PEPE Ch. 1, p. 8; Ch. 3, p. 44; DEP Ch. 1, pp. 16–22):

1. Maxim 1 (Capitalism): Payment according to the value of the contribution of one's personal labour and productive property owned (rejected due to inheritance—the "Rockefeller grandson problem"—luck, and compounding wealth accumulation).[5]
2. Maxim 2 (Market Socialism / Coordinatorism): Payment according to the value of one's personal labour contribution alone (rejected because differences in marginal productivity stem overwhelmingly from the "genetic lottery," luck, job assignment, and publicly funded training rather than personal sacrifice—illustrated by their "Mozart vs. Salieri" and "Dave Butz" examples).[6]
3. Maxim 3 (Participatory Economics): Payment according to effort and personal sacrifice expended in socially valued labour (longer hours, higher intensity, or more onerous/dangerous conditions prior to full BJC equalization).
4. Maxim 4 (Humane Solidarity): Payment according to need—incorporated into Parecon via universal free public goods (healthcare, education) and council-approved special-need allowances (PEPE Ch. 3, p. 47).

Crucially, drawing on the formal results of Quiet Revolution in Welfare Economics (1990), consulted here through PEPE's summary,[7] Albert and Hahnel argue that because human preferences are endogenous (people rationally mold their future characteristics ΔC in anticipation of future terms of supply), any institution that systematically mischarges for activities relative to true social opportunity cost causes: (1) a snowballing divergence from Pareto optimality over time, (2) systematic warping of human development trajectories, and (3) ex post invisibility of the bias to participants. Markets systematically underprice negative externalities and overprice public goods (warping preferences toward anti-social individualism), while central planning systematically under-supplies self-managed work roles (warping workers toward apathy and coordinator submission).[8]

[4] Albert and Hahnel, Political Economy of Participatory Economics, Ch. 1, pp. 8–9, 11.
[5] Hahnel, Democratic Economic Planning, Ch. 1, pp. 16–18; Albert and Hahnel, Political Economy of Participatory Economics, Ch. 3, pp. 45–46.
[6] Albert and Hahnel, Political Economy of Participatory Economics, Ch. 3, pp. 46–47.
[7] Robin Hahnel and Michael Albert, Quiet Revolution in Welfare Economics (Princeton, NJ: Princeton University Press, 1990), Chs. 4, 6, 7; summarised in Political Economy of Participatory Economics, Ch. 1, pp. 9–11.
[8] Albert and Hahnel, Political Economy of Participatory Economics, Ch. 1, pp. 12–17; Ch. 5, pp. 98–105.

3.3 Formal Traceability Matrix of Requirements

Table 3.1: Master Requirements Matrix Derived from Primary and Secondary Literature

Req ID	Requirement Specification	Classification	Primary Citations
REQ-NORM-01	Productive Commons & User-Right Fees: Abolish private ownership of productive property; all natural resources and capital stocks belong to society. Worker councils acquire temporary user rights during annual planning and pay opportunity-cost charges (P_r, P_k, P_l).	[ORIGINAL-1991] + [INTERP-LATER]	PEPE Ch. 1 (pp. 17–20), Ch. 5 (p. 85); DEP Ch. 6 (pp. 98–102), Ch. 12 (p. 244); Sandström (2020).
REQ-NORM-02	Proportional Self-Management: Decision-making input proportional to the degree affected by an economic choice.	[ORIGINAL-1991]	PEPE Ch. 1 (p. 8), Ch. 2 (p. 24); LF Ch. 1; PLAC Ch. 5; DEP Ch. 1 (p. 14).
REQ-NORM-03	Effort & Sacrifice Remuneration: Consumption budgets proportional to effort/sacrifice (Maxim 3) tempered by universal free goods and need exceptions (Maxim 4), with council average effort ratings capped by SB_j/SC_j.	[ORIGINAL-1991] + [INTERP-LATER]	PEPE Ch. 3 (pp. 44–55); DEP Ch. 6 (pp. 108–117); Sandström (2023).
REQ-NORM-04	Balanced Job Complexes (BJCs): Tasks partitioned into subsets $\{PIPA\}_i$ ordered by empowerment/desirability and combined into balanced complexes within and across workplaces via Job Complex Committees.	[ORIGINAL-1991]	PEPE Ch. 2 (pp. 25–30), Ch. 5 (pp. 82–83); LF Chs. 1–2.
REQ-ACT-01	Nested Consumer Councils & Federations: Individuals/households → Neighbourhood CC_h → Ward → City/County → State/Region → National Federation, requesting public goods matched to spatial scale.	[ORIGINAL-1991]	PEPE Ch. 3 (pp. 40–42), Ch. 5 (pp. 81–82); LF Chs. 3, 4, 8.
REQ-ACT-02	Worker Councils & Dual Federations: Democratic workplace councils federated both by industry (scale economies, R&D, SB/SC benchmarking) and by region (local production externalities).	[ORIGINAL-1991]	PEPE Ch. 2 (pp. 23–25), Ch. 4 (p. 62), Ch. 5 (p. 88); LF Chs. 1, 2, 7.
REQ-ACT-03	Six Functional Facilitation Boards: Iteration (IFB), Consumption (CFB), Production (PFB), Employment (EFB), Household (HFB), and Updating (UFB) facilitation boards performing purely informational/mediating tasks.	[ORIGINAL-1991]	LF Ch. 6 (lf6_1.htm), Ch. 9 (lf9_2.htm); PEPE Ch. 4 (p. 62).
REQ-ACT-04	Communities of Affected Parties (CAPs) & Environmental Courts: Airshed/watershed collectives participating in the Pollution Demand-Revealing Mechanism (PDRM) below hard biophysical caps.	[INTERP-LATER]	DEP Ch. 7 (pp. 138–149), Ch. 14 (pp. 261–267); PEP (2023).

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Req ID	Requirement Specification	Classification	Primary Citations
REQ-ACT-05	Stand-Alone Capital Asset Entities & Society Account: Double-entry social accounting entities tracking capital production costs, user-right fees, CAP flows, and GDP division between consumption and investment.	[INTERP-LATER]	Sandström (2020, Chs. 3–5; 2023).
REQ-DATA-01	Seven Priced Economic Categories: Private goods, public goods, intermediate inputs, natural resources, labour categories, capital stocks, and regulated pollutants.	[ORIGINAL-1991] + [INTERP-LATER]	PEPE Ch. 5 (pp. 78–88); DEP Ch. 7, Ch. 9; Szczepanczyk (2023).
REQ-DATA-02	Coarse Planning Taxonomy vs. Granular SKUs: Plan over coarse resource-equivalent product classes; extrapolate specific sizes/colors statistically and resolve at intra-year fulfilment.	[ORIGINAL-1991] + [INTERP-LATER]	PEPE Ch. 4 (p. 64); LF Ch. 6 (lf6_2.htm); DEP Ch. 8 (pp. 163–167).
REQ-DATA-03	Public Books of Blueprints & Qualitative Lists: Publish all technical input-output recipes $[A, K, R, L]$ and attach qualitative descriptive Lists to quantitative council vectors.	[ORIGINAL-1991]	PEPE Ch. 4 (pp. 61–62), Ch. 5 (p. 84), Ch. 6; PLAC Chs. 11, 17.
REQ-DATA-04	Protected Private Consumption Sphere: Personal consumption within warranted budget B_u cannot be vetoed (PEPE). Cryptographic invisibility of the bundle is a DPEP extension, not Albert and Hahnel's 1991 design (Section 7.1).	[ORIGINAL-1991] (non-veto) + [TECH-EXT] (invisibility)	PEPE Ch. 3 (p. 41), Ch. 5 (p. 80): non-veto of in-budget requests. LF Ch. 3 subjects consumption to collective criticism. PLAC Chs. 12, 18: anonymous submission against "busybody" intrusion, not cryptographic aggregation.
REQ-PLAN-01	Multi-Horizon & Collective-First Sequencing: Long-run (education, environment) and aggregate investment plans precede annual planning; federation collective goods are settled before final private consumption bundles.	[ORIGINAL-1991] + [INTERP-LATER]	LF Ch. 3 (lf3_2.htm), Ch. 8 (lf8_1.htm), Ch. 9 (lf9_1.htm); DEP Chs. 11–15.
REQ-PLAN-02	Slack Planning & Hierarchical Intra-Year Clearing: Include 3–5% planned excess supply buffer (SL^E); cancel routine consumer swaps inside neighbourhood/ward/city federations; trigger UFB replanning only when net shocks exceed slack.	[ORIGINAL-1991]	PEPE Ch. 4 (pp. 66–67), Ch. 6 (p. 113, 122); DEP Ch. 8 (pp. 167–170).

CHAPTER 4

Formal Economic Model, Accounting Identities, and Algorithms

> "It is impossible to show that a feasible nonmarket system at least approaches the productivity of the market unless (1) a rather well-developed theoretical model of the nonmarket system is available, and (2) it is demonstrated that a sufficiently productive approximation of the ideal… system described in the theoretical model is practically possible."
>
> — Allen Buchanan, Ethics, Efficiency, and the Market (1985), p. 29, quoted in Albert and Hahnel, The Political Economy of Participatory Economics (1991), Ch. 4, p. 58, and Ch. 5, p. 73[1]

4.1 Ontology of Economic Actors and Priced Objects

Let the participatory economy $\mathcal{E}$ over annual planning period T consist of the following finite sets of actors and institutional entities:[2]

- Individuals ($\mathcal{I} = \{1, \ldots, N_{\text{ind}}\}$): Natural persons participating as consumers, workers, and citizens. Each individual $u \in \mathcal{I}$ belongs to a primary neighbourhood consumer council $h(u) \in \mathcal{H}$ and, if working, a primary worker council $j(u) \in \mathcal{J}$ (plus optional secondary council $j'(u)$ for cross-workplace Balanced Job Complex rotation).[3]
- Neighbourhood Consumer Councils ($\mathcal{H} = \{CC_1, \ldots, CC_H\}$) & Nested Consumer Federations ($\mathcal{F}^{\text{C}} = \bigcup_{\ell=1}^{L} \mathcal{F}_\ell^{\text{C}}$): A rooted geographic hierarchy of depth $L = 5$ (Level 1 = Ward, Level 2 = City/County, Level 3 = State, Level 4 = Region, Level 5 = National Federation).[4]
- Worker Councils ($\mathcal{J} = \{WC_1, \ldots, WC_J\}$) & Dual Worker Federations ($\mathcal{F}^{\text{W}} = \mathcal{F}^{\text{W,ind}} \cup \mathcal{F}^{\text{W,reg}}$): Democratic workplace collectives federated both by industry ($IF_k \in \mathcal{F}^{\text{W,ind}}$) and by geographic region ($\mathcal{F}^{\text{W,reg}}$).[5]

[1] Allen Buchanan, Ethics, Efficiency, and the Market (Totowa, NJ: Rowman and Littlefield, 1985), p. 29; cited in Albert and Hahnel, Political Economy of Participatory Economics, Ch. 4, p. 58 n. 3, and Ch. 5, p. 73.

[2] Formalised from Albert and Hahnel, Political Economy of Participatory Economics, Ch. 4 (pp. 62–68) and Ch. 5 (pp. 77–95); Hahnel, Democratic Economic Planning, Ch. 7 (pp. 130–156); Sandström, Anarchist Accounting (2020), Ch. 3; and Participatory Economy Project, "User Story Mapping" (2023).

[3] Albert and Hahnel, Political Economy of Participatory Economics, Ch. 2, pp. 27–30; Ch. 5, p. 83.

[4] Albert and Hahnel, Political Economy of Participatory Economics, Ch. 3, pp. 40–42; Ch. 5, pp. 81–82.

[5] Albert and Hahnel, Political Economy of Participatory Economics, Ch. 2, pp. 23–25; Ch. 5, pp. 88, 96–97.

- Communities of Affected Parties ($\mathcal{C}^{\mathrm{CAP}} = \{CAP_1, \ldots, CAP_Z\}$): Ecological collectives of residents $\mathrm{pop}(CAP_z) \subseteq \mathcal{I}$ impacted by pollutant $z \in \mathcal{Z}$ in a specific airshed or watershed.[6]
- Six Facilitation Boards ($\mathcal{B} = \{\mathrm{IFB}, \mathrm{CFB}, \mathrm{PFB}, \mathrm{EFB}, \mathrm{HFB}, \mathrm{UFB}\}$), Stand-Alone Capital Asset Entities ($\mathcal{A}^{\mathrm{cap}}$), and the Society Account ($\mathcal{A}^{\mathrm{soc}}$).[7]

Definition 4.1 (Master Catalogue of Seven Priced Economic Categories). Following Albert and Hahnel (1991a, Chs. 5–6), Hahnel (2021, Chs. 7–9), and Szczepanczyk (2023, Sec. 4), the economy tracks supply, demand, and indicative prices across seven disjoint categories indexed by the master catalogue Ω:

$$\Omega = \mathcal{G}^{\mathrm{priv}} \sqcup \mathcal{G}^{\mathrm{pub}} \sqcup \mathcal{G}^{\mathrm{int}} \sqcup \mathcal{R} \sqcup \mathcal{L} \sqcup \mathcal{K} \sqcup \mathcal{Z} \tag{4.1}$$

with complete indicative price vector at iteration t given by:

$$P^{(t)} = \left(p^{\mathrm{priv}}, p^{\mathrm{pub}}, p^{\mathrm{int}}, p^{\mathrm{res}}, p^{\mathrm{lab}}, p^{\mathrm{cap}}, p^{\mathrm{pol}}\right) \in \mathbb{R}_{+}^{|\Omega|} \tag{4.2}$$

4.2 Primal-Dual Reference Model and Sandström's Society Account

In Chapter 5 of The Political Economy of Participatory Economics (pp. 78–79), Albert and Hahnel introduce a primal-dual linear programming reference model to clarify the meaning of shadow prices before presenting their decentralised iterative procedure.[8] Let x be the vector of production activity levels, A be the produced intermediate input matrix, K be the capital stock coefficient matrix (with available stock k^*), R be the non-producible natural resource input matrix (with available ecological supply r^*), L be the direct labour input matrix (with available labour l^*), and v be the row vector of relative social values of net outputs. The economy-wide primal and dual reference problems are:

$$\text{Primal:} \quad \max_{x \geq 0} \; v(I - A)x \quad \text{subject to} \quad Kx \leq k^*, \;\; Rx \leq r^*, \;\; Lx \leq l^* \tag{4.3}$$

$$\text{Dual:} \quad \min_{P_k, P_r, P_l \geq 0} \; P_k k^* + P_r r^* + P_l l^* \quad \text{subject to} \quad P_k K + P_r R + P_l L \geq v(I - A) \tag{4.4}$$

As Albert and Hahnel explicitly note (PEPE Ch. 5, p. 79), the shadow prices P_l solved from the static dual (4.4) reflect only the marginal physical productivities of different labour categories, not the differential disutilities or desirabilities of work from the workers' point of view; whereas the indicative prices generated by Parecon's iterative council procedure incorporate worker well-being and effort disutility directly.[9]

[6] Hahnel, Democratic Economic Planning, Ch. 7 ("The Pollution Demand Revealing Mechanism"), pp. 139–148.

[7] Albert and Hahnel, Looking Forward, Ch. 6 (lf6_1.htm); Sandström, "An Accounting System for a Libertarian Socialist Economy" (2023).

[8] Albert and Hahnel, Political Economy of Participatory Economics, Ch. 5, pp. 78–79.

[9] Albert and Hahnel, Political Economy of Participatory Economics, Ch. 5, p. 79.

4.2.1 Closing the Macro-Accounting Loop (Sandström 2020/2023)

Anders Sandström's Anarchist Accounting (2020) and 2023 technical specification formalise how opportunity-cost user fees paid by Worker Councils reconcile with effort-based consumer incomes inside the Society Account ($\mathcal{A}^{\text{soc}}$):[10]

1. Social Value of Net Final Output (GDP) at Indicative Prices P:

$$Y^{\text{GDP}}(P) = p^{\text{priv}} \cdot D^{\text{priv}} + p^{\text{pub}} \cdot D^{\text{pub}} + p^{\text{cap,new}} \cdot D^{\text{inv}} \tag{4.5}$$

   In the Society Account, credits consist of all user-right fees charged to Worker Councils for access to society's labour categories ($p^{\text{lab}} \cdot L$), natural resources ($p^{\text{res}} \cdot R$), and productive capital assets held in Stand-Alone Capital Asset Entities ($p^{\text{cap}} \cdot K$), plus Consumer Council depreciation/discount charges on public consumer assets and net WC operating surpluses.

2. Aggregate Consumption Pool, Effort-Weighted Hours (H^{eff}), and Numéraire ($w_{\text{base}} \equiv 1.0$): Let $\sigma_{\text{inv}} \in (0,1)$ be the share of GDP allocated to aggregate investment (DEP Ch. 11),[11] Y^{allow} be the ratified budget for non-working cash allowances (Maxim 4 transfers A_u^{social}), $Y^{\text{CAP}}(P) = \sum_{z \in \mathcal{Z}} p_z^{\text{pol}} E_z$ be total CAP pollution compensation (funded by WC emission charges $\sum_j p^{\text{pol}} \cdot e_j$), and $\Delta S^{\text{agg}} = \sum_{u \in \mathcal{I}} \Delta S_u$ be net aggregate household borrowing. Universal public services already appear as $p^{\text{pub}} \cdot D^{\text{pub}}$ inside Y^{GDP}; they must not also be subtracted as Y^{allow}, or the consumption pool is double-counted (Table 4.1). Effort-weighted hours $H^{\text{eff}} = \sum_{u \in \mathcal{I}^{\text{work}}} h_u \epsilon_u$ convert the worker pool into a unit wage. The identity $w_{\text{base}} \equiv 1$ is a numéraire normalisation after clearing, not an independent price:

$$\begin{aligned} Y^{\text{work}}(P) &= (1 - \sigma_{\text{inv}}) Y^{\text{GDP}}(P) - Y^{\text{allow}} - Y^{\text{CAP}}(P) - \Delta S^{\text{agg}} \\ w_{\text{base}}(P) &= \frac{Y^{\text{work}}(P)}{\sum_{u \in \mathcal{I}^{\text{work}}} h_u \epsilon_u} \quad \text{(then rescale all prices so } w_{\text{base}} = 1\text{)} \end{aligned} \tag{4.6}$$

3. Capping Both Mean Workplace Effort Rating and Total Credited Hours (DEP Ch. 6, p. 117): To prevent either collective "grade inflation" or phantom-hours inflation (INV-4B), each worker council WC_j's mean effort multiplier and total credited effort-hours are bounded by its accredited labour for the plan year—the initial endowment plus ratified net hiring, $\sum_\ell (l^0_{j,\ell} + \Delta l_{j,\ell})$, the

[10] Sandström, Anarchist Accounting (2020), Ch. 3; Sandström, "An Accounting System for a Libertarian Socialist Economy" (2023), Diagram 1 (Arrows 1–14).

[11] Hahnel and Kerkhoff, "Aggregate Investment Planning," Ch. 11 in Hahnel, Democratic Economic Planning, pp. 219–237.

same quantity charged in Eq. (4.12)—and by its social benefit-to-cost ratio:[12]

$$\frac{\sum_{u\in WC_j} h_u \epsilon_u}{\sum_{u\in WC_j} h_u} \le \min\left(\bar{\epsilon}_{\max},\ \frac{SB_j(P)}{SC_j(P)}\right)$$
$$\sum_{u\in WC_j} h_u \epsilon_u \le \left(\sum_{\ell\in\mathcal{L}} \left(l^0_{j,\ell} + \Delta l_{j,\ell}\right)\right) \min\left(\bar{\epsilon}_{\max},\ \frac{SB_j(P)}{SC_j(P)}\right) \tag{4.7}$$

Here $l_{j,\ell}$ is measured in accredited hours. Because Δl_j is itself part of the proposal, the cap is checked against the ratified roster at plan ratification and against the proposed roster in each round.

4. Individual Warranted Annual Consumption Budget (B_u): Combining Sandström's four income flows (Sandström 2023, Arrows 1–4), citizen u's warranted budget is:

$$B_u(P) = \underbrace{w_{\text{base}}(P) h_u \epsilon_u}_{\text{1: Work Effort}} + \underbrace{A_u^{\text{social}}}_{\text{2: Benefits}} + \underbrace{\sum_{z:u\in CAP_z} \frac{p_z^{\text{pol}} E_z}{|\text{pop}(CAP_z)|}}_{\text{3: CAP Compensation}} + \underbrace{\Delta S_u}_{\text{4: Loan/Need}}$$
$$\sum_{u\in\mathcal{I}} B_u(P) = Y^{\text{work}}(P) + Y^{\text{allow}} + Y^{\text{CAP}}(P) + \Delta S^{\text{agg}} = (1-\sigma_{\text{inv}}) Y^{\text{GDP}}(P) \tag{4.8}$$

These last two displays are identities once Y^{allow} is defined as cash transfers only and A_u^{social} exhausts Y^{allow}. They hold at every price vector, not only at equilibrium. Relations such as $D^{\text{priv}} = S^{\text{priv}}$ are equilibrium conditions, not identities.

Simultaneity of budgets and demand. Y^{GDP} is valued at the demanded quantities D, so budgets depend on demand and demand depends on budgets: $B_u = B_u(P, D)$. Within an iteration DPEP breaks the loop with a one-round lag, $B_u^{(t)} = B_u(P^{(t)}, D^{(t-1)})$, so councils always optimise against a published budget. The lag is consistent only at a fixed point, where $B_u^{(t)} = B_u(P^{(t)}, D^{(t)})$; whether the lagged process reaches that fixed point is part of open question P2 and is not implied by any convergence result for fixed budgets. The investment share and the investment bundle are not independent choices: at a cleared plan $\sum_u B_u = p^{\text{priv}} D^{\text{priv}} + p^{\text{pub}} D^{\text{pub}}$, so the identities force $\sigma_{\text{inv}}\, Y^{\text{GDP}} = p^{\text{cap,new}} \cdot D^{\text{inv}}$. Either the federations ratify σ_{inv} and D^{inv} adjusts in value, or they ratify a physical investment bundle and σ_{inv} is implied. DPEP takes the first option (DEP Ch. 11 treats aggregate investment as a prior collective decision), so D^{inv} is scaled each round to match.

[12] Hahnel, Democratic Economic Planning, Ch. 6 ("Capping average effort ratings"), p. 117. Using the endowment l_j^0 alone would leave newly hired workers uncredited. The Phase 1 default for $\bar{\epsilon}_{\max}$ is a political choice set out in Section 9.1.3. Vintage-equipment plants are equalised via lower $p^{\text{cap}} \cdot k_j^0$ rental charges. INV-4B does not help if the labour roster itself is falsified—that remains a physical euthynai problem.

Table 4.1: Macro balance sheet (identities vs. equilibrium conditions)

Relation	Content	Status
$Y^{\mathrm{GDP}} = p^{\mathrm{priv}} D^{\mathrm{priv}} + p^{\mathrm{pub}} D^{\mathrm{pub}} + p^{\mathrm{cap,new}} D^{\mathrm{inv}}$	Final uses at indicative prices.	Identity (definition).
$Y^{\mathrm{work}} = (1-\sigma_{\mathrm{inv}}) Y^{\mathrm{GDP}} - Y^{\mathrm{allow}} - Y^{\mathrm{CAP}} - \Delta S^{\mathrm{agg}}$	Worker remuneration pool. Y^{allow} is cash A_u^{social} only.	Identity, given that split.
$\sum_u B_u = Y^{\mathrm{work}} + Y^{\mathrm{allow}} + Y^{\mathrm{CAP}} + \Delta S^{\mathrm{agg}}$	Warranted budgets exhaust the consumption pool.	Identity at every P.
$p^{\mathrm{lab}} \cdot L + p^{\mathrm{res}} \cdot R + p^{\mathrm{cap}} \cdot K + p^{\mathrm{pol}} \cdot E =$ WC user-right charges	Society Account credits.	Accounting identity.
$D_g \le S_g$, $D_r \le r^*$, $D_z \le \min(E_z, E_z^{\max})$	One-sided feasibility.	Equilibrium / halt condition, not an identity.
$p^{\mathrm{lab}} \ge w_{\mathrm{base}}$	Labour price floor (INV-4B).	Policy constraint; can prevent labour-market clearing in surplus.

4.3 Formal Optimisation Problems of Councils and CAPs

4.3.1 Consumer Councils and Nested Federations (PEPE Ch. 5, pp. 79–82, 88)

Each consumer council $CC_h \in \mathcal{H}$ seeks to maximise its members' total economic well-being $W_h(U_h, \Delta C_h)$—incorporating both immediate preference fulfilment U_h and endogenous human characteristic development ΔC_h—subject to the equitable budget constraint that its total social cost $SC(CC_h, P)$ does not exceed its members' warranted budget $B(CC_h, P) = \sum_{u \in \mathrm{pop}(CC_h)} B_u(P)$:[13]

$$\begin{aligned} \max_{y_h^{\mathrm{priv}}, y_h^{\mathrm{pub,loc}} \ge 0} \quad & W_h\Big(U_h(y_h^{\mathrm{priv}}, y^{\mathrm{pub}}),\ \Delta C_h(y_h^{\mathrm{priv}}, y^{\mathrm{pub}})\Big) \\ \text{subject to} \quad & SC(CC_h, P) \triangleq p^{\mathrm{priv}} \cdot y_h^{\mathrm{priv}} + p^{\mathrm{pub}} \cdot y_h^{\mathrm{pub,loc}} \\ & \qquad + |\mathrm{pop}(CC_h)| \sum_{f \succ CC_h} \frac{p^{\mathrm{pub}} \cdot y_f^{\mathrm{pub}}}{|\mathrm{pop}(f)|} \le B(CC_h, P) \end{aligned} \tag{4.9}$$

where $f \succ CC_h$ denotes every higher-level ward, city, state, or national consumer federation containing CC_h. Albert and Hahnel state (PEPE Ch. 3, p. 41; Ch. 5, p. 80) that for any individual or council whose request satisfies $SC \le B$, other councils may advise but cannot veto the request; veto power applies only to exception requests exceeding warranted allowances ($SC > B$). That is a normative design rule, not a theorem.

[13] Albert and Hahnel, Political Economy of Participatory Economics, Ch. 5, pp. 77–80, 88.

4.3.2 Communities of Affected Parties (CAPs) (DEP Ch. 7, pp. 139–148)

For each regulated pollutant $z \in \mathcal{Z}$ with hard biophysical sustainability ceiling $E_z^{\max}$ (established by Long-Run Environmental Planning; DEP Ch. 14),[14] given indicative damage price p_z^{pol}, CAP_z chooses the total permissive emission supply $E_z \in [0, E_z^{\max}]$ to maximise member well-being:

$$\max_{0 \le E_z \le E_z^{\max}} \quad W_{CAP_z}\Big(p_z^{\text{pol}} E_z,\ -D_{CAP_z}(E_z)\Big) \tag{4.10}$$

At an interior optimum below $E_z^{\max}$, CAP_z sets E_z such that the sum of marginal victim damages across $u \in \text{pop}(CAP_z)$ equals p_z^{pol} (the Lindahl-Pigou efficiency condition).

4.3.3 Worker Councils (PEPE Ch. 5, pp. 85–88; DEP Ch. 7, pp. 152–156)

Each worker council $WC_j \in \mathcal{J}$ begins the planning cycle with initial labour endowment l_j^0 ($\sum_{j=1}^{J} l_j^0 = l^*$) and initial capital stock endowment k_j^0 ($\sum_{j=1}^{J} k_j^0 = k^*$), for which it is automatically charged $p^{\text{lab}} \cdot l_j^0 + p^{\text{cap}} \cdot k_j^0$.[15] In iteration t, WC_j proposes output vector $\bar{a}_j = (\bar{a}_j^{\text{priv}}, \bar{a}_j^{\text{pub}}, \bar{a}_j^{\text{int}}, \bar{a}_j^{\text{inv}})$, intermediate inputs $\underline{a}_j$, natural resources r_j, net labour additions/releases Δl_j, net capital additions/releases Δk_j, and pollutant emissions e_j from its feasible technology set $\mathcal{T}_j$. Defining Social Benefit (SB_j) and Social Cost (SC_j):

$$SB_j(P) = p^{\text{priv}} \cdot \bar{a}_j^{\text{priv}} + p^{\text{pub}} \cdot \bar{a}_j^{\text{pub}} + p^{\text{int}} \cdot \bar{a}_j^{\text{int}} + p^{\text{cap,new}} \cdot \bar{a}_j^{\text{inv}} \tag{4.11}$$

$$SC_j(P) = p^{\text{int}} \cdot \underline{a}_j + p^{\text{res}} \cdot r_j + p^{\text{lab}} \cdot (l_j^0 + \Delta l_j) + p^{\text{cap}} \cdot (k_j^0 + \Delta k_j) + p^{\text{pol}} \cdot e_j \tag{4.12}$$

Equation (4.12) correctly charges $l_j^0 + \Delta l_j$ (endowment plus net hiring). The floor $p^{\text{lab}} \ge w_{\text{base}}$ in INV-4B, however, prevents labour user-rights from falling when labour is in surplus. If prices are to be opportunity costs, a surplus of a labour category should lower p_ℓ^{lab} below w_{base}; the floor converts that surplus into open underemployment rather than a market-clearing wage cut. An underemployment rebate (paying idle accredited hours from the Society Account) can restore household budgets, but it does not restore the opportunity-cost reading of p^{lab}, and it is funded by reducing Υ^{work} or σ_{inv}. This trade-off is unresolved.

WC_j maximises its workers' fulfilment and development $W_j(U_j, \Delta C_j)$ subject to the Socially Non-Abusive Production Constraint (PEPE Ch. 5, pp. 87–88):

$$\max_{\pi_j \in \mathcal{T}_j} \ W_j\Big(U_j(\pi_j),\ \Delta C_j(\pi_j)\Big) \quad \text{subject to} \quad SB_j(P) - SC_j(P) \ge 0 \tag{4.13}$$

that is, $SB_j(P)/SC_j(P) \ge 1$.

[14] Hahnel, Democratic Economic Planning, Ch. 7, pp. 139–148, Appendix pp. 159–162, and Ch. 14, pp. 261–267.

[15] Albert and Hahnel, Political Economy of Participatory Economics, Ch. 5, pp. 85–87.

4.4 Critical Algorithmic Audit of pequod-cljs and Convergence Rules

In Chapter 9 of Democratic Economic Planning (2021, pp. 173–194) and Szczepanczyk (2023),[16] the authors simulated 40 experiments (dep1ex61..100.csv, generated via bin/bigger_gen.sh) with 30,000 Worker Councils, 30,000 Consumer Councils, and 500 categories (100 private, 100 public, 100 intermediate, 100 natural resources, 100 labour categories), alongside 50 smaller 3,000-council runs (dep1ex01..50.csv).

4.4.1 The Hahnel-Weisdorf-Szczepanczyk Price Rule and CSV Audit

In csvgen.clj (upstream lines 391–392 and 463–469), let $S_i^{(t)}$ be total supply, $D_i^{(t)}$ be total demand, and $\text{surplus}_i^{(t)} = S_i^{(t)} - D_i^{(t)}$. Defining the two-sided relative imbalance $\rho_i^{(t)}$:

$$\rho_i^{(t)} = \frac{2\left|S_i^{(t)} - D_i^{(t)}\right|}{S_i^{(t)} + D_i^{(t)}}, \qquad \tilde{\delta}_i^{(t)} = 1.05 - 0.5^{\rho_i^{(t)}} \tag{4.14}$$

the damped percentage price delta $\delta_i^{(t)}$ and multiplicative price update are:[17]

$$\begin{aligned} \delta_i^{(t)} &= \max\left(0.001,\ \tilde{\delta}_i^{(t)} \cdot \min\left(0.25, \rho_i^{(t-1)}\right)\right) \\ P_i^{(t+1)} &= P_i^{(t)}\left(1 + \text{sign}\left(D_i^{(t)} - S_i^{(t)}\right)\delta_i^{(t)}\right) \end{aligned} \tag{4.15}$$

The upstream code computes $\max(0.001, \min(\tilde{\delta}, \tilde{\delta} \cdot \min(0.25, \rho^{(t-1)})))$. Because $\tilde{\delta} \in [0.05, 0.80]$ is positive and $\min(0.25, \cdot) \leq 0.25$, the inner min always returns its second argument, so the simplified form above is identical. (A commented-out variant in the same file, $\min(\tilde{\delta}, \rho^{(t-1)})$, is the one in which the outer min would bind.) Our programmatic audit of the committed CSV files[18] reports the subsets the book actually cites. Across dep1ex61–100.csv ($n = 40$), Year 1 reaches the 5% threshold (:green) in 11.85 rounds on average and the 3% threshold (:blue) in 19.20 rounds; Year 2 reaches 5% in 6.58 rounds. Averaging all 99 files other than dep1ex100.csv yields 11.505 Year-1 5% rounds and must not be quoted as 11.85. File dep1ex100.csv is a committed large-run log; there is no matching dep1ex100.clj. The 50 dep2ex*.csv files have no corresponding .clj generators in the clone; a code comment in csvgen.clj associates a second experiment family with

[16] Hahnel, Weisdorf, and Szczepanczyk, "Computer Simulations of Participatory Planning," Ch. 9 in Hahnel, Democratic Economic Planning, pp. 173–194; Szczepanczyk, "Pseudocode and Algorithms" (2023), pp. 43–54; audited against upstream msszczep/pequod-cljs at commit 8bd46b5c233f0e48c767ad6d5069b9437a67c38b (src/clj/pequod_cljs/csvgen.clj, 836 lines; trimmed copy in research/sources/pequod_cljs_csvgen_apache2.clj, 799 lines).

[17] Szczepanczyk, "Pseudocode and Algorithms" (2023), Eq. 9, p. 48; upstream csvgen.clj lines 391–392 (get-deltas) and 463–469.

[18] Performed by research/analysis/verify_r2.py in the project repository.

toothaches, but committed dep2ex CSVs do not show the large between-section exponent jump that would mark a toothache run. Because no wall-clock timing log exists in the upstream repository, any per-iteration wall-clock figure (~ 2.5 s) is an unverified estimate. Hahnel (2021) Ch. 9 reports the Cobb–Douglas simulations; that chapter was consulted only through secondary notes and does not claim a page-level reading of increasing-returns results.

4.4.2 Audit of pequod-cljs and the "Toothache" Construction

Direct inspection of wcs.clj, ccs.clj, csvgen.clj, and all 70 dep1ex*.clj generators establishes four structural properties of pequod-cljs. Fifty files (dep1ex01–50) are stored directly; twenty (dep1ex51, dep1ex60–78) are Git LFS objects of about 160 MB each, fetched from the repository's media endpoint and checked against the SHA-256 in their pointers. Together they hold 750,000 Worker Council records.[19]

1. Unconstrained Closed-Form Cobb-Douglas Optimisation: Each WC_j maximises $\Pi(x,e) = \lambda a e^c \prod_{m=1}^{n} x_m^{b_m} - \sum_{m=1}^{n} p_m x_m - s e^k$ via symbolic Mathematica first-order conditions (solution-1..8, upstream lines 221–401) without enforcing $SB_j/SC_j \geq 1$ or labour stock conservation $\sum_j \Delta l_j = 0$.

2. "Increasing returns" (create-toothaches) and unboundedness: In wcs.clj (lines 31–70), each WC_j has $n \in \{3,\ldots,8\}$ inputs with baseline exponents $\sum_{m=1}^{n} b_m \in [0.75, 0.85]$, effort elasticity $c \in [0.05, 0.10]$, and effort disutility exponent $k \in [3,4]$ (:du). Along the ray $x_m(t) = t$, $e(t) = t^{1/k}$, revenue scales as $t^{\sum b_m + c/k}$ against linear cost, so Π is bounded with an interior maximum along that ray iff $\sum b_m + c/k < 1$. (Joint concavity of Π on (x,e) is a stronger, distinct condition and is not claimed.) create-toothaches in csvgen.clj selects 20% of councils and adds 0.1 to each b_m. Across all 750,000 records the baseline index $\sum b_m + c/k$ lies in $[0.7696, 0.8781]$; with the toothache increment it would lie in $[1.0696, 1.6652] > 1$ for every record, and hence for every selected council, so the FOC is a saddle along the scale ray and the formula predicts downward-sloping supply. That prediction is not observed in committed CSVs: -main currently sets toothaches? true, yet the largest Year-1/Year-2 exponent-sum field change found in dep1ex*.csv and dep2ex*.csv is empty (those rows carry no numeric exponent), and the largest first-green-round change between year sections is 0.03 on the files inspected.

3. Exogenous Consumer Budgets: In consume (upstream lines 188–204), each Consumer Council's budget (:income cc) is fixed at $5,000$ across all iterations.

4. Two-Sided Resource Tolerance Without Pollutants: Natural resources are given a fixed supply of $1,000$ (upstream line 168) and priced within the two-sided $\tau = 5\%$ tolerance (permitting up to $\sim 5.1\%$ overshoot), without CAP pollutant pricing or strict one-sided ceiling enforcement.

[19] Checked by research/analysis/verify_pequod_datasets.py in the project repository.

4.4.3 Why General Convergence Cannot Be Guaranteed by Heuristic Tâtonnement

Simply adding heuristic proximal regularisation and trust regions to Eq. (4.15) cannot guarantee convergence under Leontief complementarities and non-convexities, on both theoretical and empirical grounds:

- Scarf's (1960) Leontief Instability & Trust-Region Stalling: By the Sonnenschein–Mantel–Debreu theorem and Herbert Scarf's (1960) counterexample,[20] price tâtonnement can cycle indefinitely when income effects violate gross substitutability. Running Eq. (4.15) on Scarf's 3-good Leontief economy[21] confirms that over 3,000 rounds, $\max_i \rho_i^{(t)}$ never drops below 10.54% ($\beta \in \{\text{None}, 0.25, 0.10, 0.02\}$), while a decaying trust region ($\beta^{(t)} = 0.25 \cdot 0.99^t$) freezes proposals at a 56.60% imbalance—shrinking trust regions stall, they do not clear the cycle.
- Indivisible Plants vs. One-Good Shapley–Folkman Smoothing: For a single indivisible facility with $S(p) = 100 \cdot \mathbb{I}_{p \geq 10}$ and demand $D(p) = 70 - p$, no clearing price exists: the theoretical minimum imbalance is $\rho = 50.0\%$, and over 500 rounds Eq. (4.15) never falls below 50.37%. Disaggregating that capacity across $K = 20$ plants of size $100/K$ with staggered thresholds $8.5 + 3i/(K-1)$ yields a minimum imbalance of 0.40% across 500 rounds. That is the best round, not a stable gap. The last-100-round maximum at $a = 70$ is 8.4%; a sweep $a \in [60, 80]$ in steps of 0.5 has median last-100-round max 6.0%, worst 9.5%, and only 24% of cases stay $\leq 5\%$ in the last 100 rounds.[22] Identical thresholds at 10 give no smoothing ($\min \rho \approx 50\%$). Starr (1969) bounds the Hausdorff gap between the sum of non-convex sets and its convex hull; this one-good staggered-threshold test does not instantiate that theorem for interdependent Leontief markets.[23]

4.4.4 Research-Candidate Solvers (Algorithms 3a–3d)

DPEP separates four candidate coordination algorithms to be benchmarked in Phase 1. None is a production commitment. Table 4.2 places them among the method families under consideration. All four share one definition of success.

Definition 4.2 (Feasible, cleared, and socially acceptable plans). Let $u^{\text{out}} = \sum_g p_g \max(0, S_g - D_g) / \sum_g p_g S_g$ be the value-weighted unused-output share, $u^{\text{need}} = \sum_g p_g \max(0, D_g - S_g) / \sum_g p_g D_g$ the unmet-demand share, u^{lab} and u^{cap}

[20] Herbert Scarf, "Some Examples of Global Instability of the Competitive Equilibrium," International Economic Review 1, no. 3 (1960): 157–172; Hugo Sonnenschein, "Do Walras' Identity and Continuity Characterize the Class of Community Excess Demand Functions?" Journal of Economic Theory 6, no. 4 (1973): 345–354.

[21] Run with research/analysis/verify_economics.py in the project repository.

[22] Computed by research/analysis/verify_r2.py in the project repository.

[23] Ross M. Starr, "Quasi-Equilibria in Markets with Non-Convex Preferences," Econometrica 37, no. 1 (1969): 25–38.

the shares of accredited labour hours and capital services left idle, and ΔW the relative welfare or need-satisfaction gap to a reference oracle (a central LP or MILP on the same instance, Table 4.2) where one exists.

1. A plan is feasible if $D_g \leq S_g$ for every good and every hard ecological or resource cap holds ($D_r \leq r^*$, $D_z \leq E_z^{\max}$). The one-sided rule belongs here: caps must never be exceeded, and slack against a cap is not waste.
2. A plan is cleared if it is feasible and, for produced goods, labour and capital, $\max_i \rho_i \leq \tau$, $u^{\text{out}} \leq \bar{u}^{\text{out}}$, $u^{\text{lab}} \leq \bar{u}^{\text{lab}}$ and $u^{\text{cap}} \leq \bar{u}^{\text{cap}}$, net of the planned slack buffer SL^E.
3. A plan is socially acceptable if it is cleared, $u^{\text{need}} \leq \bar{u}^{\text{need}}$ for priority (Maxim 4) categories, and $\Delta W \leq \bar{\Delta}$ where an oracle exists. These thresholds are ratified politically, not derived.

Proposed defaults for Phase 1 (for author ratification): $\tau = 5\%$, $\bar{u}^{\text{out}} = \bar{u}^{\text{lab}} = \bar{u}^{\text{cap}} = 5\%$ beyond planned slack, $\bar{u}^{\text{need}} = 2\%$, $\bar{\Delta} = 5\%$ median and 15% at the 90th percentile.

The distinction matters. In the single-plant case of Section 4.4.3, Eq. (4.15) reaches one-sided feasibility at round 3 with $S = 100$ and $D = 58.48$: 42% of output is unused, and the case has no clearing price. Under the definition above the run is reported as feasible but not cleared in every one of 500 rounds. With $K = 20$ staggered plants the first feasible round (round 3) still leaves 8.4% unused, the first cleared round is round 5, and 57% of rounds are cleared. Scarf's cycle is never even feasible: some good is over-demanded in every one of 3,000 rounds.[24]

Algorithm 3a — Proximal damped tâtonnement (heuristic).

1. Input: $P^{(0)}$, $\mu > 0$, thresholds of Definition 4.2, $T_{\max}$, caps $r^*, E_z^{\max}$.
2. For $t = 1, \ldots, T_{\max}$: each council solves $\max_{\pi \in \mathcal{T}_c} W_c(\pi; P^{(t)}) - \frac{\mu}{2}\|\pi - \pi^{(t-1)}\|_2^2$ subject to $SB/SC \geq 1$ or $SC \leq B$ as appropriate; IFB updates P by Eq. (4.15) and rescales so $w_{\text{base}} = 1$.
3. Halt with success when the plan is cleared. If $T_{\max}$ is reached, publish the best feasible plan with its u^{out}, u^{lab}, u^{cap} and u^{need}, labelled feasible, not cleared, or report infeasibility. No rate is claimed.

Algorithm 3b — Proximal Jacobian ADMM (Deng, Lai, Peng & Yin 2017, Algorithm 4; fixed-budget convex case only). Deng et al. treat $\min \sum_{i=1}^{N} f_i(x_i)$ subject to $\sum_i A_i x_i = c$ with each f_i closed proper convex. DPEP's coupling constraints are inequalities $\sum_c A_c x_c \leq b$ (demand within supply and caps), so a non-negative slack block $s \geq 0$ with $f_s \equiv 0$ is added and the constraint becomes $\sum_c A_c x_c + s = b$; the number of blocks is $N = N_{\text{active}} + 1$. With $f_c = -W_c + \iota_{\mathcal{T}_c}$:

[24] Computed by research/analysis/verify_r3.py in the project repository.

1. Input: closed convex $\mathcal{T}_c$, fixed B_u, penalty $\rho > 0$, dual damping $\gamma \in (0, 2)$, proximal matrices $P_i \succ \rho\left(\frac{N}{2-\gamma} - 1\right) A_i^\top A_i$ (Deng et al. eq. 2.11; for $P_i = \tau_i I$ this is $\tau_i > \rho(\frac{N}{2-\gamma} - 1)\|A_i\|^2$, and for the prox-linear choice $P_i = \tau_i I - \rho A_i^\top A_i$ it is $\tau_i > \frac{\rho N}{2-\gamma}\|A_i\|^2$).
2. In parallel for every block i (councils and slack), with $r^k = \sum_j A_j x_j^k - b$:

$$x_i^{k+1} = \arg\min_{x_i} \; f_i(x_i) + \frac{\rho}{2}\left\| A_i x_i + \sum_{j \neq i} A_j x_j^k - b - \lambda^k/\rho \right\|^2 + \frac{1}{2}\|x_i - x_i^k\|_{P_i}^2.$$

3. Dual step: $\lambda^{k+1} = \lambda^k - \gamma\rho\,(\sum_i A_i x_i^{k+1} - b)$. With the slack block the dual needs no projection; indicative prices are $-\lambda$.
4. Halt when the plan is cleared (Definition 4.2) and $\|u^k - u^{k+1}\|$ is below tolerance; otherwise fail at $T_{\max}$.

Deng et al. prove convergence and an $o(1/k)$ rate on the successive-iterate quantity $\|u^k - u^{k+1}\|_G^2$, (primal and dual together), not a rate to optimal welfare. The theorem does not cover endogenous budgets, ε-caps that depend on prices, renormalising duals so that $w_{\text{base}} \equiv 1$ mid-run, or non-convex $\mathcal{T}_c$ (convergence results for ADMM on non-convex problems, such as Hong, Luo and Razaviyayn 2016, need structural conditions that DPEP's lumpy technologies do not meet). Writing the objective as $\sum_c W_c$ assumes cardinal, interpersonally comparable well-being. Because N includes every active council, τ_i grows with N_{active}; whether that makes convergence too slow at $N = 10^6$ is untested.

A simpler-looking iteration, $x_c^{t+1} = \arg\min -W_c(x) + \langle \lambda^t, A_c x \rangle + \frac{1}{2}\|x - x_c^t\|_{P_c}^2$ with a projected dual ascent, is easily mistaken for ADMM. It is an Arrow–Hurwicz-type primal–dual proximal iteration: it omits the augmented residual term, and the Deng et al. theorem does not apply to it. In a two-council test with linear welfare and a shared capacity of 1, it oscillates indefinitely (residual ≈ 1.0, price swinging between 0 and 2 over 20,000 rounds), while Algorithm 4 with a slack block converges to residual 0 and price 1; both converge when welfare is strictly concave.[25]

Algorithm 3d — Damped budget fixed point around a fixed-budget inner solve (candidate for endogenous income). Endogenous income makes full DPEP an equilibrium problem: budgets $B_u(P, D)$ (Eq. 4.8) depend on the prices and quantities they help determine, and the effort cap couples incomes to SB/SC. The natural formulation is a mixed complementarity problem (MCP) in (x, P, B). Phase 1 needs a solver it can actually run, so the proposed candidate is an outer loop:

1. Hold budgets B^k fixed and solve the inner fixed-budget problem with Algorithm 3b (convex instances) or 3c (lumpy instances), giving (x^k, P^k).

[25] Computed by research/analysis/reference_tests_r3.py and research/analysis/verify_r3.py in the project repository.

2. Recompute warranted budgets $\hat{B}^k = B(P^k, D^k)$ from the Society Account identities and update $B^{k+1} = (1-\alpha)B^k + \alpha\hat{B}^k$ with damping $\alpha \in (0,1]$.
3. Stop when $\|B^{k+1} - B^k\| / \|B^k\| \leq 10^{-3}$ and the inner plan is cleared; report cycling or divergence otherwise.

Assumptions and limits. Each inner solve inherits 3b's or 3c's assumptions. No convergence theorem is claimed for the outer loop; a fixed point need not be unique, and the loop can cycle when income effects are strong (the same mechanism that defeats tâtonnement on Scarf's example). As a cross-check on small instances, the MCP can be passed to a PATH-type complementarity solver; PATH is proprietary and its free licence is limited to small problems, so licensing terms must be checked before relying on it, and no open-source MCP solver of comparable robustness is assumed. Lemke's method applies only to the linear complementarity case and is used as a diagnostic.

Interaction with Algorithm 3c. The 3c objective below is written at the current budgets B_u. Under endogenous income a lumpy plant decision changes incomes, and so the demand it is meant to meet; 3c is therefore run as the inner solve of 3d, and binary decisions taken at one budget vector are re-checked after each budget update. Phase 1 tests 3c first with fixed budgets and then inside 3d.

Algorithm 3c — Two-tier federation MILP + continuous ADMM (undefined until the following are fixed).

1. What councils reveal: each WC_j posts a discrete menu of plant/shift options $z_{j,k} \in \{0,1\}$ with publicly listed input–output rows (REQ-DATA-03), not its full private technology set $\mathcal{T}_j$. Continuous operating bands around each chosen z remain local.
2. Objective (provisional): lexicographic (1) one-sided ecological and resource feasibility, (2) minimise unmet consumer demand at current B_u, (3) minimise unused output and idle labour and capital (Definition 4.2), (4) minimise a public lumpy-cost index. This objective is a federation proposal, not a social-welfare theorem. Under endogenous income 3c runs inside Algorithm 3d.
3. Decision rights: binary z are ratified by the relevant industry and regional federations plus affected CAPs; a central HiGHS MILP is an aid, not an authority (Kuch 2024).
4. Infeasibility: if the MILP is infeasible, return the IIS (irreducible infeasible subsystem), raise the relevant cap or investment debate, and do not silently relax $E_z^{\max}$. Halt and publish the IIS after $T_{\max}$ unsuccessful repairs.

4.5 Protocol State Machine and Canonical CBOR Schemas

To prevent IEEE-754 floating-point non-determinism across CPU architectures (x86_64, ARMv8 smartphones, and WebAssembly), all quantities and prices in

Table 4.2: Ten method families by problem class (research map, not a ranking)

Family	Problem class	DPEP status
Unconstrained Cobb–Douglas FOC	Smooth convex, fixed income	pequod-cljs baseline; rejected for production.
Damped tâtonnement (3a)	Heuristic, any $\mathcal{T}_c$	Candidate; fails Scarf/single-plant tests.
Proximal Jacobian ADMM (3b)	Convex, fixed budgets	Candidate; Deng et al. Algorithm 4 with slack block; successive-iterate rate only.
Arrow–Hurwicz primal–dual	Convex, fixed budgets	Resembles 3b but lacks the augmented term; oscillates on linear welfare; not used.
Fisher / Eisenberg–Gale	Linear utilities, fixed budgets	Fits fixed B_u; not endogenous income.
Negishi welfare weights	Convex equilibrium	Requires comparable W_c; open normatively.
Variational inequality / MCP	Endogenous income + prices	Leading formulation. Phase-1 candidate: Algorithm 3d (damped outer loop); PATH-type solver as small-instance cross-check, subject to licence.
Federation MILP (3c)	Discrete lumpy $z \in \{0,1\}$	Candidate; needs ratification and IIS exit.
Scarf / Lemke path-following	Linear complementarity	Diagnostic for cycles; not a planning procedure.
Central LP (Cockshott-style)	Known A, K, R, L	Rejected as workplace authority; usable as Phase-1 oracle.

DPEP are encoded as deterministic 64-bit fixed-point integers (FixedDecimal, default scale 10^{-6} micro-units):

Canonical wire encoding. Listings 4.1 and 4.2 are an IDL sketch (proto3-like field names for readability). It is not sent as protobuf. All signed payloads use RFC 8949 §4.2 Canonical CBOR (CDE) with a fixed global scale of 10^{-6} (int64 micro-units). Multiplication uses 128-bit intermediate products, round-half-to-even, saturating on overflow (the round fails closed rather than wrapping). Division by zero is a fault. The map 0.5^{ρ_i} is a degree-6 minimax polynomial on $\rho \in [0, 2]$ with published coefficients; outside that interval the update is clipped. Protocol checks verify the rounded inequalities $SB/SC \geq 1$ and $SC \leq B$ after this arithmetic. They do not prove that a council maximised a truthful W_c, and they do not exist as KKT certificates for MILPs.

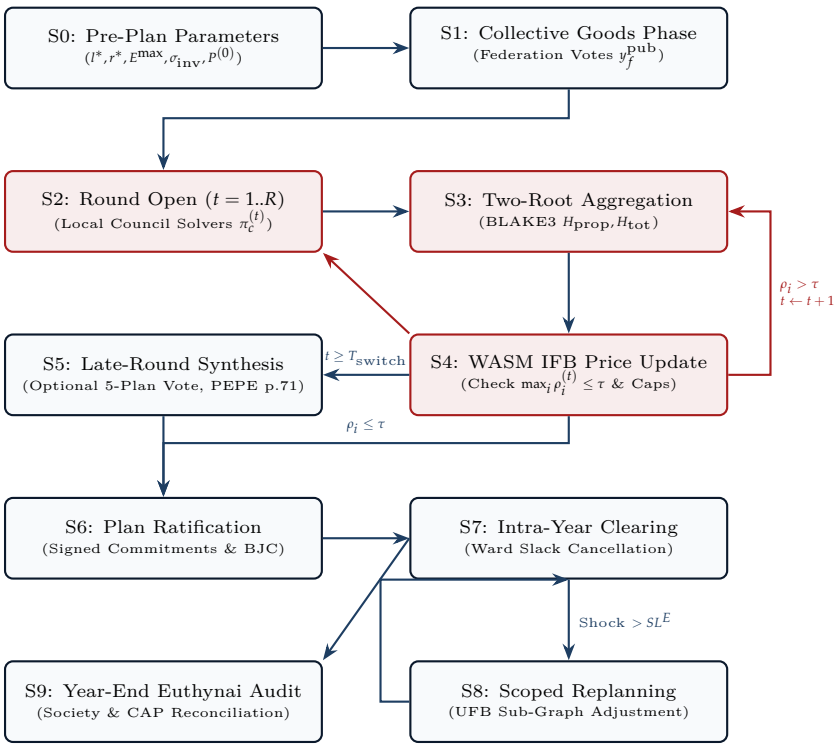


Figure 4.1: Complete 10-State DPEP Annual Planning and Intra-Year Operational State Machine

Listing 4.1: IDL sketch of DPEP item vectors and consumer-council proposals. Signed encoding is RFC 8949 Canonical CBOR, not protobuf.

```
// IDL sketch only --- not the wire format
package dpep.v1;

message SparseItemVector {
  repeated uint32 item_ids = 1;          // Strictly ascending catalog indices in Omega
  repeated int64 micro_quantities = 2;   // Strictly positive quantities (> 0) in 10^-6 micro-units
}

message QualitativeAnnotation {          // Implements PEPE Ch. 6 'Lists' with GDPR redactability
  uint32 item_id = 1;
  string summary_code = 2;
  string narrative_text = 3;
  bytes redactable_pointer_cid = 4;      // Mutable pointer supporting tombstone erasure
}

message ConsumerCouncilProposal {
  string domain_tag = 1;                 // Fixed: "DPEP-WD-v0-CC-PROPOSAL"
  uint32 plan_year = 2;
  uint32 round_number = 3;
  bytes council_id = 4;
  uint32 accredited_member_count = 5;
  SparseItemVector private_demand = 6;   // Coarse-category totals (HRU x s_g) from K=3 Prio3 aggregators
  SparseItemVector local_public_demand = 7;
  int64 total_social_cost_micro = 8;
  int64 warranted_budget_micro = 9;
  bool requests_exception = 10;
  repeated QualitativeAnnotation qualitative_list = 11;
  bytes vdaf_transcript_root = 12;       // Commits Prio3 aggregate-share commitments
  bytes signer_bitmap = 13;
  repeated bytes council_hybrid_signatures = 14; // k-of-n Ed25519+ML-DSA-65 multi-signature
}
```

Listing 4.2: IDL sketch of worker-council proposals and IFB round checkpoints (continuation of Listing 4.1).

```
message WorkerCouncilProposal {
  string domain_tag = 1;                    // Fixed: "DPEP-WD-v0-WC-PROPOSAL"
  uint32 plan_year = 2;
  uint32 round_number = 3;
  bytes council_id = 4;
  uint32 industry_federation_id = 5;
  SparseItemVector proposed_outputs = 6;
  SparseItemVector intermediate_inputs = 7;
  SparseItemVector natural_resources = 8;
  SparseItemVector total_labor_requested = 9;
  SparseItemVector total_capital_requested = 10;
  SparseItemVector pollutant_emissions = 11;
  int64 effective_effort_hours_micro = 12; // INV-4B: sum(h_u*eps_u) <= (1.0+dl)*
      min(eps_max,SB/SC)
  int64 social_benefit_micro = 13;
  int64 social_cost_micro = 14;
  int64 sb_sc_ratio_micro = 15;
  bool requests_appeal = 16;                // Subject to soft-budget cap at IF_k
  repeated QualitativeAnnotation qualitative_list = 17;
  bytes signer_bitmap = 18;
  repeated bytes council_hybrid_signatures = 19;
}

message RoundAggregationAndPrices {
  string domain_tag = 1;                    // Fixed: "DPEP-WD-v0-ROUND-
      CHECKPOINT"
  uint32 plan_year = 2;
  uint32 round_number = 3;
  repeated int64 indicative_prices_micro = 4; // Normalised so w_base_micro == 1
      _000_000
  repeated int64 total_supply_micro = 5;
  repeated int64 total_demand_micro = 6;
  repeated int64 planned_slack_micro = 7;
  repeated int64 one_sided_shortage_micro = 8; // max(0, D_i - S_i)
  int64 base_hourly_remuneration_micro = 9;    // Fixed numeraire: 1_000_000
  bool strict_one_sided_feasible = 10;
  bytes proposals_merkle_root = 11;            // 32-B BLAKE3 root over sorted
      council proposals (512-B proof)
  bytes category_totals_merkle_root = 12;      // 32-B BLAKE3 root over category
       sums
  bytes council_participation_bitmap = 13;     // 2 bits/council: 01=Active, 10=
      Partitioned, 00=Excluded
  bytes deterministic_wasm_code_hash = 14;
  bytes validator_quorum_bitmap = 15;
  repeated bytes validator_hybrid_signatures = 16;
}
```

CHAPTER 5

Distributed Systems Architecture, Consistency, and Low-Power Participation

> "Planning is computerized. All councils can access data banks of all facilitation boards and all other councils. The only thing demanding about flexible information access is that the computer system not only automatically accept but also sum and compare proposals to ascertain the status of any good."
>
> — Michael Albert and Robin Hahnel, Looking Forward (1991), Ch. 6, "Preparing First Proposals"[1]

5.1 Why Participatory Economics Is Not a High-TPS Blockchain Problem

When we examine the computational structure formalised in Chapter 4, three architectural truths emerge:

1. Annual Planning Is an Epoch-Synchronised Bulk-Synchronous Parallel (BSP) Process: Millions of participants and tens of thousands of councils draft proposals locally, aggregate them up nested geographic and industrial federation trees, and compute updated indicative prices $P^{(t+1)}$ once per planning round (10–20 rounds per year). Global state finality is required only at iteration boundaries and at plan ratification.
2. Intra-Year Execution Is Hierarchically Scoped: As Albert and Hahnel (1991a, Ch. 4, pp. 66–67) specify, routine intra-year consumer preference shifts cancel out at the neighbourhood, ward, or city federation level within planned slack buffers ($SL^E \approx 3$–5%) without altering aggregate production plans.
3. Participation Must Work Offline and on Resource-Constrained Devices: Workers and consumers in rural collectives or intermittently connected networks must be able to deliberate, optimize local proposals, and sign ballots offline.[2]

5.2 The Four-Layer Hybrid Architecture (Architecture D)

Figure 5.2 traces one planning round message by message across the layers, and Figure 5.4 (Section 5.3) shows where each node role sits.

[1] Albert and Hahnel, Looking Forward, Ch. 6 (lf6_2.htm).

[2] Martin Kleppmann, Adam Wiggins, Peter van Hardenberg, and Mark McGranaghan, "Local-First Software: You Own Your Data, in spite of the Cloud," Proceedings of the 2019 ACM SIGPLAN International Symposium on New Ideas, New Paradigms, and Reflections on Programming and Software (Onward! '19), pp. 154–178, https://doi.org/10.1145/3359591.3359737.

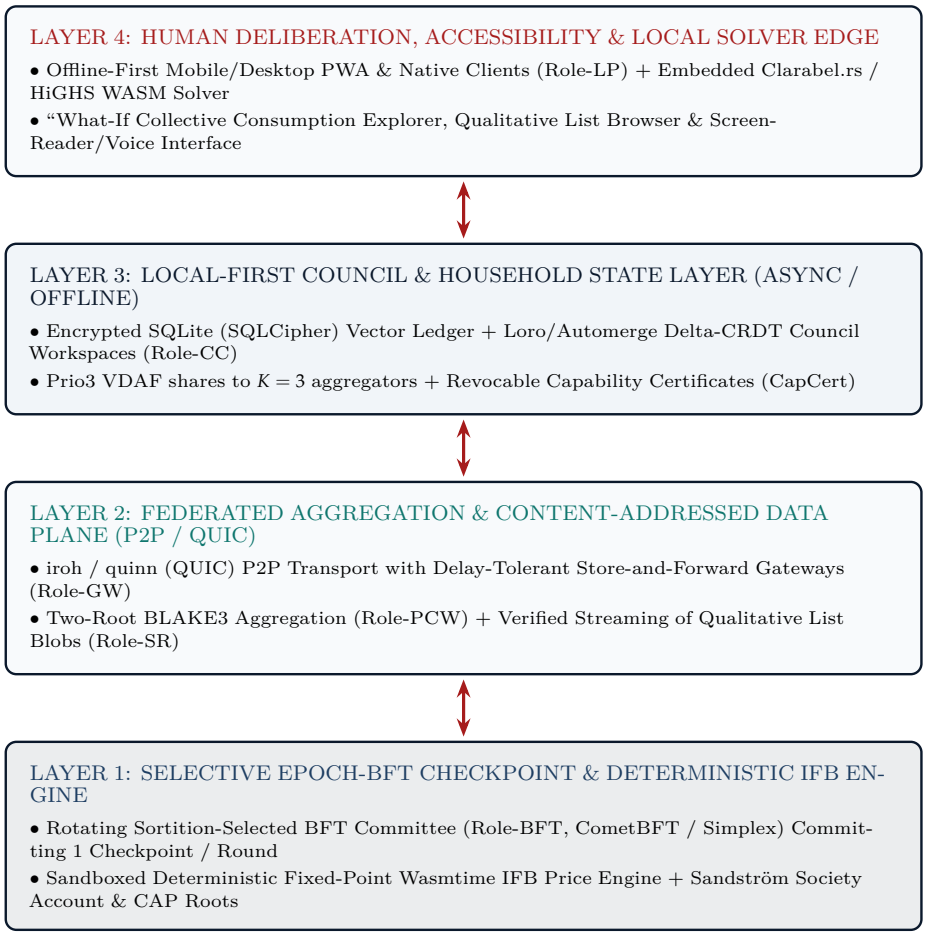


Figure 5.1: Four-Layer Architecture of the Distributed Participatory Economics Protocol (Architecture D)

5.2.1 Two-Root Authenticated Aggregation, Completeness Bitmaps, and Physical Boundaries

Naively embedding an 80 KB dense 10,000-category vector inside every internal node of a depth-16 Sparse Merkle-Sum Tree would cause a single mobile inclusion path to bloat to 16×80 KB = **1.28** MB—and still could not prevent a dishonest aggregator from fabricating off-path sibling subtrees unless auditors recomputed the entire tree from the leaves up. Moreover, a Merkle inclusion proof for council c proves membership, not completeness (whether dissident council c' was omitted) or physical truth (whether reported output physically exists).

A uniform 5.2 kB proposal size is too small: 5,200 B less one 3,373 B ML-DSA-65 signature leaves 1,827 B, about 150 sparse items at 12 B each (4-byte id, 8-byte

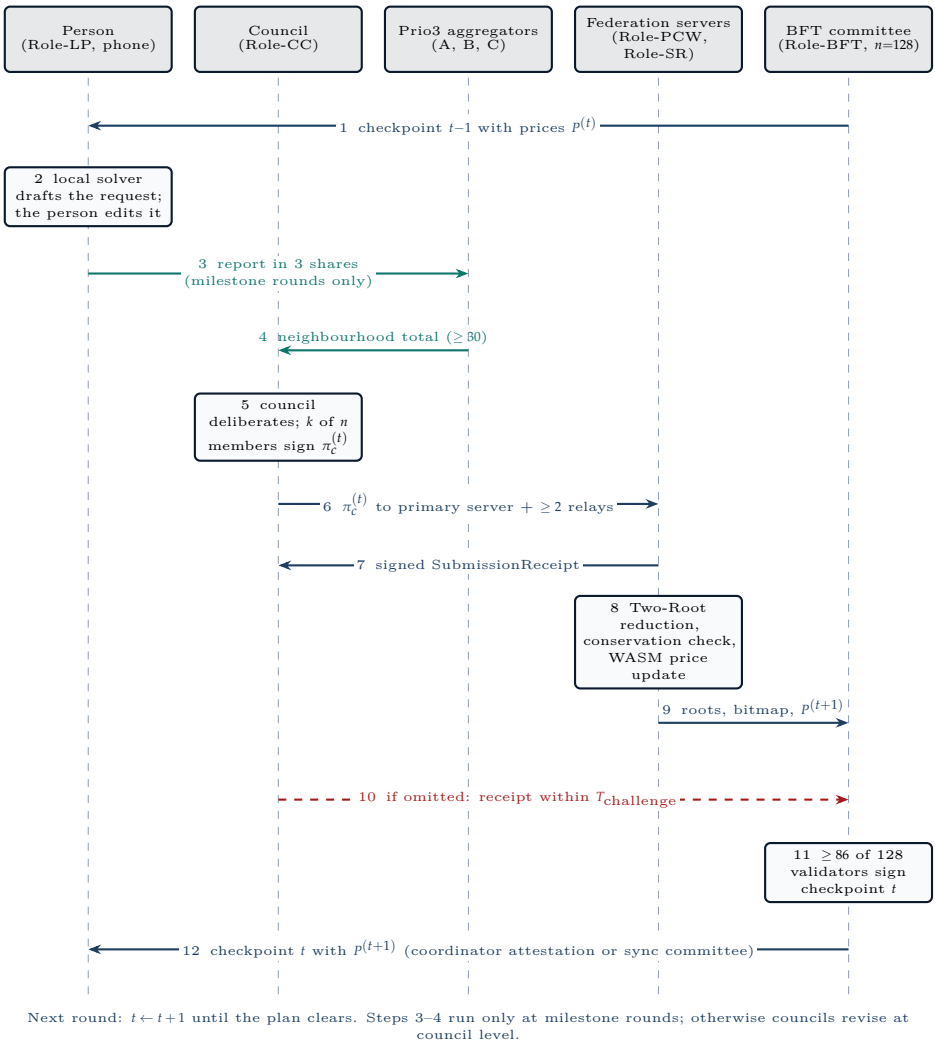


Figure 5.2: One planning round as a message sequence. Teal steps carry private household data and run only at milestone rounds; the dashed crimson step is the censorship challenge (Section 5.2.1).

quantity), while a neighbourhood's summed demand over 3,000 private categories is about 36 kB, and every extra multi-signature signer adds a further 3,373–3,437 B. Table 5.1 reworks sizes by council type. Council threshold signatures are not specified: until a standard post-quantum threshold scheme exists (open question P12), a council signs with a k-of-n multi-signature, i.e. k separate hybrid signatures plus a signer bitmap.

Table 5.1: Council proposal sizes and round archive at $N = 10^6$ (raw canonical CBOR; analytical)

Council type	1 signer	5 signers	Basis
Worker council, 150 items	5.6 kB	19.4 kB	400 B header + 150×12 B + hybrid signatures (3,437 B each)
Consumer council, 200 coarse categories	6.2 kB	20.0 kB	Proposed coarse household baseline (Section 7.4.2)
Consumer council, 3,000 categories	39.8 kB	53.6 kB	Dense neighbourhood sum
Round archive, 20k WC + 20k CC			
Coarse, single key	238 MB / 40k sig. checks		2.4 GB raw per 10-round year
Coarse, 5 signers	787 MB / 200k sig. checks		7.9 GB raw per year
Dense, single key	910 MB / 40k sig. checks		9.1 GB raw per year
Dense, 5 signers	1.46 GB / 200k sig. checks		14.6 GB raw per year

Header size (400 B) is an estimate; compression ratios for signature-dominated payloads are poor because signatures are incompressible. Source: research/analysis/verify_r3.py, section E.

Even at these sizes, a round's proposals are small enough that full replication is cheaper than heavy internal-vector proofs. DPEP therefore uses a Two-Root Authenticated Aggregation Protocol instead of internal-vector Merkle-sum proofs:

$$H^{(t)}_{\text{proposals}} = \text{BLAKE3-Root}\Big(\big\{\big(\text{id}_c,\ \text{BLAKE3}(\text{CBOR}(\pi^{(t)}_c))\big)\big\}_{c=1}^{M}\Big) \tag{5.1}$$

$$H^{(t)}_{\text{totals}} = \text{BLAKE3-Root}\Big(\big\{\big(i,\ S^{(t)}_i,\ D^{(t)}_i,\ P^{(t+1)}_i\big)\big\}_{i\in\Omega}\Big) \tag{5.2}$$

- 512-Byte Mobile Inclusion Proofs: Any mobile client (Role-LP) verifies that its council's exact proposal was included in proposals_merkle_root using a standard 16-level hash path of 16×32 B = **512** bytes.
- Full Leaf Data Availability (DA) & Conservation Verification: Auditing servers (Role-SR, Role-PCW, Role-BFT) download the round's leaf archive (0.24–1.46 GB raw; Table 5.1), verify the council signatures against the accredited roster, and check that the exact fixed-point sum equals category_totals_root before signing the checkpoint. That is 40,000–200,000 ML-DSA-65 verifications per round. Per-signature verification time on server hardware has not been benchmarked here, so whether every server can do this inside the round window is a Kill Gate 2 measurement, not an assumption; the fallback is to shard verification across servers and have each attest to its shard.

- Distinguishing Censorship from Network Partitions (council_participation_ bitmap): Each checkpoint commits a 2-bit-per-council bitmap (01 = Fresh Round-t Proposal, 10 = Timeout/Partition Carry-Forward, 00 = Excluded). Councils submit proposals to both their primary aggregator and ≥ 2 independent Role-SR relays, receiving timestamped signed SubmissionReceipts. Any council falsely marked 10 or 00 broadcasts its receipt during the pre-finality challenge window $T_{\text{challenge}}$ to force inclusion and fault the censoring aggregator.
- Explicit Cryptographic vs. Physical Boundary: Cryptography verifies digital authenticity, inclusion, and arithmetic conservation; it cannot verify that physical grain was harvested or that emissions sensors were not bypassed, which remain governed by federated physical euthynai spot-audits and counterparty delivery reconciliation.

Figure 5.3 shows the two trees, the phone's inclusion path, the participation bitmap and the challenge that turns an omission into a public fault.

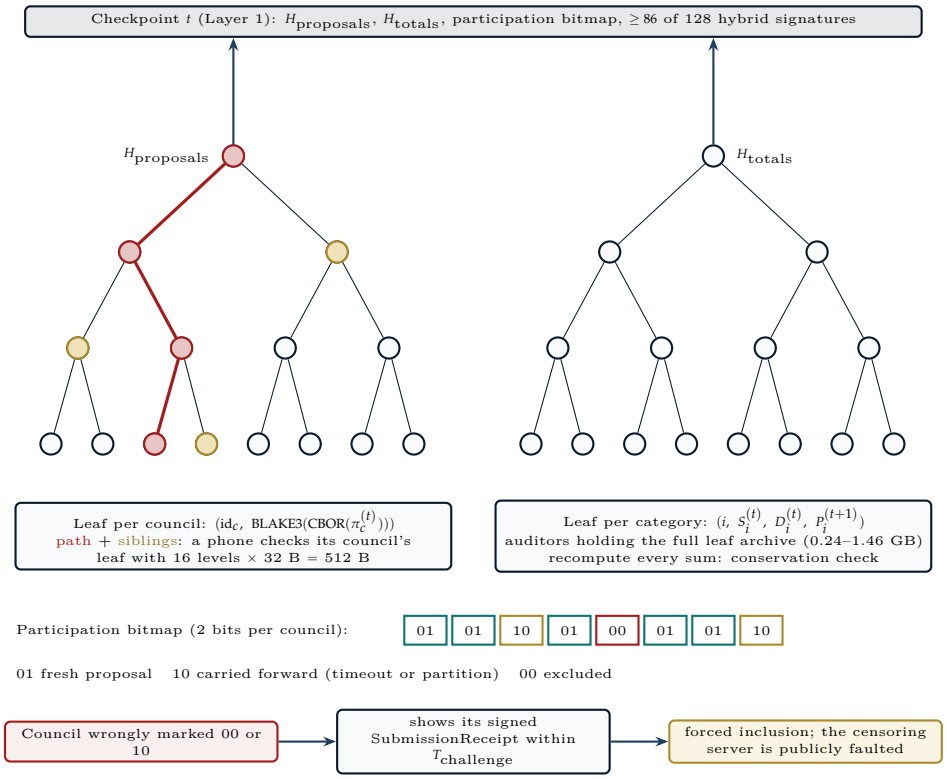


Figure 5.3: Two-Root authenticated aggregation. A phone verifies its council's leaf with a 512-byte path; auditors with the leaf archive recompute every category sum; the participation bitmap and signed receipts separate censorship from partition.

5.3 Granular Consistency Mapping and Node Role Taxonomy

Table 5.2: Consistency Models and Partition Tradeoffs Across Economic Subsystems

Economic Subsystem	Consistency Model	Latency Target	CAP Partition Behaviour	Architectural Justification
1. Proposal Drafting (S1–S2)	Strong Eventual Consistency (Delta-CRDT + SQLite)[3]	< 50 ms local	AP (Offline-First): Drafts continue on local LAN/device.	Unsubmitted drafts affect no external council; offline usability is paramount.
2. Council Ratification Vote	Council-Scoped Quorum (k-of-n multi-signature)	Seconds to hours	Locally CP on Council LAN: Independent of WAN.	Only members of WC_j or CC_h hold authority over their council's proposal.
3. Federated Round Aggregation (S3)	Deterministic Two-Root Reduction	1–10 min per round	Bounded-Wait CP + Fallback: Carries forward $\pi_c^{(t-1)}$ if WAN severed past grace window.	Prevents a single disconnected hamlet from halting national planning while preserving verifiable inclusion.
4. IFB Price Update & Final Plan (S4, S6)	Global Total Order / Linearizable Epoch BFT[4]	2–15 s per block	CP (Safety over Liveness): Requires $\geq 2f+1$ of $3f+1$ federation validators.	Every council must optimise round $t+1$ against the exact same indicative price vector $p^{(t+1)}$.
5. Intra-Year Fulfilment (S7)	Bilateral Signed Receipts + Federated Rollup	< 500 ms (offline-capable)	AP with Bounded Credit: Daily depot credit cap reconciled at Ward rollup.	Point-of-delivery pickups at neighbourhood depots must never block on global consensus.

Figure 5.4 places the six node roles of Table 5.3 in the federation structure.

[3] Marc Shapiro, Nuno Preguiça, Carlos Baquero, and Marek Zawirski, "Conflict-Free Replicated Data Types," Symposium on Self-Stabilizing Systems (SSS 2011), LNCS 6976, pp. 386–400.

[4] Ethan Buchman, Jae Kwon, and Zarko Milosevic, "The Latest Gossip on BFT Consensus," arXiv:1807.04938 (2018); Maofan Yin et al., "HotStuff: BFT Consensus with Linearity and Responsiveness," ACM PODC '19, pp. 347–356.

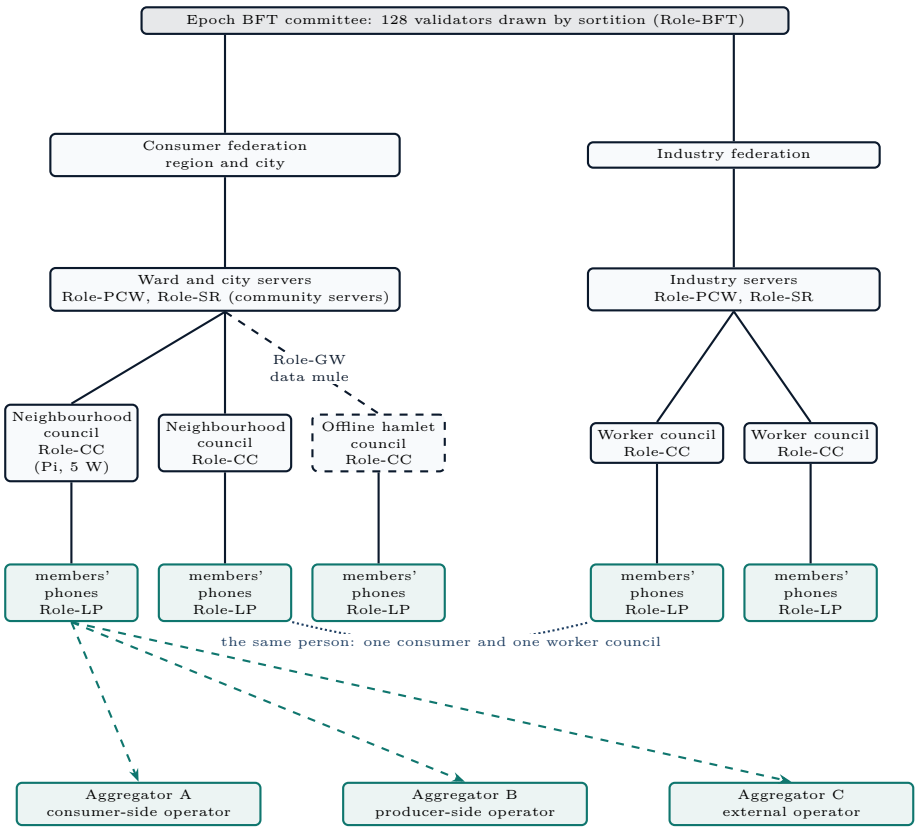


Figure 5.4: Federation topology and node placement (illustrative). Councils and servers nest by geography and industry; the three Prio3 aggregators sit outside that nesting, and the BFT committee is drawn by sortition across federations.

Table 5.3: Six-Role Hardware Realism Matrix (Supporting Legacy Smartphones and Raspberry Pis)

Node Role	Core Responsibilities	Entry Smartphone (3GB RAM ARM)?	Raspberry Pi 4/5 (4–8GB RAM)?	Community Server (8+ Cores)?
1. Lightweight Participant (Role-LP)	Local household budget QP solver, Prio3 share generation, CRDT sync, Two-Root & BFT header verifier, ballot signer.	YES (Primary) (RAM unverified)	YES	YES
2. Council Coordinator (Role-CC)	Hosts WC_j or CC_h local Wi-Fi CRDT relay, runs workplace LP/QP solver, assembles the council k-of-n multi-signature.	YES (Failover Hotspot)	YES (Ideal $50 Appliance at 5W)	YES
3. Storage Replica (Role-SR)	Pins regional content-addressed BLAKE3 blobs for qualitative Lists and audit logs.	NO (Flash wear & cell data limits)	YES (with USB SSD)	YES (Primary)
4. Planning Compute Worker (Role-PCW)	Computes Two-Root reductions, deterministic WASM price updates, and input-output analytics.	NO	PARTIAL (Up to 5,000 councils)	YES (Primary IFB Node)
5. Epoch Consensus Validator (Role-BFT)	Participates in sortition-elected BFT committee finalising round checkpoints.	NO (Requires 24/7 fixed routing)	YES (For PoC / local federations)	YES (Primary)
6. Delay-Tolerant Gateway (Role-GW)	Bridges offline mesh/LoRa/USB sneakernet bundles to federation relays.	YES (As “Data Mule”)	YES (Mesh Box)	YES

CHAPTER 6

Weighted Architecture and Implementation Stack Comparison

> "Devising and choosing from among these and other possibilities is a practical issue in implementing any actual participatory economy. Assuming the procedures chosen do not violate principles essential to participatory planning, considerations include (1) the extent to which iteration workers could bias outcomes, (2) the extent of reductions in the number of iterations required to reach a plan…"
>
> — Michael Albert and Robin Hahnel, The Political Economy of Participatory Economics (1991), Ch. 4, p. 68[1]

6.1 Weighted Comparison of the Four Architectural Paradigms

We compare the four candidate architectures across all 14 required criteria: Architecture A (Conventional Federated SQL/REST Services), Architecture B (Pure P2P Gossip + CRDTs without finality), Architecture C (Full Per-Transaction BFT Blockchain), and Architecture D (Hybrid Local-First + Two-Root Aggregation + Selective Epoch-BFT Checkpoints). Scores are expert judgements of the current design, not measured performance. Rows 4, 5, 6, 9 and 12 are scored against the measured Prio3 costs, the reworked proposal sizes (Table 5.1) and the carry-forward constraint in Section 7.4.3. A leads D on five criteria (throughput, bandwidth, hardware, implementation simplicity, energy) and D equals or beats A on the other nine.

[1] Albert and Hahnel, Political Economy of Participatory Economics, Ch. 4, p. 68.

Table 6.1: Weighted 14-criterion comparison (1–5 scale): A federated SQL/REST, B pure P2P gossip, C per-transaction blockchain, D hybrid

Criterion	Wt.	A	B	C	D	Rationale for D (judgement, not measurement)
1. Decentralisation	9%	2.0	5.0	3.0	4.5	Local autonomy plus rotating BFT; A centralises database administration, C heavy validators.
2. Trust Assumptions	10%	2.0	3.0	4.5	4.0	One honest aggregator for privacy, all for correct totals (VDAF-22 §9); correctness rests on institutions, not cryptography.
3. Censorship Resistance	8%	2.0	4.5	4.0	4.5	Signed receipts let a censored council prove omission.
4. Throughput & Latency	8%	5.0	4.0	2.0	4.0	10^6 report verifications cost each aggregator 4–61 desktop core-minutes; A has no such cost.
5. Network Bandwidth	7%	4.5	2.5	1.5	3.5	No $\mathcal{O}(N^2)$ gossip, but 290 kB quorum signatures and 0.24–1.46 GB round archives.
6. Hardware Requirements	8%	4.5	3.5	1.5	4.0	Raspberry Pi and Android are targets; no device benchmark yet.
7. Offline Capability	9%	1.5	5.0	1.0	5.0	SQLite + CRDTs + DTN store-and-forward.
8. Byzantine Fault Tolerance	8%	1.0	2.5	5.0	4.0	Detects leaf tampering given data availability; does not bind aggregator shares.
9. Privacy	9%	2.5	2.0	1.0	3.5	Coarse Prio3; no report carry-forward; credentials unspecified; differencing remains.
10. Post-Quantum Readiness	6%	4.0	3.5	2.0	4.0	FIPS 203–205 hybrids; BLAKE3 128-bit; not a uniform Level-3 claim.
11. Implementation Simplicity	6%	5.0	3.0	2.5	3.5	Four layers can be rolled out in phases, but far more code than A.
12. Operational Cost	4%	4.0	4.5	2.0	4.0	$\approx$ 298 MB/yr checkpoints, 2.4–14.6 GB/yr raw archive; audits dominate (note).
13. Energy Efficiency	4%	4.5	4.0	2.5	4.0	No proof-of-work; no sourced energy ratio claimed.
14. Maintainability	4%	4.0	3.5	2.0	4.0	Economic rules as upgradable, ratified WASM modules.
Weighted score	100%	3.10	3.59	2.54	4.07	D leads under these judgements (Section 6.1.1).

Note on row 12 (research/analysis/parameters.py, verify_r3.py; SI units, 1 kB = 1,000 B): at $N = 10^6$, $G = 10{,}000$, $n = 128$, a Layer-1 block is 5 FixedDecimal arrays (400,000 B), $\lceil 2n/3 \rceil = 86$ signatures (290,078 B), a 10,000 B participation bitmap and two 32 B roots: 700,142 B $\approx$ 700 kB. A 10-round plan has $10 + 365 + 50 = 425$ blocks/year, $\approx$ 298 MB/year; at 20 rounds (435 blocks), $\approx$ 305 MB/year. Layer-2 archive sizes come from Table 5.1; index, receipt, qualitative-List and replica overhead is unspecified. Hosting 100 federation servers at \$30–\$60/month is \$3,000–\$6,000/month and excludes audits, facilitation labour and research. All figures are unbenchmarked.

6.1.1 Sensitivity Analysis of Architecture Rankings

All figures below are computed from the cells of Table 6.1 by a script that parses the table from the book source.[2]

- Baseline weights: D 4.07, B 3.59, A 3.10, C 2.54. Equal weights ($1/14$ each): D 4.04, B 3.61, A 3.32, C 2.46.
- Weight on simplicity, throughput and cost. Put a share w of the weight on criteria 4, 11 and 12 equally and spread the rest in proportion to the baseline. A overtakes D at $w \approx 0.63$; at $w = 0.70$ the scores are A 4.08, D 3.92, B 3.75, C 2.30. The ranking of A against D therefore depends on how much a community values operational simplicity, and that is a legitimate reason to use A for Phase 1H and the earliest pilots.
- Harsher D cells (trust 3.5, bandwidth 2.5, hardware 3.0, privacy 3.0): D 3.82, still ahead of B 3.59 and A 3.10 at baseline weights.
- Score uncertainty. Perturbing every cell of every architecture independently by $U(-1, 1)$ points (clipped to $[1, 5]$; 20,000 draws) leaves D first in 99.2% of draws and B first in 0.8%; D's median lead over the runner-up is 0.50 points and its 5th-percentile lead 0.15.

D's lead at baseline weights is robust to plausible errors in individual scores, but not to a weighting that puts simplicity first.

6.2 Two-Stage Language Strategy: Python-First Research to Rust Protocol

Rather than locking the project prematurely into systems-level Rust before the non-convex economic coordination algorithms are validated, we adopt a two-stage implementation strategy:

1. Phase 1 (Economic Coordination Research) — Python Scientific Stack (NumPy/SciPy, highspy MILP/LP, Clarabel Python bindings): Minimises iteration time and maximises reproducibility for testing candidate ADMM, MILP federation decomposition, and Scarf/Leontief stress tests.
2. Phase 2 & 3 (Validated Distributed Protocol & Mobile Engine) — Rust + WebAssembly (dpep-core, dpep-node): Translates the Phase-1-validated clearing rules into memory-safe Rust (Clarabel.rs,[3] FIPS 203/204/205 implementations with independent assurance (libcrux, formally verified with hax and F*; aws-lc-rs, FIPS 140-3 validated; the unaudited RustCrypto crates only as a

[2] The script is research/analysis/verify_r3.py in the project repository.

[3] Paul J. Goulart and Yuwen Chen, "Clarabel: An Interior-Point Solver for Conic Programs with Quadratic Objectives," Mathematical Programming Computation (2026), https://doi.org/10.1007/s12532-026-00320-7; software at https://github.com/oxfordcontrol/Clarabel.rs.

WASM fallback),[4] quinn QUIC with rustls and its hybrid X25519MLKEM768 key exchange, Wasmtime, and Mozilla UniFFI). Local council solvers use IEEE-754 floating-point heuristics to draft proposals. Protocol-level checks ($SB_j/SC_j \geq 1$, $SC_h \leq B_h$, IFB price updates) are specified in deterministic 64/128-bit fixed-point arithmetic (Section 4.5). Those checks verify constraint satisfaction after rounding, not a truthful objective or MILP optimality.

Table 6.2: Programming Language Comparison Across Core Protocol Requirements (1–5 Scale)

Technical Dimension	Rust	Go	TypeScript	Kotlin	Swift	WASM (from Rust)
Memory Safety & Zero GC Pauses	5.0	3.5	2.5	3.0	4.0	5.0 (Sandboxed linear memory)
NIST PQC (libcrux, aws-lc-rs, RustCrypto)	5.0	4.0	2.0	2.5	2.5	5.0 (Direct compilation)
Local-First & P2P (iroh, CRDT, SQLite)	5.0	4.5	4.0	3.0	3.0	4.5 (Browser worker target)
Operations Research (Clarabel.rs, HiGHS)	4.5	2.5	2.0	2.5	2.5	5.0 (Runs on every phone/PC)
Cross-Platform Mobile Reuse (UniFFI)	5.0	3.0	4.5	3.0	1.5	5.0 (Universal execution)
Accessible UI & Screen-Reader Tooling	2.5	1.5	5.0	4.5	4.5	Paired with TypeScript PWA UI

6.3 Component Maturity

A component can be mature at one level and not at another, so Table 6.3 rates three levels separately: the algorithm or protocol, the library this book names, and the DPEP combination (the way DPEP uses it). The scale runs from M5 (ubiquitous: final standard, several independent implementations, billions of users, ten or more years of public analysis) through M4 (production-proven by several major operators, with audits or formal verification), M3 (early production: a new standard, an Internet-Draft, or few operators), M2 (pre-production: unaudited, pre-1.0 or beta), M1 (research) to M0 (unspecified). Ratings are judgements as of November 2026, not measurements. Evidence, production users, the researchers behind each component and sources are in the project's maturity dossier and register.[5]

[4] The RustCrypto ml-kem documentation states that the implementation "has never been independently audited" (https://docs.rs/ml-kem); libcrux-ml-kem is verified with hax and F* (https://docs.rs/libcrux-ml-kem) and its generated C is vendored in Mozilla NSS; AWS-LC FIPS 3.0 is a FIPS 140-3 validated module including ML-KEM (https://aws.amazon.com/blogs/security/aws-lc-fips-3-0-first-cryptographic-library-to-include-ml-kem-in-fips-140-3-validation). Checked November 2026.

[5] reports/STACK_MATURITY.md and research/stack_maturity_register.csv in the project repository.

Table 6.3: Component maturity at three levels (M0–M5; judgement as of November 2026)

Component	Algorithm	Library	DPEP use	Main gap
X25519 / Ed25519	M5	M4–M5	M4	None material.
Hybrid X25519 + ML-KEM-768	M4	M4	M2	Default in major browsers, OpenSSH 10 and Apple OS 26 TLS; the gap is DPEP's own use.
Hybrid Ed25519 + ML-DSA-65	M3	M2–M3	M2	Composite format is a draft; 3.4 kB per signature.
SLH-DSA-SHA2-192s	M3	M2–M3	M3	Little production use; acceptable for rare root signing.
BLAKE3	M3	M4	M3	Not standardised; SHA3-256 registered as alternative.
Prio3 VDAF / DAP	M3	M3	M1	Budget circuit unspecified; library tracks draft 18.
Post-quantum HPKE for shares	M2	M2	M1	No standard suite yet.
Two-Root aggregation	M5	M4	M1	Merkle logs mature (Certificate Transparency); DPEP protocol original.
One-report-per-person credentials	M4 / M1[a]	—	M1	Interim classical (Section 7.2.1); B_{iu} binding open.
Sortition beacon	M4 / M1[a]	—	M1	Interim threshold beacon; key generation and custody unspecified.
Capability certificates (CapCert)	M2	M2	M0	Format not chosen (UCAN 1.0 release candidate, Biscuit).
QUIC: quinn + rustls	M5	M4	M3	Hybrid key exchange must be enabled explicitly.
iroh	M3	M3	M2	1.0 released June 2026; relays must be self-hosted.
SQLite / SQLCipher; CRDTs	M5 / M4	M3–M5	M3	CRDT schema evolution and metadata growth.
CometBFT	M4–M5	M4 / M2[b]	M2	Post-quantum keys are pure ML-DSA, not hybrid.
Fixed-point WASM on Wasmtime	M4	M4	M2	Pin a long-term-support release; fix NaN and fuel settings.
HiGHS / Clarabel	M4	M4 / M2[c]	M2	No phone benchmark.
Mozilla UniFFI	M3–M4	M3–M4	M2	Interface still changing.

[a] Classical / post-quantum. [b] Ed25519 keys / ML-DSA-65 keys (v0.40). [c] Native / WASM on a phone.

Nine DPEP combinations have no deployment precedent, so they inherit no maturity: the Prio3 budget circuit, Two-Root aggregation, the hybrid signature format, CapCerts, credentials, the beacon, the light-client policy (Table 7.8), non-nested aggregator governance, and ratified WASM economic rules. Two-Root aggregation, WASM rules and server-side signature verification are tested at Kill Gate 2. The budget circuit, hybrid format, CapCerts, credentials, light-client policy and aggregator governance are tested at Kill Gate 3, whose external audit covers them. The beacon has an interim construction and stays an open question. Post-quantum deployment is moving quickly, so the hybrid-signature, Prio3, share-encryption, credential, beacon, iroh and CometBFT rows are re-checked before each gate.

CHAPTER 7

Post-Quantum Cryptography, Democratic Membership, and Privacy

> "Decisions about what individuals wish to consume will be subject to collective criticism by fellow council members, though with specific guarantees for preserving individual freedoms and privacy."
>
> — Michael Albert and Robin Hahnel, Looking Forward (1991), Ch. 3, "Consumption Norms"[1]

7.1 What the Source Tradition Requires of Consumption Privacy

The epigraph is the 1991 design: consumption is subject to collective criticism, with unspecified "guarantees" for freedom and privacy. PEPE Ch. 3 (p. 41) and Ch. 5 (p. 80) add a non-veto rule for in-budget requests. Albert (2003, Chs. 12, 18) argues for anonymous submission against busybody intrusion. None of those texts specifies cryptographic aggregation. DPEP's Prio3 invisibility is a technical extension: neighbours cannot see or comment on the anonymous bundle, so the 1991 criticism channel is lost. That is a genuine normative cost, not a proven improvement. REQ-DATA-04 therefore splits: non-veto is [ORIGINAL-1991]; invisibility is [TECH-EXT].

7.2 Post-Quantum Cryptography (NIST FIPS 203, 204, 205)

To protect multi-decade household records against harvest-now-decrypt-later adversaries, DPEP uses cryptographic agility (CryptoSuiteId) and hybrid classical + NIST PQC primitives. This is not a uniform NIST Level-3 claim, and it is not a "zero lattice" claim: ML-KEM and ML-DSA are lattice schemes. BLAKE3's specification targets 128-bit security for all goals. VDAF-22 makes no post-quantum claim. Table 7.2 states the status of each component.[2]

[1] Albert and Hahnel, Looking Forward, Ch. 3 (lf3_2.htm); Michael Albert, Parecon: Life After Capitalism (Verso, 2003), Ch. 12 ("Consuming") and Ch. 18 ("Privacy / Frenzy: A Busybody Economy?"), where Albert specifies that individual consumption proposals within warranted effort-rated budgets are submitted anonymously to prevent "busybody" intrusion; see also Jan Philipp Dapprich, "Democratic Theory and Economic Planning: A Critique of Participatory Economics," Erasmus Journal for Philosophy and Economics 17, no. 2 (Winter 2024), https://ejpe.org/journal/article/view/867.

[2] NIST, FIPS 203 (ML-KEM), FIPS 204 (ML-DSA), and FIPS 205 (SLH-DSA) (13 August 2024). Byte sizes in Table 7.1 are specification constants. The latency columns are placeholders carried from an unsourced estimate; in particular $\sim 80\ \mu$s for a hybrid X25519+ML-KEM-768 operation looks optimistic for an entry-level phone. They must be replaced by device benchmarks (Kill Gate 3) before use.

Two formats are still in standardisation. Hybrid signatures follow the composite ML-DSA Internet-Draft of Ounsworth et al. (2026), which defines an Ed25519 combination but is not yet an RFC; until it is, DPEP's hybrid wire format is DPEP-defined under CryptoSuiteId. Plain ML-DSA in X.509 certificates is standardised (Massimo et al. 2025). Share encryption: DAP encrypts report shares with HPKE, whose standard suites are classical, so the hybrid share encryption in Table 7.1 needs a post-quantum HPKE suite (ML-KEM or the X-Wing combiner), which is still in IETF drafting.

Table 7.1: Post-quantum primitives: specification byte sizes and placeholder ARMv8 latencies (unbenchmarked)

Primitive & Standard	Security	Pub Key	Cipher/Sig	ARM Sign	ARM Verify	Assigned Role in DPEP
X25519 + ML-KEM-768 (FIPS 203 Hybrid KEM)	NIST L3 + Classical	1,216 B	1,120 B	$\sim 80\ \mu s^{P}$	$\sim 85\ \mu s^{P}$	Point-to-point QUIC channels & Prio3 aggregator share encryption.
Ed25519 + ML-DSA-65 (FIPS 204 Hybrid Sig)	NIST L3 + Classical	1,984 B	3,373 B	$\sim 230\ \mu s^{P}$	$\sim 175\ \mu s^{P}$	Workhorse Signature for council proposals, CapCerts & receipts.
SLH-DSA-SHA2-192s (FIPS 205 Hash-Based)	NIST L3 (Hash Only)	48 B	16,224 B	unverified	unverified	Level-3 Root Anchor for Federation accreditation & WASM rule signing. Signing is slow by design; device timings not benchmarked.
BLAKE3 (BLAKE3 spec) / SHA3-256 (FIPS 202)	BLAKE3: 128-bit (all goals). SHA3-256: 128-bit collision.	N/A	32 B	$< 1\ \mu s/kB^{P}$	$< 1\ \mu s/kB^{P}$	Two-Root Merkle trees & content CIDs. BLAKE3 targets 128-bit security for preimage as well as collision resistance; quantum (Grover/BHT) reductions are not claimed.

[P] Placeholder, not measured. Council signatures are k-of-n multi-signatures of the hybrid scheme (3,437 B per signer); no post-quantum threshold signature is assumed.

7.2.1 Which Properties Need Post-Quantum Protection Now, and Crypto Agility

Post-quantum protection is urgent for one property and not for the others. Confidentiality must be post-quantum now: ciphertext recorded today can be decrypted

Table 7.2: Component-by-component post-quantum status

Component	Primitive	Claim type
Share encryption	Hybrid X25519 + ML-KEM-768	Standards-based (FIPS 203 + classical). Lattice. NIST L3 for ML-KEM-768.
Council / validator signatures	Hybrid Ed25519 + ML-DSA-65	Standards-based (FIPS 204 + classical). Lattice.
Root / WASM signatures	SLH-DSA-SHA2-192s	Standards-based (FIPS 205). Hash-based NIST L3.
Transcript hashes / Merkle	BLAKE3 (SHA3-256 registered as alternative)	Spec: 128-bit for all goals. Not NIST Level 3. Collision classically ≈ 128 bits. Not standardised by NIST or the IETF, so SHA3-256 (FIPS 202) is registered under CryptoSuiteId for checkpoint roots.
FLP soundness / joint randomness	Finite-field FLP + XofTurboShake128	Inferred from VDAF-22; no PQ theorem. Joint randomness is a known attack surface if the XOF is broken.
B_u credentials / sortition beacon	Interim classical (Section 7.2.1)	Not post-quantum. Accepted as interim because forgery and bias need a live CRQC; post-quantum replacement is open research.
Prio3 as a whole	Mix of the above	VDAF-22 Informational draft; no PQ claim.

once a CRQC exists, so household reports, share encryption and transport use hybrid ML-KEM from the first deployment and are never downgraded. Authenticity, Sybil resistance and randomness need post-quantum protection before a CRQC exists, not before deployment: a CRQC can forge future signatures and bias future draws, but it cannot alter a checkpoint whose history is already anchored under a hash-based signature, or redo a committee draw already made.

Design Principle 7.1 (Interim Classical Constructions Under Crypto Agility (adopted)). Where a post-quantum construction is standardised and practical, DPEP uses it in hybrid. Where none exists (one-report-per-person credentials, the sortition beacon) or the consensus engine does not yet support the hybrid (Layer-1 validator keys), DPEP uses the most mature available construction, on four conditions:

1. Suite inside the signature. Every signed or encrypted object carries its CryptoSuiteId inside the signed bytes, next to its domain-separation tag, so a suite cannot be stripped or swapped.
2. No downgrade below a ratified floor. The minimum suite for each object type is fixed in the checkpoint and changed only by the dual-federation ratification used for WASM rules (Section 7.6); objects below the floor are rejected, never negotiated.
3. Published migration triggers. A suite is replaced when the first of these occurs: the replacement is a final standard (composite ML-DSA RFC; FIPS 206; a standardised post-quantum credential or VRF); the software in use supports

it; or a credible public warning of CRQC progress. Each migration has a stated window with dual signing during the transition.

4. History survives the switch. Checkpoint history signed under an outgoing suite is re-anchored under SLH-DSA before that suite is withdrawn; per-epoch credentials and the beacon's threshold key are re-issued, not converted.

Table 7.3: Post-quantum urgency, current construction and upgrade path by function

Function	PQ needed	Construction now	Upgrade path and trigger
Transport, report-share encryption	Now	Hybrid X25519 + ML-KEM-768	Already post-quantum; add HQC as a backup KEM once standardised (final expected 2027).
Council signatures, receipts, CapCerts	Before CRQC	Hybrid Ed25519 + ML-DSA-65 (DPEP-defined format)	Composite ML-DSA RFC; FN-DSA once FIPS 206 is final (smaller signatures).
Layer-1 validator keys	Before CRQC	Strongest suite the engine supports at deployment	Hybrid once a composite standard and engine support exist. Sizes in this book assume the hybrid, the worst case.
Root anchors, WASM rules, re-anchoring	Before CRQC (long-lived)	SLH-DSA-SHA2-192s	None needed (hash-based).
One-report-per-person credentials	Before CRQC (forgery)	Classical anonymous credential issued jointly by k-of-n councils, with a per-epoch nullifier	Lattice credentials once they fit the phone budget of Kill Gate 3.
Sortition beacon	Before CRQC (bias)	Threshold beacon run by the federations (threshold BLS)	A post-quantum VRF or VDF beacon once one is standardised.
Binding B_u to the credential	—	Unspecified (Section 7.4.4)	Open research.

Figure 7.1 draws the same table as migration lanes.

Credentials. Threshold issuance, in which no single council can mint or link a credential, follows Coconut (Sonnino et al. 2019), a classical pairing-based construction. Lattice alternatives exist as research (Bootle et al. 2023), but reported sizes, under 80 kB per credential in one recent construction,[3] do not fit beside a 55 kB report in a 150 kB round. Selection criterion for the interim scheme: its unlinkability should not rest on an assumption a CRQC breaks (statistically zero-knowledge presentation proofs, statistical blinding). This must be verified for the chosen scheme; it is not a property of every classical credential. RFC 9474, for example, attributes the blindness of RSA blind signatures to a separate proof without stating its assumptions, and states that the protocol is not post-quantum

[3] Sven Argo et al., "Practical Post-Quantum Signatures for Privacy," ACM CCS 2024, https://doi.org/10.1145/3658644.3670297 (IACR ePrint 2024/131).

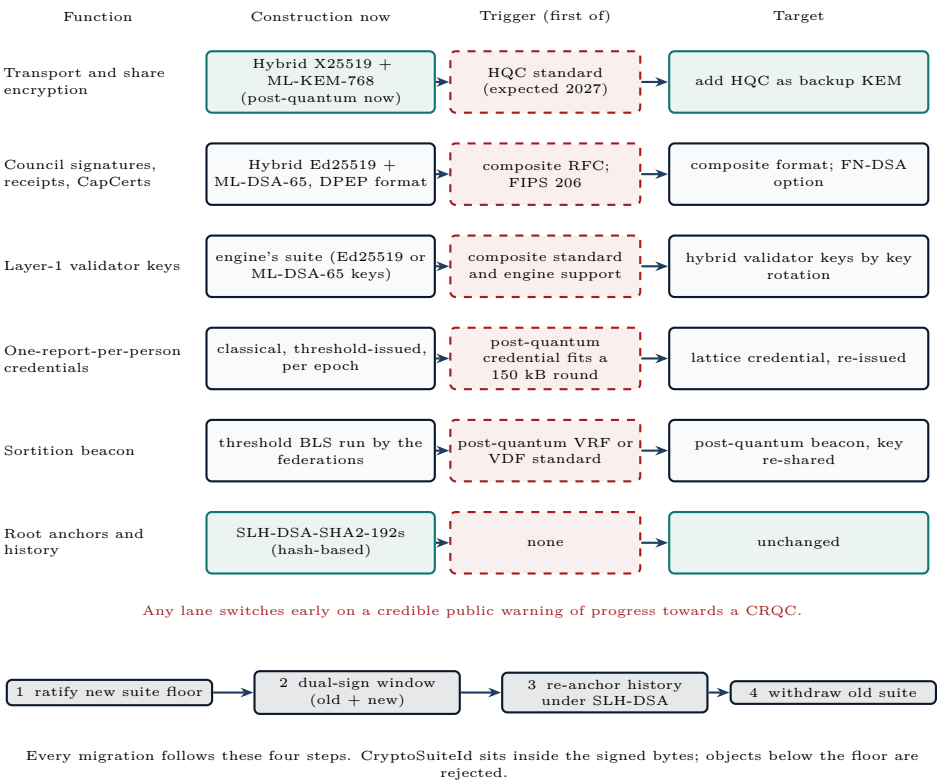


Figure 7.1: Crypto-agility lanes. Each function moves from its current construction to its target when the first trigger occurs, through the same four-step migration.

ready. If no scheme meets the criterion, a future CRQC could link past credential showings to identities. That reveals who submitted, not what was submitted, since report contents stay secret-shared under ML-KEM.

Beacon. The interim beacon uses threshold-BLS randomness of the kind produced by the open-source drand software,[4] with the threshold key shared among the federations rather than delegated to an outside network. It has no last-revealer problem, and a CRQC could bias only draws made after it exists. Its distributed key generation and share custody are not yet specified.

Layer 1. CometBFT remains the reference engine (provisional). Its v0.40 release accepts ML-DSA-65 consensus keys through key rotation, as a pure rather than hybrid scheme, and the Cosmos documentation estimates about 331 kB of signatures per block for 100 validators, consistent with the 290–296 kB for 86 hybrid

[4] https://drand.love (checked November 2026).

signatures in Table 7.8.[5] The validator suite therefore moves to ML-DSA by key rotation and to the hybrid when a composite standard and engine support exist, without a change of engine. A Rust engine with a pluggable signing interface (Malachite) is the fallback if a hybrid consensus signature becomes mandatory earlier; it is alpha software without an external audit.

7.3 Democratic Membership, Sybil Resistance, and Civil Liberties

Design Principle 7.2 (Four-Way Separation of Identity, Eligibility, Accountability, and Legitimacy). DPEP strictly separates: (1) Cryptographic Identity (control of a keypair), (2) Institutional Eligibility (accredited membership in a council), (3) Economic Accountability (correspondence of reported claims to physical reality), and (4) Democratic Legitimacy (self-management proportional to degree affected).

Figure 7.2 shows how the separation is carried into membership, delegation and private reporting.

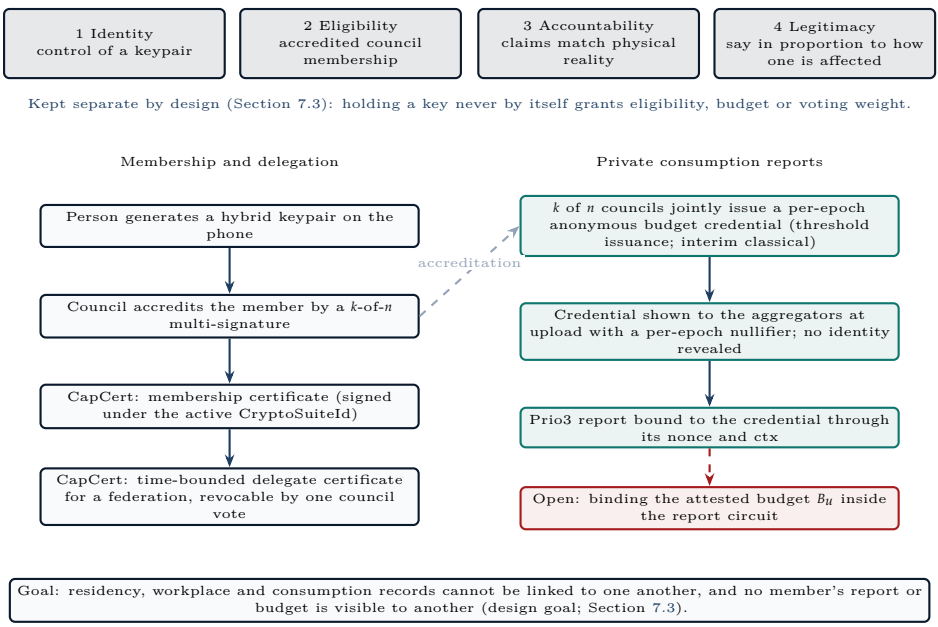


Figure 7.2: Identity, eligibility and the two credential paths. Membership and delegation are attributed and revocable; private consumption reports use a separate, unlinkable credential whose budget binding is still open.

While phantom digital accounts in CC_h cannot mint adult work income without

[5] Cosmos SDK documentation, "Post-quantum keys" (requires Cosmos SDK 0.55 and CometBFT 0.40), https://docs.cosmos.network/sdk/latest/keys/post-quantum-keys; release dates not confirmed (checked November 2026).

WC_j effort certificates capped by SB_j / SC_j, four deeper Sybil and collusion vectors must be closed explicitly:

1. Closing Workplace Hours Inflation (INV-4B): Capping only the mean effort rating $\bar{\epsilon}_j$ fails if a WC_j fabricates phantom workers to inflate total hours $\sum h_u$. Equation (4.7) closes this by capping total credited effort-hours $\sum_{u \in WC_j} h_u \epsilon_u \leq \sum_\ell (l^0_{j,\ell} + \Delta l_{j,\ell}) \min(\bar{\epsilon}_{\max}, SB_j / SC_j)$ (accredited hours) and flooring labour user-right charges at $p^{\text{lab}}_\ell \geq w_{\text{base}}$.

2. Detecting Collusive "Carousel Fraud" Between Fake Producer and Consumer Councils: If a colluding WC_j and CC_h sign fictitious delivery receipts for non-existent services to mint effort credits, cryptography alone cannot detect the lie. DPEP pairs automated input-output ratio anomaly flags (detecting closed-loop service claims without corresponding physical/intermediate inputs) with mandatory Industry Federation physical euthynai spot-audits prior to accrediting capacity.

3. Preventing Vote & Sortition Dilution: Sortition tickets and federation referenda are weighted by cross-audited council accreditation quotas rather than raw public keys.

4. Civil-Liberties Protection Against a Coercive Population Register (design goal, not yet constructed): To prevent housing (HFB) and workplace (EFB) registries from forming a surveillance database or trapping victims of domestic coercion, the design aims to let each person turn council attestations into unlinkable per-epoch budget credentials, separating residency from workplace and consumption. No such construction is specified (Section 7.4.4), so this protection does not yet exist. Budgets and reports are individual: the index u in Section 7.4 is one budget-holding person, never a household head. People who share a household and wish to plan jointly each submit their own report against their own budget; no member's report or budget is visible to, or controlled by, another. Unaffiliated migrants and transients receive unconditional Maxim-4 baseline provisioning through community clinics.

7.4 Private Household Aggregation via Prio3 VDAFs

A discarded design in which each household u generates a standalone zero-knowledge budget proof Π_u on a commitment C_u while separately submitting a pairwise-masked vector $\tilde{y}_u$ has three flaws: (1) nothing binds C_u to $\tilde{y}_u$; (2) if even one of many neighbours drops offline, pairwise masks fail to cancel; (3) aggressive FRI/STIR/WHIR parameterisations rely on Reed–Solomon proximity-gap conjectures that Crites and Stewart (IACR ePrint 2025/2046) disproved at up-to-capacity rates.[6] Those constructions are not part of Architecture D.

[6] Elizabeth Crites and Alistair Stewart, "On Reed–Solomon Proximity Gaps Conjectures," IACR Cryptology ePrint Archive 2025/2046 (December 2025), https://eprint.iacr.org/2025/2046.pdf.

Catalogues. Ω with $G = |\Omega| = 10{,}000$ is the full priced catalogue, and $\mathcal{G}^{\text{priv}}$ with $G_{\text{priv}} = 3{,}000$ the private-consumption categories at national scale. Household reports need not use $\mathcal{G}^{\text{priv}}$ directly. Section 7.4.2 proposes a coarser reporting catalogue $\mathcal{G}^{\text{rep}}$ of 100–250 categories, consistent with REQ-DATA-02.

VDAF construction and what it checks. Each budget holder u (one person; Section 7.3) encodes its request as a vector $q_u \in \{0, \dots, 2^b - 1\}^{|\mathcal{G}^{\text{rep}}|}$ in Household Reporting Units (Section 7.4.1) and splits it into $K = 3$ additive Field128 shares, encrypted with hybrid X25519MLKEM768 to three aggregators. Standard Prio3SumVec proves only that every coordinate lies in $[0, 2^b - 1]$ (VDAF-22 §7.4.3). The budget predicate

$$\sum_{g \in \mathcal{G}^{\text{rep}}} \pi_g^{(t)} q_{u,g} + \text{PubShare}_u \le B_u, \qquad \pi_g^{(t)} = p_g^{(t)} s_g \tag{7.1}$$

needs a custom validity circuit, which is not specified. Two facts constrain it. First, the prices $\pi^{(t)}$ must enter the circuit as public constants, so each price vector defines a different VDAF instance: a report proves the predicate for the round's prices and for no other. The application context string ctx should bind the report to $(\text{epoch}, t, \text{BLAKE3}(P^{(t)}), \text{aggregator set})$, but ctx enters only the domain separation of the XOF, not the circuit's arithmetic. Second, B_u must reach the circuit either as a public value (which identifies the person) or as a secret-shared input whose correctness is vouched for by a credential; the second option depends on the open credential problem (Section 7.4.4). Reference: IETF CFRG draft-irtf-cfrg-vdaf-22 (14 August 2026; Informational), full text.[7]

Privacy versus correctness (VDAF-22 §9). These must not be collapsed. Privacy of a measurement requires at least one honest aggregator. Correctness of the aggregate requires all aggregators: "An Aggregator can, without detection, manipulate the aggregate result by modifying its own aggregate share." Any one aggregator can inflate or deflate demand and so move prices.

Figure 7.3 shows the asymmetry.

How much the public budget check limits a cheating aggregator. The check $p \cdot \sum_u y_u \le \sum_u B_u$ is cheap and should be run, but it constrains only one number. A malicious aggregator can (i) shift demand between categories while keeping the value total fixed, which is undetectable and moves relative prices; (ii) inflate the

[7] Henry Corrigan-Gibbs and Dan Boneh, "Prio: Private, Robust, and Scalable Computation of Aggregate Statistics," USENIX NSDI '17, pp. 259–282; Dan Boneh et al., "Zero-Knowledge Proofs on Secret-Shared Data via Fully Linear PCPs," CRYPTO 2019; Richard Barnes et al., Verifiable Distributed Aggregation Functions (VDAFs), IETF CFRG draft-irtf-cfrg-vdaf-22 (14 August 2026), https://datatracker.ietf.org/doc/html/draft-irtf-cfrg-vdaf-22. Security analysis: Hannah Davis, Christopher Patton, Mike Rosulek, and Phillipp Schoppmann, "Verifiable Distributed Aggregation Functions," PoPETs 2023, no. 4, pp. 578–592; cite the full version (IACR ePrint 2023/130), whose 2024 revision corrects the proof of Theorem 1 with a weaker bound and a larger PRG seed.

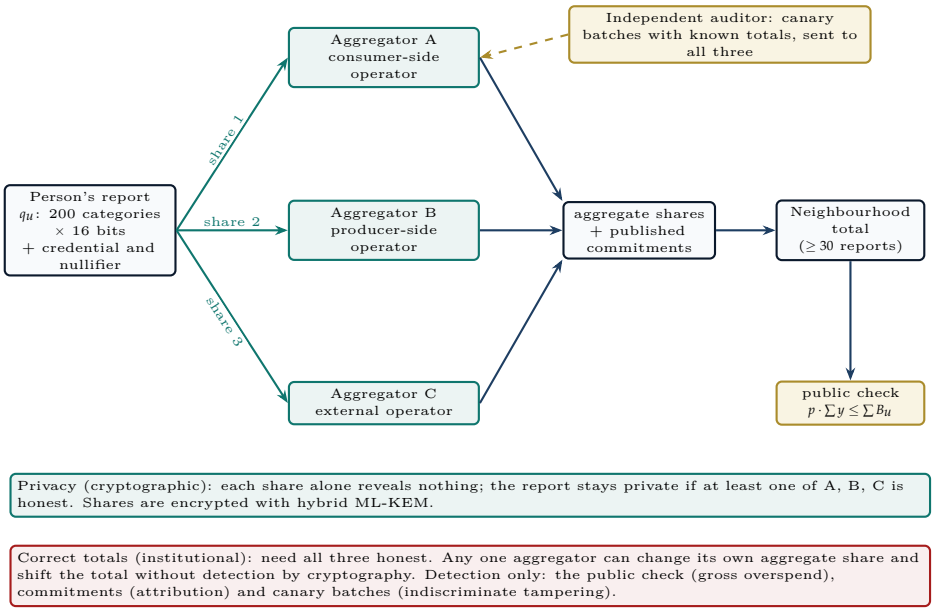


Figure 7.3: Household report flow and the privacy/correctness asymmetry. Privacy needs one honest aggregator; correct totals need all three, and the remaining safeguards detect rather than prevent manipulation.

total up to the slack between actual spending and $\sum_u B_u$ (if households leave 10% of budgets unspent, about 10% of value can be added undetected); and (iii) deflate demand without limit, since the inequality has no lower bound. It is a tripwire for gross overspend, not a correctness guarantee.

Recommended correctness model for Phase 3 (proposed, for author ratification). Use three non-nested aggregators run by organisations with different interests (for a co-op pilot: the consumer side, the producer side, and an external operator), plus the public budget check and published commitments to each aggregator's aggregate share, so that a dispute can at least be attributed. Nested CC/Ward/Federation aggregators should not be the default. Add canary batches run by an auditor independent of all three aggregators. Cryptography in this model guarantees privacy, not correct totals: correctness rests on institutional independence plus detection, and no mature construction removes that trust while keeping Prio3's privacy. Robust MPC is deferred until a pilot shows that aggregator manipulation is a real rather than a hypothetical threat. Trusted execution environments are rejected for this role, because they would make the chip vendor the party trusted for correctness. This choice cannot be made final before Phase 3 because it depends on which organisations exist in a pilot.

Table 7.4: Aggregator-correctness options (experimental except the first row)

Option	Guarantee	Cost / institutional demand
Status-quo Prio3 ($K = 3$), nested CC / Ward / Federation	Privacy if ≥ 1 honest; correct totals only if all honest.	Cheapest; but the three organs belong to one federation, so their independence is doubtful.
Non-nested aggregators ($K = 3$)	Same guarantees; independence more plausible.	E.g. consumer federation, an industry federation, and an external civil-society or academic operator.
Replicated aggregators ($K' > 3$)	Privacy if one honest; correctness still needs all shares.	More KEMs and operational load; no correctness gain without robust aggregation.
Robust / maliciously secure MPC	Detects or bounds a cheating aggregator.	Super-linear communication; not Prio3; unaudited for DPEP.
Public check $p \cdot \sum y \leq \sum B_u$	Detects gross aggregate overspend only (see text).	Cheap; complements every option.
Canary batches	Detects indiscriminate tampering with aggregate shares, with probability rising with the canary share.	An independent auditor submits whole batches with known totals. Needs per-batch aggregate shares and batches the aggregators cannot tell apart; does not catch tampering aimed at batches known to be real.
Committee-run aggregation	Ties aggregation to the $n = 128$ BFT set.	Concentrates privacy-breaking power in the body that finalises prices.
Differential privacy on released totals	Bounds one-household differencing; outside VDAF.	Bias against planning accuracy; parameters unspecified.

7.4.1 Report Sizes, Units, and Measured Costs

Household Reporting Unit (HRU). At the protocol's fixed 10^{-6} scale, a b-bit coordinate can express at most $(2^b - 1)/10^6$ units: 1.05 units at 20 bits, which is useless, and a full 64-bit field makes reports enormous. Household reports therefore use a per-category Household Reporting Unit: the catalogue fixes for each reporting category g an integer scale s_g (micro-units per HRU, for example 100,000 for 0.1 kg), and a report carries $q_{u,g} \in \{0, \ldots, 2^{16} - 1\}$. The aggregate in protocol units is

$$D_g^{\text{micro}} = s_g \sum_u q_{u,g},$$

which is exact integer arithmetic; no rounding occurs after the report is made. Rounding occurs once, when a person's intended quantity is rounded to whole HRUs (round-half-to-even): at most half an HRU per category per person, so at most $Ns_g/2$ micro-units in aggregate and, for independent errors, a standard deviation of $s_g\sqrt{N/12}$ (for $s_g = 0.1$ unit and $N = 10^6$, about 29 units against a total in the hundreds of thousands). Sixteen bits gives 0–65,535 HRU per category per report; categories whose range does not fit choose a larger s_g.

What the measurements change. A dense 3,000-category report is about 1 MB, thirteen times the 75 kB budget. A 50-coordinate "sparse" report cannot be used

Table 7.5: Prio3SumVec with 3 aggregators: measured sizes and desktop timings (libprio 0.18.1)

Length	Bits	Leader share	Client upload	Shard	Verify (leader)	Verify (helper)	Fits 75 kB?
100	20	35.1 kB	35.3 kB	0.55 ms	0.21 ms	0.23 ms	Yes
100	32	54.9 kB	55.1 kB	0.65 ms	0.21 ms	0.22 ms	Yes
100	64	107.7 kB	108.0 kB	1.13 ms	0.30 ms	0.32 ms	No
230	20	78.0 kB	78.3 kB	0.81 ms	0.23 ms	0.25 ms	No
230	32	123.6 kB	123.8 kB	1.28 ms	0.33 ms	0.35 ms	No
230	64	243.3 kB	243.6 kB	2.52 ms	0.82 ms	0.86 ms	No
3,000	20	975.8 kB	976.0 kB	10.6 ms	3.46 ms	3.64 ms	No
3,000	32	1,556.3 kB	1,556.5 kB	16.2 ms	4.85 ms	5.21 ms	No
3,000	64	3,100.4 kB	3,100.7 kB	33.3 ms	12.0 ms	12.7 ms	No
Computed with the same size formula (exact on all nine measured rows); not timed							
150	16	41.7 kB	41.9 kB				Yes
200	16	54.9 kB	55.1 kB				Yes
250	16	68.1 kB	68.3 kB				Yes
10,000	64	10,292.8 kB	10,293.0 kB				No

Medians of 30 runs, one core, Apple M4 Pro, macOS 26.6, rustc 1.98.1, –release, chunk_length chosen by libprio's optimal_chunk_length. "Verify" is verify_init (proof-share computation); combining verifier shares and verify_next take under 5 μs. Helper shares are 64 B seeds; the public share is 96 B. libprio 0.18.1 implements VDAF draft 18, not draft 22; the encoded sizes follow the same Prio3SumVec layout. These are desktop figures: phones and Raspberry Pi boards will be slower by an unmeasured factor. Code and commands: research/analysis/bench_prio3/.

to send 50 items chosen from 3,000: Prio3SumVec sums fixed coordinates, so a short vector either covers only 50 fixed categories or reveals which categories each coordinate stands for, the same leak that dense reports exist to prevent. No private-sparse construction is specified here; any such design would be experimental. Proof checking is not negligible either: verifying one dense report costs each aggregator, helper as well as leader, about 3.5 ms on a fast desktop core, so 10^6 reports cost about 61 core-minutes per aggregator per fresh-report round (3.8 minutes on 16 cores), before network transfer and on hardware much faster than a neighbourhood Raspberry Pi. A coarse 200-category, 16-bit report costs about 0.25 ms, roughly 4 core-minutes per aggregator at 10^6, and a neighbourhood of 200 people verifies in about 50 ms of desktop time. The illustrative "1 ms per report" used before measurement was optimistic for dense reports even on a desktop.

7.4.2 A Working Privacy Baseline (proposed)

Recommendation (proposed, for author ratification). Adopt option (ii) with $|\mathcal{G}^{\text{rep}}| = 200$ and 16-bit HRUs (55 kB upload) as the working baseline for household cryptography, and option (iii) for Phase 1H and the first Phase 3 cooperative pilots. Household-level cryptography is therefore not required for early pilots: with 10–30 consenting cooperatives a disclosed tallier, data minimisation and a data-protection impact assessment give adequate protection at a fraction of the audit cost, and the pilot can test whether members value bundle secrecy enough

Table 7.6: Privacy baselines compared at $N = 10^6$ (sizes per fresh report; CPU per aggregator per fresh-report round)

Option	Upload	Agg. CPU (desktop)	Hidden from councils	Costs and risks
(i) Dense Prio3, 3,000 categories, 20 bits	976 kB	≈ 61 core-min	Every category choice	Exceeds mobile budgets; 20 bits too narrow for many goods.
(ii) Coarse Prio3, 100–250 categories, 16-bit HRU	28–68 kB	≈ 4 core-min	Choice among coarse categories; not the product within a category	Loses detail; more reliance on disaggregation to products (P5) and on local intra-year fulfilment; Hayek's objection bites harder.
(iii) Council-level trust, no household cryptography	Plain request to council tallier	Negligible	Bundle hidden from neighbours by policy only; tallier sees it	Trusted tallier can leak or alter; fine for small consenting pilots with data-protection safeguards.
(iv) Pilot-selectable modes	(ii) or (iii)	–	As chosen	More code paths to audit; mode choice must itself be democratic and disclosed.

to justify option (ii). Option (i) is not a baseline at any scale considered here. The coarse baseline ties three open questions together: P5 (how well coarse totals disaggregate to products, now with 200 rather than 3,000 reported categories), P13 (the minimum batch $m_{\min} = 30$ now applies per coarse category, which makes small-hamlet anonymity easier), and Kill Gate 3's bandwidth and verification caps (Table 9.2). The resulting mobile budget per fresh-report round is about 55 kB up plus the light-client download in Section 7.6.

7.4.3 Prices, Carry-Forward, and Replay

What an offline household can prove. Eq. (7.1) uses the round's prices. A report made at round t proves the budget predicate at $P^{(t)}$ and nothing about $P^{(t+1)}$; scaling an old report proportionally does not create a new proof. VDAF-22 adds a harder limit: "Prio3 only permits a report to be aggregated once" (§7.2.3), because re-aggregation can reuse correlated randomness and compromise confidentiality (§9.4). Carrying an individual report forward into a later round's batch is therefore not available as specified.

Proposed rule (for author ratification). Combine (b) and (c). Households submit fresh, budget-checked reports at the three milestone rounds of the hybrid schedule ($t = 1, 4, 8$; P3). Between milestones, each neighbourhood's demand is held at its last milestone aggregate: aggregators reuse their aggregate shares as constants,

Table 7.7: Handling households between fresh reports

Option	What is verified	Cost
(a) Fresh report every round	Budget at every round's prices	10 uploads and verifications per year; excludes anyone offline in a round.
(b) Fresh reports at milestone rounds only; aggregate carried between	Budget at milestone prices; between milestones, only the aggregate check at current prices	Household demand frozen between milestones; councils adjust at council level.
(c) Bounded overspend	Aggregate $p^{(t)} \cdot \sum y \le (1+\kappa) \sum B_u$	Individual overspend between milestones is not detected.
(d) Exclude stale reports	Nothing stale	Biases totals against people with poor connectivity.

with no individual report re-verified or re-aggregated. The public aggregate check is applied at current prices with a tolerance κ (proposed 5%), and price-driven revisions between milestones happen at council level. A person offline at a milestone may submit within a grace window; otherwise they are counted at their council's per-capita average, flagged as imputed. Each released total at a milestone is a fresh aggregate of at least $m_{\min} = 30$ reports, which limits but does not remove differencing between milestones (minimum batches, coarser release and DP noise remain the mitigations). This raises the per-person upload to three fresh reports a year: about 0.17 MB at the coarse baseline, against 9.8 MB for ten dense reports.

Figure 7.4 places the milestone rounds in the planning season and operating year.

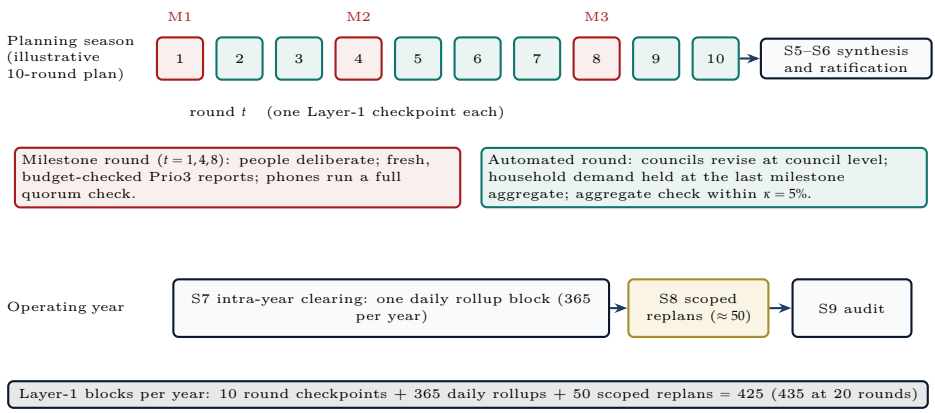


Figure 7.4: Annual calendar for an illustrative 10-round plan: three human milestone rounds with fresh reports, automated rounds in between, then the operating year and its Layer-1 block count.

Replay, re-runs and retention. Aggregators keep the nonce of every accepted report for the epoch and reject duplicates (VDAF-22 §9.2 recommends the nonce

for replay protection); ctx binds epoch, round, price-vector hash and aggregator set, so a report cannot be replayed into another round or instance. A round cannot be re-run from stored reports with a replacement aggregator: the stored shares are encrypted to the failed aggregator, and re-verifying them would aggregate them a second time. Recovery therefore means failing that milestone and requesting fresh reports, or carrying the previous milestone's aggregate. Aggregators retain encrypted input shares, nonces and aggregate-share commitments only until the challenge window $T_{\text{challenge}}$ closes (proposed 14 days), to settle inclusion disputes, and then delete them; that bounded retention is what "ephemeral shares" means in Section 11.2. People keep their own reports locally.

7.4.4 Budget Credentials and One-Report-Per-Person

The protocol needs each eligible person to submit at most one report per milestone, against their own warranted budget, without a register linking reports to identities. No construction is specified, and Table 7.1 lists no post-quantum blind or anonymous credential. VDAF-22 notes that clients may be anonymous and that Sybil and rate-limit enforcement is outside the VDAF. Three points narrow the problem. (1) BBS-style anonymous credentials are pairing-based and offer no post-quantum security; lattice-based anonymous credentials are an active research area without a standard, and none is named here. (2) Checking the credential inside the FLP is not realistic: the FLP validates low-degree arithmetic over the secret-shared measurement, while verifying a signature or hash chain needs a very large circuit, and the credential itself would have to be secret-shared. The credential should instead be shown to the aggregators at upload, outside the VDAF, with a per-epoch nullifier to block double submission, and bound to the report through its nonce and ctx. (3) What remains open is binding a secret-shared B_u inside Eq. (7.1) to the credential's attested budget. Until that is solved, one-report-per-person rests on council accreditation and rate limits, which is why option (iii) above is the early-pilot default. Section 7.2.1 adopts a classical credential with threshold issuance as the interim construction for the one-report-per-person part; the binding of B_u remains open.

7.5 Threat Model, Equivocation, Recovery, and Ballots

Figure 7.5 summarises what each part of the design rests on; the list below gives the detail.

Holds under stated assumptions. Hybrid KEMs/signatures as in FIPS 203/204/205; BLAKE3 collision/preimage as in the BLAKE3 spec (128-bit target); Two-Root inclusion if a verifier has the leaf archive; Prio3 measurement privacy if ≥ 1 aggregator is honest.

Detected, not prevented. Leaf omission given receipts and a challenge window; BFT equivocation given ≥ 1 honest validator recording both votes; aggregate

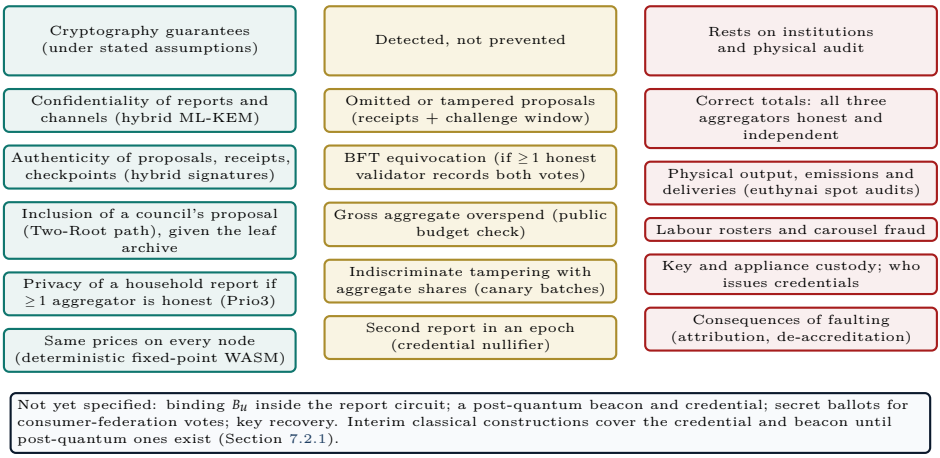


Figure 7.5: Trust map: what cryptography guarantees, what is only detected, and what rests on institutions and physical audit.

overspend if the public $p \cdot \sum y \leq \sum B_u$ check is implemented.

Depends on social institutions. Physical output, emissions, labour rosters, carousel fraud, aggregator independence, sortition appliance custody, and whether "faulting" an aggregator has consequences. There is no stake to slash; faulting means public attribution, dual-federation de-accreditation, and exclusion from the next sortition pool.

Aggregator compromise. Replace the aggregator and either request fresh reports for the affected milestone or carry the previous milestone's aggregate. Stored reports cannot be re-verified with a replacement aggregator (Section 7.4.3). Old committee keys are a long-range risk: checkpoints must be weakly subjective (users pin a recent checkpoint) or re-signed by the new committee.

Key compromise. Individual keys: revoke by council k-of-n decision and re-issue. No specified recovery UX or secure-element/mobile-storage profile; community-hosted appliances may make the host the real validator unless keys are in a citizen-held token. Until Android exposes post-quantum keys in hardware, keys use split hybrid custody: the classical half of each hybrid key stays in the platform keystore (Android Keystore or StrongBox, Apple Secure Enclave) and the ML-DSA half in software, encrypted under a hardware-held key. Until a CRQC exists a forger needs both halves, so the hybrid inherits the hardware's protection. Apple's Secure Enclave can hold ML-DSA keys directly from OS 26.[8] Forward-secure erasure of committee keys is an operational promise until a

[8] Apple, "Enhancing your app's privacy and security with quantum-secure workflows," CryptoKit documentation, https://developer.apple.com/documentation/cryptokit/enhancing-your-app-s-privacy-and-security-with-quantum-secure-workflows (checked November 2026). No equivalent Android hardware profile was found.

ratchet (e.g. per-round key derivation with deletion) is specified.

Replay and domain separation. Every signed object includes epoch_id, round, role, and a domain-separation tag of the form DPEP-WD-v0-<OBJECT> (Listing 4.1); the version component changes when the specification leaves working-draft status. Unspecified objects are not to be accepted.

Secret ballot. Multi-signature bitmaps reveal who voted yes. Binding council votes may stay attributed; consumer-federation ballots that must be secret need a separate coercion-resistant scheme (open).

7.6 Constitutional Governance and Sortition

To prevent software maintainers or server-rich councils from becoming a new coordinator class:

1. Dual-federation ratification of WASM rules, catalogues, and caps: Changes to wasm_rule_hash, product categories Ω, or 5-plan voting grids require a $\geq 2/3$ dual supermajority of the National Consumer Federation and Congress of Industry Federations with a 60-day reproducible-build audit window, and cannot be altered mid-plan.
2. Sortition beacon (unresolved). A beacon "anchored" on outgoing SLH-DSA-192s signatures is grindable if those signatures are randomised (FIPS 205 permits this). Commit-reveal has a last-revealer problem. A verifiable delay function could address grinding but is not specified here. A post-quantum beacon remains an open cryptographic problem, and there is no VRF in this draft. The interim construction is a threshold beacon run by the federations (Section 7.2.1); its key generation and custody are unspecified.
3. Custody: Sortition draws human delegates and provisions Role-BFT appliances. If keys live on the appliance, the host is the validator. Required: citizen-held tokens, or mark custody unresolved.
4. Committee size $n = 128$: $f = 42$, quorum $\lceil 2n/3 \rceil = 86$. Exact binomial tails[9] for a single annual draw, corrupt share p: at $p = 15\%$, $1.3 \times 10^{-5}\%$ ($n = 128$) vs. $9.5 \times 10^{-3}\%$ ($n = 64$); at $p = 20\%$, 0.022%; at $p = 25\%$, 1.82%; at $p = 30\%$, 21%. Larger committees give little protection at a 30% corrupt pool. An annual draw means a captured committee lasts a year; 1–4 rotations/year reduce but do not eliminate that window.

Light clients. At $n = 128$ a checkpoint carries 86 hybrid signatures, about 290–296 kB, which would dominate a phone's per-round traffic. Table 7.8 sets out what each lighter option trusts.
Proposed default (for author ratification): coordinator attestation every round, plus a full quorum check at each milestone round and whenever a member's client

[9] Computed by research/analysis/verify_r2.py in the project repository.

Table 7.8: Light-client verification options for Role-LP (per checkpoint)

Option	Download	What the phone trusts
Fetch and verify all 86 signatures	≈ 296 kB	Only the BFT assumption (≤ 42 faulty validators).
Council coordinator verifies and attests	≈ 3.4 kB	The council's own Role-CC; a corrupt coordinator can feed a false checkpoint until a member checks.
Random sample of 8 signatures	≈ 27 kB	Probabilistic: if only the 42 corrupt validators really signed a checkpoint claiming 86, all 8 samples pass with probability $(42/86)^8 \approx 0.3\%$. Not a proof, and it fails if the sample is predictable to the adversary.
Full check at milestone rounds only	296 kB ×3/yr	BFT at milestones; coordinator attestation in between.
Rotating sync committee: 24 validators drawn by the beacon, quorum 16	≈ 55 kB	That fewer than 16 of the 24 are faulty. With 42 faulty validators among 128, a draw yields a capturing committee with probability 0.017% (16 members, quorum 11: ≈ 38 kB, 0.19%). A captured committee lasts until the next rotation, and the draw is only as good as the beacon.

is on Wi-Fi, with every client able to opt into full checks. The sync-committee row, the pattern used by Ethereum light clients,[10] replaces trust in a member's own coordinator with trust in a rotating, randomly drawn committee, and is the preferred upgrade once the interim beacon (Section 7.2.1) is running. The resulting mobile target, written into Kill Gate 3 (Table 9.2), is ≤ 150 kB per ordinary round (55 kB report at milestones, price deltas, attestation, 512 B inclusion proof) and ≤ 450 kB in a round with a full quorum check.

[10] Ethereum consensus specifications, Altair light-client sync protocol, https://github.com/ethereum/consensus-specs/blob/dev/specs/altair/light-client/sync-protocol.md. Probabilities are exact hypergeometric tails from research/analysis/verify_sync_committee.py.

CHAPTER 8

Accountable AI, Mathematical Optimisation, and Failure Modes

> "Finally, we point out to those who fear that computers could become the new dictators, the computer programmes are human products, socially evaluated and improved each year… In short, neither consumers nor producers need accept computer facilitated alternatives."
>
> — Michael Albert and Robin Hahnel, Looking Forward (1991), Ch. 9, "Revising Data in Subsequent Iterations"[1]

8.1 Architectural Rule: Deterministic Core, Advisory Edge

AI must never become an unaccountable central planner or surrogate coordinator class. DPEP enforces a strict boundary:

- Core Clearing & Pricing Engine (non-AI by specification): All price updates, SB/SC ratios, and Society Account settlements are specified as deterministic fixed-point arithmetic. Local IEEE-754 solver output, including any KKT certificate, checks stationarity of the programme the council chose to solve. It does not prove a truthful objective, and MILPs have no KKT certificate of global optimality (Section 4.5).[2]
- Edge Advisory & Accessibility Support: Classical hierarchical time-series models (ETS/ARIMA) forecast SKU size/color distributions; local QP solvers draft initial proposals from last year's bundle (eliminating David Schweickart's "consumption list burden" objection);[3] and optional local open-weight LLMs are restricted to translation, voice/screen-reader accessibility, and summarising qualitative Lists with mandatory links to raw council text.

8.2 Substantive Engagement with Major Economic & Democratic Critics

This section engages the main theoretical critiques of Participatory Economics across philosophy, economics and mechanism design: the Winter 2024 special

[1] Albert and Hahnel, Looking Forward, Ch. 9 (lf9_2.htm).

[2] Stephen Boyd, Neal Parikh, Eric Chu, Borja Peleato, and Jonathan Eckstein, "Distributed Optimization and Statistical Learning via the Alternating Direction Method of Multipliers," Foundations and Trends in Machine Learning 3, no. 1 (2011): 1–122.

[3] David Schweickart, Against Capitalism (Cambridge: Cambridge University Press, 1993), pp. 329–334.

Table 8.1: Evaluation of key decision-support capabilities against verifiability and override criteria

Capability	Best Method (Is AI Needed?)	Local?	Verifiability & Challengeability	Override & Power Concentration Safeguards
1. SKU Demand Forecasting	No LLM needed: Hierarchical ETS/ARIMA + demographics.	Yes (< 50 ms)	Published MAPE error bounds & open parameters.	Non-binding baseline; councils edit freely.
2. Workplace Input Optimisation	No AI needed: Conic QP / LP (Clarabel.rs / HiGHS).	Yes (WASM time unverified)	KKT certificates, where they exist, witness stationarity of the local convex programme, not a truthful W_j or MILP optimum.	Presents 3 Pareto-frontier options for worker council vote.
3. Household Proposal Drafting	No AI needed: Deterministic diff + constrained utility QP.	Yes (on device)	Line-by-line diff against prior year's bundle.	Mandatory review UI highlights top changes; zero data leaves user's device.
4. Tradeoff Explanation	Deterministic Leontief Cost-Push Attribution ($(I-A)^{-1}$).	Yes (< 50 ms)	Exact algebraic decomposition of ΔP_i.	Avoids LLM hallucination by rendering exact resource/labour/pollution cost shares.
5. Qualitative List Translation	Yes: Local Neural Machine Translation + TTS.	Yes (Role-CC)	Original source text & BLAKE3 CID always linked alongside.	Bilingual delegates can post verified overrides; expands cross-regional inclusion.
6. Late-Round 5-Plan Synthesis	No AI needed: Parametric LP over 5 fixed weight vectors.	Yes (< 2 s)	Deterministic WASM LP verifiable by any council.	Councils can vote to reject all 5 plans and continue iterating (LF Ch. 9 lf9_2.htm).

issue of the Erasmus Journal for Philosophy and Economics (John O'Neill, Hannes Kuch, Jan Philipp Dapprich), read in the open-access issue PDF;[4] David Laibman (2014), consulted through the publisher's abstract, with Robin Hahnel's (2014)

[4] Erasmus Journal for Philosophy and Economics 17, no. 2 (Winter 2024), issue PDF at https://ejpe.org/journal/issue/download/35/67: John O'Neill, "Pluralism, Ecology and Planning," pp. 27–48, https://doi.org/10.23941/ejpe.v17i2.884; Hannes Kuch, "Centralization Tendencies in Participatory Planning," pp. 49–72, https://doi.org/10.23941/ejpe.v17i2.885; Jan Philipp Dapprich, "Democratic Theory and Economic Planning: A Critique of Participatory Economics," pp. 73–93, https://doi.org/10.23941/ejpe.v17i2.867; John O'Neill, "Incommensurability, Environment and Planning: A Response to Hahnel's Reply," pp. 117–124, https://doi.org/10.23941/ejpe.v17i2.933.

response;[5] Paul Cockshott and Allin Cottrell (1993);[6] János Kornai (1980, 1986); Martin Weitzman (1974, 1980); and David Schweickart (1993).

1. Dapprich (2024) on democratic theory, liberty, and general procedures: Dapprich asks whether three justifications of democracy (protecting interests, revealing preferences, and liberty in the neo-Roman republican sense) support citizen participation in concrete plans, and argues that they do not: citizens should instead choose general planning procedures, an "algorithmic planning process" in his words, which protects them from the arbitrary power of coworkers and neighbours as well as of central planners. He also argues that most participatory input in ParEcon actually comes from individual consumption requests, leaving consumer councils with little to vote on, and that without constitutional constraints a worker is exposed to the arbitrary will of the majority on their own council.

   - Where Dapprich is right: Neighbour veto of in-budget personal consumption is a liberty problem; DPEP's non-veto rule follows PEPE, and cryptographic invisibility goes further (Section 7.1). His point about workplace majorities is also well taken: DPEP needs entrenched individual rights inside councils (safety, exit to another council through the EFB, BJC protections), not only majority procedures.
   - Where this monograph disagrees: Workplace pace, safety and emissions, and city-scale public goods, are collective activities whose terms cannot be fixed in advance by a general procedure without someone writing the procedure's parameters, which re-creates a planning authority one level up. DPEP therefore keeps councils as decision-makers for collective activities and uses general procedures (WASM rules ratified by dual supermajority, Section 7.6) as constraints on them. This answers Dapprich only in part.

2. Kuch (2024) on centralisation tendencies: Kuch argues that a participatory economy needs more central institutions than Hahnel suggests, for strong normative reasons: industry federations act as gatekeepers, appellate bodies and adjustment agencies; national federations decide investment and society-wide public goods; disciplinary bodies may be needed to counter the procedure's false incentives; and conflicts between federations may require a national legislature or something like a ministry of planning. He doubts that council-democratic remedies resolve this. DPEP's responses (Algorithm 3c's federation MILP as an aid rather than an authority, sortition, the Innovation Jury

[5] David Laibman, "Horizontalism and Idealism in Socialist Imagination: An Appraisal of the Participatory Economy," Science & Society 78, no. 2 (2014): 207–234, https://doi.org/10.1521/siso.2014.78.2.207; Robin Hahnel, "Response to David Laibman's 'Appraisal of the Participatory Economy'," Science & Society 78, no. 3 (2014): 379–389, https://doi.org/10.1521/siso.2014.78.3.379. Only the abstract of Laibman's article and the bibliographic record of the response were consulted.

[6] W. Paul Cockshott and Allin Cottrell, Towards a New Socialism (Nottingham: Spokesman Books, 1993).

below, dual-federation ratification) are institutional proposals that Kuch's argument puts directly in question; they are tested in P7 and are not shown to work.

3. Cockshott & Cottrell (1993) on direct LP and scalar labour-time: Cockshott and Cottrell propose replacing iterative council negotiation with central LP in embodied labour hours. Two objections are raised here (from the bibliographic record and the secondary literature, not a full reading of Towards a New Socialism): (a) scalar labour-time omits the independent scarcity of non-producible natural resources (r^*), ecological sinks ($E_z^{\max}$) and fixed capital (k^*), and differential workplace disutility; (b) a central LP that needs each workplace's full private $\mathcal{T}_j$ collides with Hayek (1945) on tacit knowledge and with worker self-management. Hayek's point also applies to DPEP: coarse categories, an annual cycle and published technical recipes (REQ-DATA-03) still lose local know-how, and Algorithm 3c's federation MILP is a limited central solve over posted menus, not over hidden $\mathcal{T}_j$. Treating 3c as a Phase-1 oracle and a ratified aid, not an authority, is the intended boundary; it remains a tension, not a resolution.

4. Laibman (2014), Kornai (1980) and Weitzman (1974, 1980) on equilibrium, soft budgets and the ratchet effect:

   - The Walrasian resemblance and dynamic entry (Laibman 2014; Kuch 2024): Laibman's abstract argues that the Albert–Hahnel model is designed in an "abstractly speculative" way, that its fear of authority and hierarchy has roots in classical anarchism, and that the result bears an "uncanny resemblance to Walrasian competitive market equilibrium." Hahnel replied in the following issue. This monograph does not attribute to Laibman any argument beyond his abstract. One practical consequence of the Walrasian reading is that an iterative procedure over a fixed list of producers says nothing about how new producers enter. DPEP's proposal: an accredited worker group refused accreditation by an Industry Federation (Kuch's gatekeeping concern) can appeal to a Cross-Industry Innovation Jury, selected by sortition and co-ratified by the Consumer Federation, backed by a dedicated R&D capital slice. This is untested (P7).
   - Hardening Soft Budget Constraints (Kornai 1980, 1986): If councils with $SB_j/SC_j < 1$ could obtain unlimited requests_appeal waivers, budget discipline would collapse.[7] DPEP limits appeal waivers to one planning cycle (requiring a physical retooling audit) and funds them from a capped Industry Transition Pool, so granting a waiver directly reduces peer councils' discretionary industry reserve.

[7] János Kornai, Economics of Shortage (Amsterdam: North-Holland, 1980); Kornai, "The Soft Budget Constraint," Kyklos 39, no. 1 (1986): 3–30; Martin L. Weitzman, "Prices vs. Quantities," Review of Economic Studies 41, no. 4 (1974): 477–491; Weitzman, "The 'Ratchet Principle' and Performance Incentives," Bell Journal of Economics 11, no. 1 (1980): 302–308.

- Ratchet effect (Weitzman 1980) — open: Basing endowment charges on physical assets rather than last year's output removes one Soviet-style ratchet. It does not remove the fear that revealed capacity will feed next year's benchmarks, public blueprints (REQ-DATA-03), or SB/SC comparisons. Weitzman (1980) is cited at bibliographic level. Status: open.

5. Effort rating, surveillance, ecology and limits (O'Neill 2024): O'Neill makes three criticisms. First, distributing income by a metric of effort would involve pervasive surveillance and potential humiliation; this bears directly on the effort default discussed in Section 9.1.3. Second, a single price metric of opportunity cost and cost–benefit analysis cannot address value pluralism and incommensurability; he defends rational choice without commensurability and in-kind analysis in the tradition of Neurath and Kapp. Third, environmental limits must be taken more seriously than Hahnel does: substitution is necessary but not sufficient, and he proposes consumption and production "corridors" above sufficiency for all but below excess. DPEP's intended responses: (a) strict one-sided biophysical ceilings $(r^*, E_z^{\max})$, which are in-kind constraints rather than prices, set by long-run environmental planning, with an intergenerational trusteeship mandate that living cohorts cannot relax $E_z^{\max}$ without a supermajority plus a sortitioned youth/future-persons panel; (b) a per-capita exposure floor so that no neighbourhood's modelled dose exceeds a published health threshold even if the regional CAP accepts the aggregate E_z; (c) a plume sub-zone veto for the most-impacted $CC_h \subset CAP_z$; (d) earmarking $\geq 50\%$ of $p_z^{\text{pol}} E_z$ to remediation and preventative health. Rules (c) and (d) create hold-out and earmark-gaming problems (a sub-zone can veto to extract transfers; 50% earmarks invite recategorising ordinary spending), and DPEP has no counterpart yet to an upper consumption corridor. Status: working rules, not solved mechanism design.

6. Governance of solver defaults and deliberative fatigue (Devine 1988; Albert 2003, Ch. 18): Even without AI, local QP solver defaults could become de facto decisions if citizens rubber-stamp them. DPEP bans silent auto-submission of $> 5\%$ effort or category shifts, requires active choice among ≥ 3 options at milestone rounds, and measures rubber-stamping, comprehension and the distribution of influence in Phase 1H (Table 9.2).

7. Black-market arbitrage and wage labour (Albert 2003, Ch. 20): Prio3 is intended to keep personal bundles private and to reject reports that violate the encoded budget predicate if that circuit is specified and sound. An unauthorised firm still cannot draw intermediate inputs through the planning tree. Off-protocol barter remains a social question. No "mathematical prevention" of wage labour is claimed.

CHAPTER 9

Three-Phase Experimental Programme, Kill Gates, and Benchmarks

> "The easiest 'experiments' to implement are a variety of computer simulations in which various features of the system and assumptions about actors' behaviours can be altered to see how outcomes change… Conceivably one could run a 'shadow' participatory economy within an existing economy."
>
> — Michael Albert and Robin Hahnel, The Political Economy of Participatory Economics (1991), Ch. 6, "Feasibility," pp. 108, 129[1]

9.1 Why DPEP Uses Gated Phases Instead of a Single Monolithic Build

Attempting to build non-convex solvers, QUIC, aggregation, WASM, PQC, Prio3, and mobile UIs in one monolithic prototype would conflate mechanism validation with distributed-systems engineering. Each phase therefore ends in a kill gate: a test with a pre-registered pass condition, a fail condition, and a stated consequence. A gate that cannot fail is not a gate.

Table 9.1: Staged experimental programme (budgets as in Table 1.2)

Phase & Stack	Core experiments	Duration & staff
Phase 1: Economic coordination (Python, NumPy, HiGHS, Clarabel)	Pre-registered instance set (below); Algorithms 3a–3d against a central LP/MILP oracle; scaling from 50 to 1,000 councils.	4–6 months; 1.0 FTE OR/economics
Phase 1H: Low-tech human probe (disclosed trusted server + web UI)	30–60 volunteers in 3 worker and 3 consumer councils run three milestone rounds of a simplified plan. Ethics review and consent first.	3–4 months; 1.0 FTE UX/dev; in parallel with Phase 1
Phase 2: Distributed coordination (Rust dpep-node, CBOR, QUIC, Wasmtime)	WAN testbed at $\geq 10^4$ simulated participants; fault injection (censorship, equivocation, 20% partition); Prio3 verification and signature checks in the round.	4–6 months; 2.5 FTE
Phase 3: PQC, Prio3 & pilot (FIPS 203–205, Prio3, mobile PWA)	Device benchmarks; external audit; 10–30 cooperative field pilot, starting with the council-level trust model (Section 7.4.2).	9–12 months; 4.0 FTE + audit

[1] Albert and Hahnel, Political Economy of Participatory Economics, Ch. 6, pp. 108, 129.

9.1.1 Kill Gates as Tests

Table 9.2 states each gate. All numeric thresholds are proposed, for author ratification, and each carries its reason. Consequences use four outcomes: proceed; narrow scope (continue only for the problem class or setting that passed); redesign and rerun (once, on a freshly pre-registered set); stop (publish the negative result and fund no later phase on that basis).

Table 9.2: Kill gates: pass and fail conditions, consequences and reasons (thresholds proposed)

Gate	Pass (all required)	Fail (any)	Consequence and reason
1 Economic	For the best candidate: (a) $\geq 90\%$ of convex and Leontief instances cleared (Definition 4.2) within 25 rounds; (b) $\geq 70\%$ of staggered-lumpy instances cleared; (c) on Scarf and identical-lumpy instances, either cleared or correctly reported feasible, not cleared or infeasible with diagnostics in 100% of runs; (d) 100% of infeasible instances reported infeasible; (e) $\geq 70\%$ of endogenous-income instances cleared by 3d; (f) median welfare gap to the oracle $\leq 5\%$, 90th percentile $\leq 15\%$.	Fewer than 50% of convex and Leontief instances cleared; or any infeasible instance reported feasible; or median welfare gap $> 15\%$; or endogenous-income clearance $< 40\%$.	Proceed if all pass. Narrow scope if (a), (d) and (f) pass but (b) or (e) does not: Phase 2 then covers only the passing classes, with lumpy decisions left to federation MILP and ratification. Redesign and rerun once if between pass and fail. Stop if the fail condition holds after the rerun. Reasons: convex instances have known convergent methods, so failure there signals a design or implementation fault; 70% for lumpy instances reflects the one-good test, in which 57% of rounds clear; 25 rounds is about twice the 11.85-round pequod-cljs baseline; reporting an infeasible plan as feasible is never acceptable.
1H Human probe	(a) $\geq 80\%$ of participants complete all three milestones; (b) median score $\geq 70\%$ on a pre-registered comprehension quiz (prices, budgets, what an override does); (c) $\leq 25\%$ of solver drafts accepted without opening the change view; (d) of coded override reasons, $\geq 50\%$ are substantive (need, preference, local knowledge) rather than confusion; (e) the most active 20% of participants originate $\leq 60\%$ of adopted amendments.	Completion $< 60\%$; or median comprehension $< 50\%$; or $> 50\%$ of drafts accepted unopened; or the most active 20% originate $> 80\%$ of adopted amendments.	Proceed if all pass. Redesign and rerun (interface, facilitation, schedule) once if between. Stop human-facing claims and keep simulator research only if the fail condition holds after the rerun. Reasons: override rate alone rewards bad defaults, so it is paired with comprehension, reasons and the spread of influence; (e) is a direct test for an emerging technical or articulate elite. Limits: 30–60 self-selected volunteers cannot show generalisation to a population, long-run fatigue, or power dynamics at scale; with $n = 40$ a 95% interval on a proportion near 0.5 is about ± 15 points.

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Gate	Pass (all required)	Fail (any)	Consequence and reason
2 Distributed	(a) 100% of injected omissions and tamperings detected within $T_{\text{challenge}}$ over $\geq 1{,}000$ fault-injection trials; (b) automated round time (excluding human deliberation), including Prio3 verification and signature checks, p95 ≤ 10 min at the tested scale ($\geq 10^4$ participants), and ≤ 60 min when extrapolated to 10^6 from measured per-item costs with ≤ 64 cores per aggregator; (c) total bytes and CPU $\leq 10\times$ an Architecture A baseline on the same workload.	Any undetected tampering; or p95 > 30 min at the tested scale; or overhead $> 50\times$ the A baseline.	Proceed if all pass. Narrow scope (smaller committees, coarse reports only, regional rather than national deployment) or redesign and rerun if between. Stop the distributed build and pilot on Architecture A if the fail condition holds. Reasons: detection is the point of Two-Root aggregation, so one miss fails it; 10 min fits the 1–10 min target of Table 5.2; a tenfold overhead is a reasonable price for removing a trusted operator, fiftyfold is not.
3 Crypto, mobile, pilot	(a) External audit with no unresolved critical or high findings before field use; (b) on a 3 GB-RAM Android device: ≤ 150 kB per ordinary round and ≤ 450 kB in a full-verification round (Section 7.6), peak RAM ≤ 150 MB, coarse report shard ≤ 1 s; (c) pilot: $\geq 70\%$ of member councils still participating at the end, no unresolved privacy incident, and members' rated legitimacy of the plan not below their rating of the pre-pilot process.	Unresolved critical audit finding; or any device metric more than $2\times$ its cap; or pilot retention $< 40\%$.	Proceed to wider pilots if all pass. Narrow scope to the council-level trust model (no household cryptography) if (b) fails but (c) passes. Stop field use of custom circuits on an unresolved critical finding. Reasons: the bandwidth caps follow from the coarse 55 kB report plus the light-client options; 150 MB leaves room on entry-level phones; retention and legitimacy test whether people want to keep using the process, which no simulation can show.

Pre-registration. Before each gate, the fast-moving rows of the component maturity table (Table 6.3) are re-checked and the result is recorded with the gate's pre-registration. Before any candidate solver is run on them, the instance generators, random seeds, instance counts, metrics, thresholds and analysis scripts are committed to a pre-registration directory in the project repository,[2] tagged with a signed Git tag, and the tag's commit hash is deposited with an independent registry (for example the Open Science Framework). Results are reported against that tag. Deviations are logged with reasons in the same directory, and any re-run uses a freshly registered set. Phase 1H registers its quiz, coding scheme for override reasons, and influence measure in the same way, together with its ethics approval.

Pre-registered Phase 1 instance set. At least 20 instances in each family, at 50, 200 and 1,000 councils: (1) convex baselines (Cobb–Douglas and CES with fixed budgets); (2) Leontief input–output economies; (3) Scarf-style cycles; (4) identical lumpy plants; (5) staggered lumpy plants; (6) deliberately infeasible plans

[2] research/preregistration/.

(caps below minimum requirements); (7) endogenous income through the Society Account; (8) binding ecological caps; (9) strategic misreporting, in which a fixed share of councils understate capacity by a fixed fraction, measuring the resulting price distortion and the misreporters' gain (reported as a diagnostic; no pass threshold is set until a baseline exists).

What a 1,000-council result can and cannot show. Phase 1 can show whether rounds-to-clear grow with the number of councils between 50 and 1,000 (a published log–log slope) and whether failure modes depend on scale. A flat or falling slope makes extrapolation plausible; a rising slope is evidence against national-scale feasibility. A pass at 1,000 councils does not show convergence at 10^6 participants: heterogeneity, the number of distinct goods, strategic behaviour and communication costs all change with scale, and Phase 2 measures the last of these separately.

9.1.2 Ethics, Consent and Data Protection

Phases 1H and 3 involve human participants and personal data. Before recruitment each requires: independent ethics review (a university research ethics committee if an academic partner is involved, otherwise an independent ethics panel whose membership is published); informed, written consent with a plain-language description of what is recorded, who can see it, and the right to withdraw and have data deleted; data minimisation and a retention schedule; a data-protection impact assessment where GDPR or UK GDPR applies (Article 35); and fair compensation for participants' time. Phase 3 adds a data-processing agreement with each pilot cooperative and a named contact for complaints. These add 4–10 weeks of elapsed time before each phase and the non-labour costs shown in Table 1.2.

9.1.3 The Effort-Rating Default (proposed)

Equation (4.7) has two parameters with different meanings. $\bar{\epsilon}_{\max}$ caps a council's mean effort multiplier and so prevents council-wide grade inflation; the individual spread $\epsilon_{\max}$, with $|\epsilon_u - 1| \leq \epsilon_{\max}$, decides whether two people working the same hours in the same council can be paid differently for intensity. Maxim 3 asks for that difference. O'Neill (2024) argues that rating it means pervasive surveillance and potential humiliation, and cooperatives often prefer flat hourly pay. This is a political choice, not a technical one. Proposed default (for author ratification): $\bar{\epsilon}_{\max} = 1.0$ in every council; individual spread $\epsilon_{\max} = 0$ (flat pay per accredited hour, with onerousness handled through job-complex balancing and hours) unless a council votes to allow peer-rated spread, capped at $\epsilon_{\max} = 0.15$ and reviewed after one plan year. Phase 1 simulates both settings; P6 tests them with cooperatives. Under this default DPEP honours Maxim 3 through hours and job balancing but not, by default, through intensity, and the monograph says so.

9.2 Multi-Scale Analytical Projections (10^2 to 10^6 Participants)

Methodological note: Only the pequod-cljs CSV convergence counts, the Python stress tests and the Prio3 benchmark have been executed.[3] Table 9.3 uses a 10-round annual plan with three fresh household-report rounds; Kill Gate 1's 25-round bound is a halt criterion, not the table's round count. Archive rows use the coarse single-key proposal sizes of Table 5.1; multi-signature councils multiply them by up to 3.3. All rows assume an $n = 128$ committee; small federations could use smaller committees.

Table 9.3: Analytical scaling projections across five population scales (10-round plan; SI units)

Metric (Category B analytical estimate)	100	1,000	10,000	100,000	1,000,000
Total councils & federations (M)	24	51	430	4,120	40,600
Priced categories ($G = \|\Omega\|$)	30	80	500	2,000	10,000
Council proposals / round	20	40	400	4,000	40,000
Raw leaf archive / round	119 kB	238 kB	2.4 MB	24 MB	237 MB
Raw leaf archive / year	1.2 MB	2.4 MB	24 MB	237 MB	2.4 GB
Mobile traffic / round (Role-LP), target	$\leq$ 150 kB ordinary round; $\leq$ 450 kB with full quorum check (Kill Gate 3)				
Layer-1 checkpoints / year (425 blocks, $n = 128$)	124 MB	125 MB	132 MB	158 MB	298 MB
Household-report CPU / aggregator / fresh round (desktop, coarse)	< 0.1 s	0.25 s	2.5 s	25 s	4 core-min
Mobile peak RAM target (Role-LP)					$\leq$ 150 MB (unmeasured)
Automated round time	Not estimated; measured in Phase 2 against Kill Gate 2				
Server hosting (excl. operations and audits)	\$0–\$5/mo	\$10–\$25/mo	\$80–\$150/mo	\$400–\$800/mo	\$3,000–\$6,000/mo

[3] Scripts and benchmark code are in research/analysis/ in the project repository.

CHAPTER 10

Open Theoretical, Institutional, Economic, and Technical Questions

> "A simulation incorporates more detail than a differential economic model, but still misses the richness of living interaction. As a result, the full characteristics of participatory economics and the diverse solutions to institutional problems associated with it will ultimately only reveal themselves in actual operations."
>
> — Michael Albert and Robin Hahnel, The Political Economy of Participatory Economics (1991), Ch. 6, p. 129[1]

We conclude our study by ranking the 15 most important unresolved questions across theoretical economics, institutional sociology, and distributed systems engineering, specifying how each will be investigated through either our simulation programme ([SIM]) or human-in-the-loop cooperative pilots ([HUMAN]):

Table 10.1: Ranked Research Agenda of Unresolved Questions (Priorities 1–15)

Rank	Unresolved Question	Domain	Experimental Investigation Protocol
P1	Convergence under non-convexities, setup costs & Leontief complementarities	[ECON / OR] ([SIM])	Replace unconstrained Cobb-Douglas in a new simulator (not by patching pequod-cljs FOCs); benchmark Algorithms 3a–3c on pre-registered instances.
P2	Stability of coupled endogenous income	[ECON] ([SIM])	Couple Society Account updates into active rounds with Algorithm 3d (damped budget fixed point; damping $\alpha \in (0,1]$ is a relaxation, not a second numéraire, which remains $w_{\text{base}} \equiv 1$). Report cycling, multiple fixed points and the effect of the one-round budget lag.
P3	Deliberative Complexity & Participation Fatigue in Multi-Round Planning	[INST / HCI] ([HUMAN])	Test the "3-Milestone Hybrid Schedule" (human rounds at $t = 1, 4, 8$ + automated intermediate solver rounds); measure NASA-TLX load.
P4	Biophysical Incommensurability vs. PDRM Pollution Pricing (O'Neill 2024)	[ECOLOGY] ([SIM/GOV])	Verify hard barrier convergence when aggregate demand hits absolute biophysical caps $(r^*, E_z^{\max})$ above local CAP PDRM pricing.

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[1] Albert and Hahnel, Political Economy of Participatory Economics, Ch. 6, p. 129.

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Rank	Unresolved Question	Domain	Experimental Investigation Protocol
P5	Coarse-to-Granular SKU Disaggregation & Slack Buffer Sizing	[ECON / OR] ([SIM])	Simulate 200 coarse reporting categories (the proposed privacy baseline, Section 7.4.2) mapped to 3,000 planning categories and 20,000 SKUs under stochastic preference drift, against slack $\eta \in [1\%, 10\%]$; measure the unmet-demand share it causes.
P6	Workplace Effort-Rating Dynamics: Solidarity vs. Surveillance or Parity	[INST] ([HUMAN])	Pilot with worker cooperatives comparing scaled effort multipliers vs. flat hourly parity ($\epsilon_u \equiv 1.0$) under BJC rotation.
P7	Preventing Industry Federation Cartelization & Gatekeeping (Kuch 2024)	[INST / ECON] ([HUMAN])	Test "Dual-Path Startup Accreditation" allowing rejected innovators to appeal to a sortition Innovation Jury + Consumer Federation.
P8	Calibrating Balanced Job Complexes Across Disparate Industries	[INST / OR] ([SIM/HUM])	Model cross-workplace BJC rotation as a bipartite matching problem in EFB accounting for travel & skill constraints.
P9	Inter-Regional Investment Apportionment Across Unequal Endowments	[ECON] ([SIM])	Compare per-capita vs. Porto Alegre need-weighted regional investment matrices over 10-year horizons.
P10	Reproductive & In-Home Caring Labour Integration (DEP Ch. 10)	[FEMINIST ECON] ([HUMAN])	Convene Women's and Caregivers' Caucuses to define care-allowance rules and household BJC norms.
P11	Byzantine Resilience Under Prolonged Regional Network Partitions	[DIST-SYS] ([SIM])	Benchmark Regional Proxy Extrapolation accuracy when 10% of councils are partitioned for multiple rounds.
P12	Standardisation of Compact Post-Quantum Threshold Signatures	[CRYPTO] ([SW-ARCH])	Councils use k-of-n multi-signatures (3,437 B per signer) until a standard post-quantum threshold signature exists; CryptoSuiteId agility allows the switch. NIST opened a call for multi-party threshold schemes in January 2026 (NIST IR 8214C), a reference process rather than a standards competition. Proposal sizes depend on k (Table 5.1).
P13	Privacy anonymity sets in small hamlets ($m < 30$)	[PRIVACY] ([SW-ARCH])	Apply $m_{\min} = 30$ per coarse reporting category and aggregate rare categories at Ward level with $m_{\text{rare}} = 250$; measure differencing leakage between milestone releases under the carry-forward rule of Section 7.4.3.

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Rank	Unresolved Question	Domain	Experimental Investigation Protocol
P14	Endpoint Security vs. Open Modifiability on User Smartphones	[SECURITY] ([SW-ARCH])	Use local hardware keystore unlock + social anomaly verification without mandatory proprietary OS attestation lock-in; split hybrid custody until Android holds post-quantum keys in hardware (Section 7.5).
P15	Bootstrapping a "Parallel Economy" Inside Surrounding Markets	[ADOPTION] ([HUMAN])	Implement a Dual-Price Fiat Reserve Bridge in Sandström's Society Account for federated cooperatives (PEPE Ch. 6 p. 129).

CHAPTER 11

Practical Adoption Strategy, Cooperative Pilots, and Theory of Transition

> "Conceivably one could run a 'shadow' participatory economy within an existing economy. This could occur in a subset of the whole economy, whether geographic or by industrial sector… not only [to] serve as a laboratory to test participatory planning, but also as a school, both educating the public and developing the means to implement participatory planning without undue disruption."
>
> — Michael Albert and Robin Hahnel, The Political Economy of Participatory Economics (1991), Ch. 6, "Experiment 2: Developing a Parallel Economy," pp. 129–130[1]

11.1 Distinguishing Research Simulation from Practical Cooperative Coordination

To avoid equating small-scale cooperative software with a full post-capitalist economy, DPEP separates two distinct applications:

1. The Scientific Research Platform (Phases 1–2): A computational laboratory for testing whether multi-tier iterative planning algorithms converge under realistic industrial non-convexities, input-output networks, and strategic incentives.
2. The Practical Cooperative Coordination Suite (Phase 3): Modular local-first tools allowing existing worker cooperatives, consumer buying clubs, community land trusts (CLTs), and participatory budgeting municipalities to coordinate procurement, shared equipment, and inter-cooperative production within a surrounding capitalist market economy.

11.2 Theory of Transition: Market Boundaries, Voluntary Exit, and Legal Exposure

Any voluntary federation of cooperatives experimenting with participatory planning inside a surrounding market economy must navigate four structural realities of a pilot, plus two that remain open even for a universal economy:

1. Market-priced inputs, external debts, and taxes (the dual-price boundary): Participating cooperatives cannot pay external steel suppliers, commercial

[1] Albert and Hahnel, Political Economy of Participatory Economics, Ch. 6, pp. 129–130.

Table 11.1: Six Early Practical Applications and Their Institutional Requirements

Application	DPEP Modules Used	Viable Inside Market Economy?	Institutional / Ownership Changes Needed for Full Parecon Semantics
1. Cooperative Joint Procurement	Layer 2/3 demand aggregation across food co-ops & worker co-ops.	Yes (Immediately): Legal B2B purchasing cooperative.	Requires pooled working-capital reserves to buffer seasonal supplier price swings.
2. Shared Equipment & Tool Trusts	Stand-Alone Capital Asset ledger charging user-right fees (P_k).	Yes: Via a Cooperative Equipment Trust or CLT.	Member co-ops transfer title or long-term lease rights of heavy machinery/kitchens to the shared trust.
3. Community Mutual-Aid Distribution	Layer 3/4 Maxim-4 need allocation & non-transferable credits.	Yes: Non-commercial mutual-aid association.	Bounded by the volume of goods produced by volunteer or grant-funded community labour.
4. Inter-Cooperative Production Planning	Quarterly ADMM/tâtonnement clearing across 10–50 federated co-ops.	Partially (Dual-Price Regime): Needs Fiat Reserve Bridge.	External raw materials, rent, debt service, and taxes must still be settled in fiat currency (Eq. 11.1).
5. Municipal Participatory Budgeting	Layer 4 "What-If" explorer, Need Matrix & Prio3 voting.	Yes: Municipal public capital budgeting.	Vulnerable to municipal coalition reversals and fiscal austerity (as in Porto Alegre's 2017 suspension).
6. Regional Co-op Supply Networks	Multi-year investment (σ_{inv}) & EFB worker reallocation.	Partially (e.g., Mondragón / Cecosesola).	Full Parecon requires abolishing enterprise profit retention and equalising BJC empowerment across plants.

landlords, bank mortgages, or national tax authorities in internal effort credits. Every pilot federation must enforce an explicit External Fiat Clearing Constraint alongside its internal Society Account:

$$\sum_{j \in \mathcal{J}} P_{\text{ext}}^{\text{fiat}} \cdot \bar{a}_j^{\text{export}} + D_{\text{fiat}}^{\text{dues}} \geq \sum_{j \in \mathcal{J}} P_{\text{ext}}^{\text{fiat}} \cdot \underline{a}_j^{\text{import}} + F^{\text{taxes}} + F^{\text{debt}} \tag{11.1}$$

where external imports $\underline{a}_j^{\text{import}}$ carry an internal shadow price $\lambda_{\text{fiat}} P_{\text{ext}}^{\text{fiat}}$ reflecting the scarcity of external fiat revenue.

2. Strict Voluntarism, Right of Exit, and Non-Participants (Learning from Aragon 1936): Informed by the conflict between Aragonese collectives and "individualist" smallholders (Casanova 2005), membership in any DPEP pilot or transitional federation must be strictly voluntary with a constitutionally codified Right of Exit: exiting individuals or cooperatives settle their net savings credits and refundable capital shares according to pre-agreed cooperative bylaws without confiscation, and non-participating artisans/households trade with the federation across transparent external interfaces.

3. Graduated sanctions without coercive labour discipline: Following Ostrom (1990), failure to meet council commitments triggers: (a) automatic SB_j / SC_j accounting adjustment, (b) UFB peer mediation, and (c) for persistent bad-faith breach, revocation of federation accreditation and access to shared federation inputs/credit—never forced labour. Losing access to federation inputs is mild in a voluntary pilot with an external market to exit into, and severe in a universal economy where exit is hard. The same rule therefore has two political meanings; a full-economy constitution needs a due-process and subsistence-access clause that a pilot can omit.

4. Jurisdiction-Specific Legal Exposure (Informational Analysis, Not Legal Advice):

   - Competition / Antitrust Law (US Sherman Act §1; EU TFEU Art. 101; UK Competition Act 1998): Independent corporations that coordinate prices and output quotas through an IFB risk prosecution as a horizontal price-fixing cartel unless structured under statutory cooperative/agricultural exemptions (e.g., US Capper-Volstead Act, EU producer-organisation rules) or incorporated as operating divisions of a single secondary cooperative / common-ownership consortium.
   - Taxation of Multilateral Barter & Clearing (US IRS Form 1099-B; EU-/UK VAT): Multilateral credit clearing is generally taxable at fair market value; pilot deployments must generate statutory fiat-equivalent tax reports where required by domestic law.
   - Data Protection & Right to Erasure (EU/UK GDPR Art. 17): DPEP is designed to avoid immutable personal ledger data by keeping personal profiles local, deleting Prio3 input shares when the challenge window closes

(Section 7.4.3), and storing qualitative Lists behind mutable council pointers with tombstone erasure. Pilots additionally need the ethics and data-protection steps of Section 9.1.2.

Open even for a universal economy. Foreign trade (how a DPEP polity prices and rations extra-federation goods, manages a foreign-exchange constraint, and avoids reproducing centre–periphery extraction) and defence (who commands organised coercion, how it is paid, and how it is made compatible with sortition and dual-federation ratification) are not specified for either a pilot or a full economy. They remain open political questions, not protocol features.

Six-Discipline Critical Examination and Remaining Weaknesses

Table 12.1 lists the strongest remaining criticisms, grouped by discipline. For each it gives a severity, the evidence that would settle it, the phase that would produce that evidence, and whether it could stop the project. Severity has three levels: Critical (if confirmed, the programme as conceived should stop or be fundamentally redesigned), Major (forces narrowing of scope or a significant redesign), and Moderate (a known kind of fix exists, but it is not yet specified or tested). Nothing in the table is marked resolved.

Table 12.1: Strongest remaining criticisms: severity, settling evidence, and phase

Criticism	Severity	Evidence that would settle it	Phase	Could stop?
Participatory economics and democratic theory				
Participation at scale: ordinary people will not sustain multi-round planning; turnout and attention will collapse after the novelty fades.	Critical	Completion, comprehension and fatigue (NASA-TLX) in Phase 1H; retention over a full plan year in pilots.	1H, 3	Yes
A technical elite forms around the software: facilitators, solver experts and articulate members set defaults and dominate amendments, re-creating a coordinator class.	Critical	Share of adopted amendments originated by the most active 20%; who sets solver defaults; rotation of facilitation roles (Kill Gate 1H(e)).	1H, 3	Yes
Federations become central planning authorities in disguise (Kuch 2024).	Major	Count and scope of federation decisions in pilots; outcomes of Innovation Jury appeals (P7).	3	No (narrows)
Flat effort pay by default ($\epsilon_{\max} = 0$) honours Maxim 3 only through hours and job balancing; rating intensity invites surveillance (O'Neill 2024).	Moderate	Cooperative pilots comparing flat and peer-rated pay (P6).	3	No
Cryptographic invisibility removes the 1991 channel of collective criticism of consumption.	Moderate	Members' stated preferences in pilots that offer both modes.	3	No
Operations research				
No candidate solver is shown to clear Leontief, lumpy and endogenous-income economies; 3b's theorem covers only fixed-budget convex cases, and 3d has no convergence result.	Critical	Kill Gate 1 on the pre-registered set, against an oracle.	1	Yes

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Criticism	Severity	Evidence that would settle it	Phase	Could stop?
One-good smoothing results do not transfer to interdependent markets.	Major	Lumpy Leontief families in the instance set.	1	No (narrows)
Rounds-to-clear may grow with the number of councils.	Major	Log–log scaling slope from 50 to 1,000 councils.	1	Yes
Distributed systems				
Round cost at 10^6 (0.24–1.46 GB archives, 40k–200k signature checks, 10^6 report verifications per aggregator) may not fit a usable round time.	Major	Kill Gate 2 measurements at $\geq 10^4$ with extrapolation from measured per-item costs.	2	Yes (for D; A remains)
Annual committee capture lasts a year; a 30% corrupt pool gives a 21% chance of an unsafe draw.	Major	Rotation frequency; operation of the interim threshold beacon (Section 7.2.1); a post-quantum beacon (open). Not settled by an experiment.	2	No
Cryptography				
Aggregate correctness needs all aggregators; the public budget check limits only gross overspend.	Major	Pilot use of non-nested aggregators; audit.	3	No
No construction for budget credentials and one-report-per-person; the budget circuit is unspecified and price-dependent.	Critical (for household crypto only)	A specified, audited construction: an interim classical credential with threshold issuance covers one report per person (Section 7.2.1); budget binding and the circuit remain open. Until then the council-level trust model.	3	Yes (household mode only)
Carry-forward cannot re-aggregate reports; milestone-only revision freezes household demand between milestones.	Moderate	Planning-accuracy cost in simulation (P5) and pilots.	1, 3	No
Cooperative practice				
30–60 self-selected volunteers say little about a general population.	Major	Larger, more representative pilots after Phase 1H.	3	No
Sanctions that cut federation inputs are mild in a pilot and severe in a universal economy.	Moderate	Constitutional due-process and subsistence clause (open).	–	No
Sceptical economist				
Ratchet effects and incentive limits (Weitzman, Hurwicz) remain; public blueprints do not remove them.	Major	Strategic-misreporting family in Phase 1 (diagnostic); pilots.	1, 3	No
The labour floor $p^{\text{lab}} \geq w_{\text{base}}$ blocks surplus clearing.	Moderate	Phase 1 instances with labour surplus.	1	No
Foreign trade and defence are unspecified.	Major	Not settled by any experiment here; a political design question.	–	No (scope limit)

APPENDIX A

Scientific Claims, Evidence, and Validation Status Register

Each claim has eight separate fields: identifier, claim, classification, status (first line) and evidence, assumptions, limitations and required validation (second line). Classifications: Established (peer-reviewed theorem or standard), Source-supported (checked in primary literature, audited code or a measurement), Analytical (derived from models or arithmetic in this monograph), Conjectural (hypothesis or judgement), Experimental target (a goal to be tested). Statuses: Verified, Partially verified, Unverified, Refuted.

Table A.1: Master Register of Scientific and Engineering Claims

Claim ID / Evidence	Claim / Assumptions	Classification / Limitations	Status / Validation
CLM-ECON-01	Economic convergence (reference): the continuous-time FMPE gradient process converges.	Class: Established	Status: Verified (as a cited theorem; not a DPEP result)
Evidence: Arrow–Hurwicz–Uzawa (1958); Heal (1973), Tier 3.	Assumptions: Strict convexity; gross substitutability.	Limitations: Fails on Scarf-type and non-convex economies.	Validation: Reference only.
CLM-ECON-02	dep1ex61–100.csv: Year 1 reaches 5% in 11.85 rounds and 3% in 19.20; Year 2 5% in 6.58.	Class: Source-supported	Status: Verified
Evidence: Audit script[1] on pequod-cljs @ 8bd46b5c.	Assumptions: Fixed \$5,000 incomes; Cobb–Douglas FOCs.	Limitations: Other subsets differ (01–50: 11.18); no wall-clock log.	Validation: Reproduce in the Phase 1 simulator.
CLM-ECON-03	Eq. (4.15) plus trust regions does not clear Scarf's cycle or a single lumpy plant.	Class: Analytical	Status: Verified
Evidence: Simulation script[2]: min imbalance 10.5% and 50.4%.	Assumptions: The stress-test economies.	Limitations: Not a general impossibility proof.	Validation: Instance families 3–4 in Phase 1.
CLM-ECON-04	With the toothache increment, $\sum b + c/k > 1$ for every one of 750,000 worker-council records in the 70 generators.	Class: Source-supported	Status: Partially verified
Evidence: Dataset audit script[3]; LFS objects SHA-256 checked.	Assumptions: 20% of councils selected at run time.	Limitations: Perverse supply predicted, not observed in CSVs; Clojure not run.	Validation: Run pequod-cljs with toothaches.

[1] research/analysis/verify_r2.py in the project repository.
[2] research/analysis/verify_economics.py in the project repository.
[3] research/analysis/verify_pequod_datasets.py in the project repository.

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Claim ID / Evidence	Claim / Assumptions	Classification / Limitations	Status / Validation
CLM-ECON-05	One good, $K = 20$ staggered plants: min imbalance 0.40% over 500 rounds; last-100-round max 8.4%.	Class: Analytical	Status: Partially verified
Evidence: Simulation script[4]; sweep median 6.0%, worst 9.5%.	Assumptions: $a \in [60, 80]$; thresholds in $[8.5, 11.5]$.	Limitations: One good; not Leontief.	Validation: Lumpy Leontief family in Phase 1.
CLM-ECON-06	$\sum_u B_u = (1 - \sigma_{\text{inv}})Y^{\text{GDP}}$, and $\sigma_{\text{inv}} Y^{\text{GDP}} = p \cdot D^{\text{inv}}$ at a cleared plan.	Class: Analytical	Status: Partially verified
Evidence: Algebra in Section 4.2 and Table 4.1.	Assumptions: Y^{allow} is cash transfers only.	Limitations: No numerical test on a full Society Account ledger.	Validation: Phase 1 ledger test.
CLM-ECON-07	Algorithm 3b (Deng et al. Algorithm 4 with slack) converges on fixed-budget convex problems.	Class: Established	Status: Partially verified
Evidence: Deng et al. (2017) Thm. 2.3–2.4 (read in full); test B in the analysis script.[5]	Assumptions: Convex $\mathcal{T}_c$; fixed budgets; P_i condition (2.11).	Limitations: Not endogenous income, non-convexity or dual renormalisation.	Validation: Phase 1 convex families.
CLM-ECON-08	Some candidate (3a–3d) clears full DPEP instances (Leontief, lumpy, endogenous income, strategic).	Class: Experimental target	Status: Unverified
Evidence: None yet.	Assumptions: Pre-registered instance set.	Limitations: Toy tests only; 3d has no convergence result.	Validation: Kill Gate 1.
CLM-ECON-09	One-sided feasibility ($D \leq S$) is an adequate success criterion.	Class: Analytical	Status: Refuted
Evidence: Single plant: feasible at round 3 with 42% unused output (reference test script)[6].	Assumptions: –	Limitations: –	Validation: Replaced by Definition 4.2.
CLM-TECH-01	Leaf transport plus BFT takes 9–27 s per round at $N = 10^6$.	Class: Analytical	Status: Refuted
Evidence: Assumed 5.2 kB proposals; reworked sizes give 0.24–1.46 GB per round (Table 5.1).	Assumptions: –	Limitations: Omitted report verification and signature checks.	Validation: Replaced by Kill Gate 2 thresholds.

[4] research/analysis/verify_r2.py in the project repository.
[5] research/analysis/verify_r3.py in the project repository.
[6] research/analysis/reference_tests_r3.py in the project repository.

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Claim ID / Evidence	Claim / Assumptions	Classification / Limitations	Status / Validation
CLM-TECH-02	Computational scalability: Prio3 verification costs ≈ 0.25 ms (coarse) to 3.5 ms (dense) per report per aggregator.	Class: Source-supported	Status: Partially verified
Evidence: libprio 0.18.1 benchmark (Table 7.5).	Assumptions: One desktop core; VDAF draft 18 code.	Limitations: No phone or Raspberry Pi measurement; network excluded.	Validation: Device benchmarks, Kill Gate 3.
CLM-TECH-03	Prio3SumVec report sizes follow the VDAF size formula.	Class: Source-supported	Status: Verified
Evidence: The size model[7] matches libprio byte-for-byte on 9 configurations.	Assumptions: libprio chunk-length rule.	Limitations: Custom budget circuit sizes unknown.	Validation: Recompute when the circuit is specified.
CLM-TECH-04	Mobile feasibility: ≤ 150 kB per ordinary round and ≤ 450 kB with a full quorum check; RAM ≤ 150 MB.	Class: Experimental target	Status: Unverified
Evidence: Byte budget analytical (Section 7.6).	Assumptions: Coarse 55 kB report; coordinator attestation.	Limitations: No device test.	Validation: Kill Gate 3.
CLM-TECH-05	Operating cost: hosting 100 servers $\approx$ \$3k–\$6k/month.	Class: Analytical	Status: Unverified
Evidence: \$30–\$60 per server-month (assumed).	Assumptions: Commodity servers.	Limitations: Excludes audits, facilitation, research and energy.	Validation: Phase 2 and 3 cost accounting.
CLM-TECH-06	Architecture D outranks A, B and C at baseline weights.	Class: Conjectural	Status: Partially verified
Evidence: Expert scores; arithmetic and perturbation test in the analysis script.[8]	Assumptions: Table 6.1 scores.	Limitations: A overtakes D when simplicity, throughput and cost carry $\geq 63\%$ of the weight.	Validation: Phases 2–3 replace judgements with measurements.
CLM-TECH-07	Component maturity: every function except threshold signatures, anonymous credentials, the beacon, capability-certificate formats and post-quantum HPKE has a standardised, deployed primitive.	Class: Source-supported	Status: Partially verified
Evidence: Table 6.3; maturity dossier[9] (web checks, November 2026).	Assumptions: Ratings are judgements on the M0–M5 scale.	Limitations: Primitive maturity is not library or combination maturity; nine combinations are original.	Validation: Re-check before each gate; Kill Gate 3 audit.

[7] research/analysis/parameters.py in the project repository.
[8] research/analysis/verify_r3.py in the project repository.
[9] reports/STACK_MATURITY.md in the project repository.

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Claim ID / Evidence	Claim / Assumptions	Classification / Limitations	Status / Validation
CLM-SEC-01	Privacy: Prio3 hides an individual report if at least one aggregator is honest.	Class: Source-supported	Status: Partially verified
Evidence: VDAF-22 §9 (full text).	Assumptions: Non-colluding aggregators; $m_{\min} = 30$.	Limitations: Budget circuit unspecified; differencing between releases.	Validation: Phase 3 audit.
CLM-SEC-02	Censorship resistance: Two-Root aggregation with receipts detects omitted or tampered proposals.	Class: Analytical	Status: Unverified
Evidence: Design argument (Section 5.2.1).	Assumptions: ≥ 1 honest relay; clients keep receipts.	Limitations: Does not bind aggregator shares.	Validation: Kill Gate 2(a).
CLM-SEC-03	Post-quantum security: the FIPS 203/204/205 primitives are standardised.	Class: Established	Status: Verified
Evidence: NIST FIPS 203–205 (2024).	Assumptions: Module-LWE/SIS and hash assumptions.	Limitations: No whole-system PQ claim; BLAKE3 128-bit; VDAF no PQ claim.	Validation: Phase 3 review.
CLM-SEC-04	Cryptographic integrity: aggregate correctness requires every aggregator to be honest.	Class: Source-supported	Status: Verified
Evidence: VDAF-22 §9 quotation.	Assumptions: Standard Prio3.	Limitations: The public budget check limits only gross overspend.	Validation: Non-nested aggregators in pilots.
CLM-SEC-05	Sybil resistance (partial): INV-4B bounds credited hours given a true roster.	Class: Analytical	Status: Partially verified
Evidence: Eq. (4.7).	Assumptions: Roster and SB/SC honest.	Limitations: Falsified rosters and carousel fraud need physical audit.	Validation: Pilot audits.
CLM-SEC-06	Sybil resistance (household): one report per person without a population register.	Class: Conjectural	Status: Unverified
Evidence: No post-quantum construction (Section 7.4.4); interim classical credential adopted (Section 7.2.1).	Assumptions: An anonymous credential with threshold issuance and nullifiers.	Limitations: Budget binding open; interim scheme not post-quantum.	Validation: Research; council-level trust meanwhile.
CLM-SEC-07	An $n = 128$ annual draw with $p = 20\%$ corrupt gives a 0.022% unsafe committee.	Class: Analytical	Status: Verified
Evidence: Exact binomial computation.[10]	Assumptions: i.i.d. draws; failure at $k \geq 43$.	Limitations: $p = 30\%$ gives 21%; capture lasts a year.	Validation: Beacon design.

[10] research/analysis/verify_r2.py in the project repository.

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Claim ID / Evidence	Claim / Assumptions	Classification / Limitations	Status / Validation
CLM-SEC-08	Prio3 reports cannot be carried forward into later rounds' aggregates.	Class: Source-supported	Status: Verified
Evidence: VDAF-22 §7.2.3, §9.4.	Assumptions: Standard Prio3.	Limitations: Aggregate-level carry-forward is still possible.	Validation: –
CLM-SEC-09	An off-the-shelf BFT engine accepts post-quantum validator keys.	Class: Source-supported	Status: Verified
Evidence: Cosmos SDK documentation (CometBFT 0.40, Cosmos SDK 0.55).	Assumptions: Releases as documented.	Limitations: Pure ML-DSA-65, not hybrid; ≈ 331 kB of signatures per block at 100 validators; release dates not confirmed.	Validation: Kill Gate 2.
CLM-SEC-10	An interim classical credential keeps past showings unlinkable after a CRQC.	Class: Conjectural	Status: Unverified
Evidence: None for a chosen scheme; RFC 9474 states no assumption for blindness.	Assumptions: Statistically zero-knowledge presentation and statistical blinding.	Limitations: If false, a CRQC reveals who submitted, not what.	Validation: Scheme selection before Phase 3.
CLM-SEC-11	Canary batches detect indiscriminate tampering with aggregate shares.	Class: Analytical	Status: Unverified
Evidence: Argument in Section 7.4.	Assumptions: Per-batch aggregate shares; batches indistinguishable to aggregators.	Limitations: Misses tampering aimed at batches known to be real.	Validation: Fault injection in Phase 3.
CLM-POL-01	Democratic scalability: ordinary participants complete, understand and genuinely shape multi-round planning.	Class: Experimental target	Status: Unverified
Evidence: None yet.	Assumptions: Phase 1H design.	Limitations: $n = 30$–60 self-selected volunteers.	Validation: Kill Gate 1H.
CLM-POL-02	No technical elite dominates proposals and defaults.	Class: Experimental target	Status: Unverified
Evidence: None yet.	Assumptions: Influence measure of Kill Gate 1H(e).	Limitations: Small, short probe.	Validation: Kill Gate 1H and pilots.
CLM-POL-03	Neighbourhood councils of ~ 50–200 are workable.	Class: Conjectural	Status: Unverified
Evidence: Speaking-time arithmetic.	Assumptions: One shared meeting hour.	Limitations: Not a finding of Mansbridge or Dahl & Tufte.	Validation: Pilots.

Comprehensive Source Bibliography

Each entry's bibliographic identity was checked in November 2026 against Crossref (by DOI or bibliographic query), a publisher or journal page, or the IETF Datatracker.[11] That is not a claim that every page citation was re-read; access levels are in the front-matter notice.

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